

Online Appendix to Product Switching and Young Firm Dynamics

Seula Kim

Karam Jo

September 28, 2026

This Online Appendix contains the full theoretical proofs and supporting empirical and quantitative material. The main paper includes concise proof sketches and supporting theoretical derivations.

OA.A Theory Proofs

OA.A.1 Proof of Lemma 1

Proof. Fix the true θ_{ip} , an observation window of length τ , and the information-processing path. Expectations average over all potential signal histories within this window, without conditioning on endogenous product retention. Taking expectations over the noise terms in the observed outcomes and additional signals in (2.3) yields

$$\begin{aligned}\mathbb{E}_{\varepsilon,v} [\hat{\theta}_{ipt} \mid \theta_{ip}] &= \hat{\sigma}_{ipt}^2 \left[\frac{\theta_0}{\sigma_0^2} + \frac{\tau + \sum_{\tilde{t}=t-\tau+1}^t \psi_{i\tilde{t}}}{\sigma_\varepsilon^2} \log \theta_{ip} \right] \\ &= \log \theta_{ip} + \frac{\hat{\sigma}_{ipt}^2}{\sigma_0^2} (\theta_0 - \log \theta_{ip}),\end{aligned}$$

where the second equality follows from (2.4). Thus, the expected posterior mean lies between the prior mean θ_0 and the true $\log \theta_{ip}$.

Holding earlier information-processing abilities fixed,

$$\frac{\partial \hat{\sigma}_{ipt}^2}{\partial \psi_{it}} = -\frac{(\hat{\sigma}_{ipt}^2)^2}{\sigma_\varepsilon^2} < 0.$$

Consequently,

$$\frac{\partial \mathbb{E}_{\varepsilon,v} [\hat{\theta}_{ipt} \mid \theta_{ip}]}{\partial \psi_{it}} = \frac{(\hat{\sigma}_{ipt}^2)^2}{\sigma_\varepsilon^2 \sigma_0^2} (\log \theta_{ip} - \theta_0).$$

Therefore, lower ψ_{it} increases posterior variance and magnifies the downward conditional bias when $\log \theta_{ip} > \theta_0$ and the upward conditional bias when $\log \theta_{ip} < \theta_0$. When $\log \theta_{ip} = \theta_0$, the conditional bias is zero regardless of information-processing ability. $\square$

OA.A.2 Destination-Product Assignment

This appendix describes the assignment of destination products across simultaneously searching product lines and establishes that, although destinations are drawn without replacement across lines, each searching line is marginally equally likely to draw any feasible product. This result allows us to suppress expectations over destination-product draws in the product-line value functions under symmetric product markets.

Consider firm i with J_{it}^s product lines searching simultaneously. The firm's exclusion set contains its currently operated products and the most recent destination associated with each vacant line, leaving $N - n(\varphi_i)$ feasible destination products. Since $J_{it}^s \leq n(\varphi_i) \leq \lfloor N/2 \rfloor$, we have $J_{it}^s \leq N - n(\varphi_i)$ for every combination of line-level search decisions. Destinations are drawn uniformly without replacement across the J_{it}^s searching lines.

Fix a searching line k and a feasible destination product p' . One way to view the assignment procedure is to first draw J_{it}^s distinct products from the feasible set and then randomly assign them across the J_{it}^s ordered product lines. The probability that p' is included among the selected products is

$$\frac{J_{it}^s}{N - n(\varphi_i)},$$

and, conditional on being selected, its probability of being assigned to line k is $1/J_{it}^s$. Hence,

$$\Pr(p'_k = p') = \frac{J_{it}^s}{N - n(\varphi_i)} \frac{1}{J_{it}^s} = \frac{1}{N - n(\varphi_i)}. \quad (\text{OA.A.1})$$

Thus, every feasible product is marginally equally likely to be assigned to any given searching line.

Equivalently, consider drawing destinations sequentially for the J_{it}^s ordered searching lines. Any particular ordered assignment of J_{it}^s distinct products occurs with probability

$$\frac{1}{N - n(\varphi_i)} \frac{1}{N - n(\varphi_i) - 1} \cdots \frac{1}{N - n(\varphi_i) - J_{it}^s + 1}.$$

There are

$$J_{it}^s! \binom{N - n(\varphi_i)}{J_{it}^s} = \frac{[N - n(\varphi_i)]!}{[N - n(\varphi_i) - J_{it}^s]!}$$

possible ordered assignments. Multiplying the number of assignments by the probability of each assignment gives one, confirming that the procedure assigns equal probability to every ordered set of distinct destinations. Equation (OA.A.1) then follows for any individual line.

Since search succeeds independently with probability x , the probability that line k successfully adopts a particular feasible product p' is

$$\Pr(\text{line } k \text{ adopts } p') = \frac{x}{N - n(\varphi_i)}.$$

Therefore, for any destination-specific continuation value $W(p')$,

$$\mathbb{E}_{p'_k}[W(p'_k)] = \frac{1}{N - n(\varphi_i)} \sum_{p' \in \mathcal{P}_{it}} W(p'),$$

where $\mathcal{P}_{it}$ denotes the firm's feasible destination set. Under symmetric product markets, $W(p')$ is identical across feasible destinations, so the expectation equals the common continuation value. Expectations over destination-product draws can therefore be omitted from the product-line value functions.

OA.A.3 Threshold Properties

This appendix establishes the monotonicity and single-crossing properties underlying the search and product-drop thresholds, proves Proposition 1, and derives additional implications of the threshold conditions. We take stationary aggregate prices and quantities as given. Firm age, information-processing ability, and posterior beliefs continue to evolve as specified in the model.

OA.A.3.1 Monotonicity and Single Crossing

Lemma OA.1 (Monotonicity and Single Crossing). *For fixed product tenure τ , firm-level states, and aggregate prices and quantities, and $0 < x < 1$, $V_{it}^{ns}(\hat{\theta}, \tau)$ and $V_{it}^{cs}(\hat{\theta}, \tau)$ are continuous and strictly increasing in $\hat{\theta}$, while $V_{it}(\hat{\theta}, \tau)$ is continuous and weakly increasing. Moreover, $V_{it}^{cs}(\hat{\theta}, \tau) - V_{it}^{ds}(\emptyset)$ and $V_{it}^{ns}(\hat{\theta}, \tau) - V_{it}^{cs}(\hat{\theta}, \tau)$ are strictly increasing in $\hat{\theta}$.*

Proof. The proof proceeds in three steps. First, because flow profits are unbounded in $\hat{\theta}$, we bound their growth and show that discounting makes the infinite-horizon tail vanish. Second, we establish continuity and monotonicity using finite-horizon approximations. Finally, we use the Bellman equations to establish single crossing. This follows the recursive approach of [Stokey and Lucas Jr \(1989\)](#).

Step 1: Finiteness and finite-horizon approximation. Let $r \equiv \eta - 1 > 0$ and define

$$H_{it}(\hat{\theta}, \tau) \equiv \exp \left\{ r\hat{\theta} + \frac{r^2}{2} \hat{\sigma}_{ipt}^2 \right\}.$$

Since $\hat{\theta}' \mid (\hat{\theta}, \tau) \sim N(\hat{\theta}, \hat{\sigma}_{ipt}^2 - \hat{\sigma}_{ip,t+1}^2)$, the normal moment-generating function implies

$$\mathbb{E}_{it} \left[H_{it+1}(\hat{\theta}', \tau + 1) \right] = H_{it}(\hat{\theta}, \tau). \quad (\text{OA.A.2})$$

Here, expectations condition on the current product-line and firm-level states. Thus, H_{it} is a martingale under the posterior-belief transition for a retained product.

Moreover,

$$\mathcal{B}_{it}(\hat{\theta}, \tau)^\eta = \exp \left\{ r\hat{\theta} + \frac{r^2}{2\eta} (\hat{\sigma}_{ipt}^2 + \sigma_\varepsilon^2) \right\} \leq \exp \left\{ \frac{r^2}{2\eta} \sigma_\varepsilon^2 \right\} H_{it}(\hat{\theta}, \tau).$$

With stationary aggregates and fixed firm productivity, (2.10) therefore implies that there exists a finite constant K , uniform over dates, such that

$$\left| \bar{\pi}_{it+1}(\hat{\theta}, \tau) \right| \leq K \left[1 + H_{it}(\hat{\theta}, \tau) \right]. \quad (\text{OA.A.3})$$

Search costs are also bounded.

Let $H^0 \equiv \exp\{r\theta_0 + r^2\sigma_0^2/2\} < \infty$. Retaining the current product preserves H in expectation by (OA.A.2), while successful switching resets beliefs to the initial prior. Consequently, expected absolute payoffs j periods ahead are bounded, under any feasible policy, by a finite multiple of $1 + H_{it}(\hat{\theta}, \tau) + jH^0$. For some finite K' , the expected absolute discounted payoff from period T onward is therefore bounded by

$$K' \sum_{j=T}^{\infty} \beta^j \left[1 + H_{it}(\hat{\theta}, \tau) + jH^0 \right] \longrightarrow 0 \quad \text{as } T \rightarrow \infty. \quad (\text{OA.A.4})$$

The convergence follows from $\beta^T \rightarrow 0$ and $T\beta^T \rightarrow 0$ for $0 < \beta < 1$. Thus, discounting dominates the growth of the unbounded payoffs, and finite-horizon approximations converge to the infinite-horizon value. The convergence is uniform on compact subsets of $\hat{\theta}$.

Step 2: Continuity and monotonicity. Let $V^{(T)}$ denote the product-line value obtained by truncating the problem after T periods and assigning zero terminal continuation value. We show by induction that $V^{(T)}$ is continuous and weakly increasing in $\hat{\theta}$. The result is immediate for $T = 0$. Suppose it holds for $V^{(T)}$.

Consider $\hat{\theta}^H > \hat{\theta}^L$, holding tenure and firm-level states fixed. Since the transition variance is independent of the current posterior mean, the two transitions can be coupled using the same $Z \sim N(0, 1)$:

$$\hat{\theta}'^H = \hat{\theta}^H + \sqrt{\Delta} Z, \quad \hat{\theta}'^L = \hat{\theta}^L + \sqrt{\Delta} Z,$$

where $\Delta \equiv \hat{\sigma}_{ipt}^2 - \hat{\sigma}_{ip,t+1}^2$. Thus, $\hat{\theta}'^H > \hat{\theta}'^L$ for every realization of Z . By the induction hypothesis,

$$V_{it+1}^{(T)}(\hat{\theta}'^H, \tau + 1) \geq V_{it+1}^{(T)}(\hat{\theta}'^L, \tau + 1)$$

for every realization, so expected continuation values are weakly increasing in the current posterior mean.

Since $\mathcal{B}_{it}$ and $\bar{\pi}_{it+1}$ are strictly increasing in $\hat{\theta}$, the no-search action value generated from $V^{(T)}$ is strictly increasing. For $0 < x < 1$, the retain-and-search action value is also strictly increasing because the firm retains the current product following failed search with positive probability. The drop-and-search action value is independent of the current $\hat{\theta}$. Taking the maximum therefore yields a $V^{(T+1)}$ that is weakly increasing in $\hat{\theta}$.

Continuity is preserved by the Bellman operator. Flow profits are continuous in $\hat{\theta}$; the Gaussian transition and payoff bounds make expected continuation values continuous; and the maximum of finitely many continuous action values is continuous. Hence, $V^{(T)}$ is continuous and weakly increasing for every T .

By (OA.A.4), $V^{(T)}$ converges to V uniformly on compact subsets. The limiting value V is therefore continuous and weakly increasing. Evaluating the no-search and retain-and-search Bellman equations at V establishes that $V_{it}^{ns}(\hat{\theta}, \tau)$ and $V_{it}^{cs}(\hat{\theta}, \tau)$ are continuous and strictly increasing in $\hat{\theta}$.

Step 3: Single crossing. Define

$$\mathcal{V}_{it}(\hat{\theta}, \tau) \equiv \bar{\pi}_{it+1}(\hat{\theta}, \tau) + (1 - \delta)\mathbb{E}_{it}V_{it+1}(\hat{\theta}', \tau + 1).$$

Let $\mathcal{V}_{it}^0$ denote this expression evaluated at the common prior $(\hat{\theta}^0, 0)$, and define $\mathcal{V}_{it}^\emptyset \equiv (1 - \delta)V_{it+1}^{ds}(\emptyset)$. By Step 2, $\mathcal{V}_{it}(\hat{\theta}, \tau)$ is continuous and strictly increasing in $\hat{\theta}$.

Using the Bellman equations and the normalization $V_{it+1}^\delta = 0$ gives

$$\begin{aligned} V_{it}^{cs}(\hat{\theta}, \tau) - V_{it}^{ds}(\emptyset) &= (1-x)\beta \left[\mathcal{V}_{it}(\hat{\theta}, \tau) - \mathcal{V}_{it}^\emptyset \right], \\ V_{it}^{ns}(\hat{\theta}, \tau) - V_{it}^{cs}(\hat{\theta}, \tau) &= w_t f^s + x\beta \left[\mathcal{V}_{it}(\hat{\theta}, \tau) - \mathcal{V}_{it}^\emptyset \right]. \end{aligned}$$

Since $0 < x < 1$, both differences are continuous and strictly increasing in $\hat{\theta}$, establishing single crossing. $\square$

OA.A.3.2 Proof of Proposition 1

We first state the low-belief condition used in the proposition.

Assumption OA.1 (Interior Thresholds). *For each relevant combination of product tenure and firm-level states, $0 < x < 1$ and, for all sufficiently low $\hat{\theta}$,*

$$V_{it}^{ns}(\hat{\theta}, \tau) < V_{it}^{cs}(\hat{\theta}, \tau) < V_{it}^{ds}(\emptyset).$$

This condition ensures that both pairwise indifference conditions have finite solutions. Their ordering requires the separate condition in (2.18). Both conditions can be checked in the computed equilibrium.

Proof of Proposition 1. Fix product tenure and the suppressed firm-level states. By Lemma OA.1, the differences $V_{it}^{cs}(\hat{\theta}, \tau) - V_{it}^{ds}(\emptyset)$ and $V_{it}^{ns}(\hat{\theta}, \tau) - V_{it}^{cs}(\hat{\theta}, \tau)$ are continuous and strictly increasing in $\hat{\theta}$. Assumption OA.1 makes both differences negative for sufficiently low $\hat{\theta}$.

From Step 3 of the lemma's proof, these differences are $(1-x)\beta[\mathcal{V}_{it}(\hat{\theta}, \tau) - \mathcal{V}_{it}^\emptyset]$ and $w_t f^s + x\beta[\mathcal{V}_{it}(\hat{\theta}, \tau) - \mathcal{V}_{it}^\emptyset]$, respectively. Expected operating profits diverge as $\hat{\theta} \rightarrow \infty$, while the continuation value is bounded below by the feasible drop-and-search alternative. Thus, $\mathcal{V}_{it}(\hat{\theta}, \tau) \rightarrow \infty$. Because $0 < x < 1$, both value differences tend to $+\infty$. The intermediate value theorem and strict monotonicity therefore give unique finite zeros, $\hat{\theta}^{ds}$ and $\hat{\theta}^{cs}$.

Since $V_{it}^{ns} - V_{it}^{cs}$ is strictly increasing and equals zero at $\hat{\theta}^{cs}$,

$$\hat{\theta}^{cs} > \hat{\theta}^{ds} \iff V_{it}^{ns}(\hat{\theta}^{ds}, \tau) < V_{it}^{cs}(\hat{\theta}^{ds}, \tau) = V_{it}^{ds}(\emptyset).$$

Equation (2.14) also implies

$$V_{it}^{cs}(\hat{\theta}, \tau) - V_{it}^{ns}(\hat{\theta}, \tau) = -w_t f^s + x \left[V_{it}^{ns}(\hat{\theta}^0, 0) - V_{it}^{ns}(\hat{\theta}, \tau) \right].$$

Evaluating this expression at $\hat{\theta}^{ds}$ gives

$$\hat{\theta}^{cs} > \hat{\theta}^{ds} \iff V_{it}^{ns}(\hat{\theta}^0, 0) - V_{it}^{ns}(\hat{\theta}^{ds}, \tau) > \frac{w_t f^s}{x},$$

which establishes (2.18).

When this condition holds, $V^{ds} > V^{cs} > V^{ns}$ below $\hat{\theta}^{ds}$; V^{cs} exceeds both alternatives between $\hat{\theta}^{ds}$ and $\hat{\theta}^{cs}$; and $V^{ns} > V^{cs} > V^{ds}$ above $\hat{\theta}^{cs}$. The stated policy follows, up to indifference at the thresholds. Search after dropping remains conditional on firm continuation, as specified in the product-line problem.

Finally, $\mathcal{B}_{it}(\hat{\theta}, \tau)$ is strictly increasing in $\hat{\theta}$. The ordered posterior-mean cutoffs therefore uniquely define $\mathcal{B}_{it}^{ds}(\tau) < \mathcal{B}_{it}^{cs}(\tau)$. $\square$

The drop-threshold condition. At the drop threshold, the firm is indifferent between retaining the product while searching and dropping it. Subtracting (2.15) from (2.14) cancels the common search costs and successful-search payoffs. Since $0 < x < 1$, indifference requires

$$V_{it}^{ns}(\hat{\theta}^{ds}, \tau) = \beta \left[(1 - \delta) V_{it+1}^{ds}(\emptyset) + \delta V_{it+1}^{\delta} \right].$$

Substituting (2.13) and cancelling the common exogenous-exit payoff yields

$$\bar{\pi}_{it+1}(\hat{\theta}^{ds}, \tau) + (1 - \delta) \mathbb{E} \left[V_{it+1}(\hat{\theta}', \tau + 1) \mid \hat{\theta}^{ds}, \tau \right] = (1 - \delta) V_{it+1}^{ds}(\emptyset), \quad (\text{OA.A.5})$$

where the expectation also conditions on the suppressed firm-level states. At this margin, expected operating profit and continuation value from retaining the product equal the continuation value of a vacant line.

Product values and expected demand. At the search threshold, (2.16) implies that a new product has a higher no-search value than the marginal existing product whenever $f^s > 0$.

This need not imply a higher expected demand term $\mathcal{B}$, because values also reflect posterior uncertainty and future learning.

For an existing product with tenure $\tau_{ip_k t}$ and posterior variance $\hat{\sigma}_{ip_k t}^2$, equation (2.8) gives

$$\mathcal{B}_{it}(\hat{\theta}_{ip_k t}^0, 0) > \mathcal{B}_{it}^{cs}(\tau_{ip_k t}) \iff \hat{\theta}_{ip_k t}^{cs} < \theta_0 + \frac{\eta - 1}{2\eta} [\sigma_0^2 - \hat{\sigma}_{ip_k t}^2].$$

For positive product tenure, $\hat{\sigma}_{ip_k t}^2 < \sigma_0^2$. Thus, $\hat{\theta}_{ip_k t}^{cs} \leq \theta_0$ is sufficient for the new product's expected demand term to exceed the search threshold. The comparison holds firm age and fundamental types fixed: adopting a new product resets product-specific beliefs and tenure, not the firm's information-processing ability.

OA.A.4 Proof of Lemma 2

Proof. The value of product line k is determined by flow profits, product-switching decisions, and the firm's exit decision. Conditional on the common firm-level states $(a_{it}, \psi_{it}, \varphi_i, m_i)$, flow profits and the evolution of posterior beliefs depend only on the line's own state $(\hat{\theta}_{ip_k t}, \tau_{ip_k t})$.

When the product in line k is dropped, its continuation value depends on whether the firm exits or continues to search, where the endogenous-exit decision depends on the states of the firm's other product lines. By assumption, both outcomes yield the same line-level continuation value $V_{it}^{ds}(\emptyset)$. Therefore, although endogenous exit depends on posterior beliefs in other product lines, this dependence does not affect the value of line k . Since the same payoff equivalence applies at every future decision date, it removes portfolio dependence from the recursive continuation value of the line.

It follows that the value of product line k depends only on $(\hat{\theta}_{ip_k t}, \tau_{ip_k t}; a_{it}, \psi_{it}, \varphi_i, m_i)$ and not on posterior beliefs in the firm's other product lines. This establishes independence across product lines. $\square$

OA.A.5 Product-Line Distribution

Because tracking the joint portfolio Θ_{it} is computationally infeasible, we instead evolve marginal product-line measures. This representation is exact except for the portfolio event determining whether a vacant or newly dropped line reaches the search stage, which we approximate using

the marginal distribution of non-focal product lines.

For computation, we extend μ_{pt} to include vacant lines. The measure $\mu_{pt}(\hat{\theta}, \tau, a)$ denotes the mass of active product lines associated with market p , while $\mu_{pt}(\emptyset, a)$ denotes the mass of vacant lines whose most recent search destination is p . Initially vacant lines of age-zero startups have not yet drawn a destination and are therefore handled directly in their first search. To simplify notation, the firm-level states (ψ, φ, m) are suppressed throughout this section; all measures, transition densities, and policy regions are understood to be conditional on their appropriate values. We retain age explicitly because the transition differs for new entrants, age-one startups, and older incumbents. Dependence on φ is shown when scope $n(\varphi)$ enters directly. We write $g(\hat{\theta}' | \hat{\theta}, \tau, a)$ for the transition density of the posterior mean from a line with state $(\hat{\theta}, \tau, a)$.

Approximate portfolio-continuation probability. For incumbent firms transitioning to age $a' \geq 2$, let $q_{t+1}^\emptyset(a')$ denote the marginal probability that a non-focal product line is vacant after product-drop decisions and before search. Define the normalized product-line distribution conditional on the suppressed firm-level states by

$$\bar{\mu}_{qt}(\emptyset, a) \equiv \frac{\mu_{qt}(\emptyset, a)}{\mu_{qt}(\emptyset, a) + \sum_{\tau \geq 0} \int_{\mathbb{R}} \mu_{qt}(\hat{\theta}, \tau, a) d\hat{\theta}},$$

and

$$\bar{\mu}_{qt}(\hat{\theta}, \tau, a) \equiv \frac{\mu_{qt}(\hat{\theta}, \tau, a)}{\mu_{qt}(\emptyset, a) + \sum_{\tilde{\tau} \geq 0} \int_{\mathbb{R}} \mu_{qt}(\tilde{\theta}, \tilde{\tau}, a) d\tilde{\theta}}.$$

Thus,

$$\begin{aligned} q_{t+1}^\emptyset(a') = \frac{1}{N-1} \sum_{q \neq p} & \left[\bar{\mu}_{qt}(\emptyset, a' - 1) + \sum_{\tau \geq 0} \int_{\mathbb{R}} \int_{\mathbb{R}} \mathbf{1}_{\mathcal{R}_{t+1}^{ds}}(\hat{\theta}', \tau + 1, a') \right. \\ & \left. \times g(\hat{\theta}' | \hat{\theta}, \tau, a' - 1) \bar{\mu}_{qt}(\hat{\theta}, \tau, a' - 1) d\hat{\theta} d\hat{\theta}' \right]. \end{aligned} \quad (\text{OA.A.6})$$

The first term captures lines already vacant before the current product-drop decisions, while the second captures active lines whose products are dropped.

Approximating the vacancy indicators of the $n(\varphi) - 1$ non-focal lines as conditionally inde-

pendent, we set

$$p_{t+1}^e(a') \equiv \left[q_{t+1}^\emptyset(a') \right]^{n(\varphi)-1}, \quad a' \geq 2. \quad (\text{OA.A.7})$$

Thus, $p_{t+1}^e(a')$ approximates the probability that all non-focal lines are vacant immediately after product-drop decisions. If the focal line is vacant or is itself dropped, $1 - p_{t+1}^e(a')$ is the probability that at least one other line remains active and the firm reaches the search stage. No adjustment is required for a focal line in $\mathcal{R}_{t+1}^{cs}$, since its current product remains active through search.

The first adjustment decision of a startup is treated separately. Let $\mathcal{R}_{t+1}^{ns,e}$, $\mathcal{R}_{t+1}^{cs,e}$, and $\mathcal{R}_{t+1}^{ds,e}$ denote the corresponding age-one decision regions for its initial product. For an initially vacant focal line, all non-focal lines are vacant after the drop stage if and only if the startup drops its initial product. Hence,

$$p_{t+1}^e(1) \equiv \int_{\mathbb{R}} \mathbf{1}_{\mathcal{R}_{t+1}^{ds,e}}(\hat{\theta}', 1, 1) g(\hat{\theta}' | \hat{\theta}^0, 0, 0) d\hat{\theta}'. \quad (\text{OA.A.8})$$

Therefore, $1 - p_{t+1}^e(1)$ is the probability that the initial product is retained and the startup reaches the search stage. Unlike (OA.A.7), (OA.A.8) is exact and does not use the conditional-independence approximation.

Continuing product lines. For incumbent product lines transitioning to age $a' > 1$, a line with tenure τ at the end of period t remains active in the same market with tenure $\tau + 1$ if it does not search or if it searches while retaining its product and the search fails:

$$\begin{aligned} \mu_{p,t+1}(\hat{\theta}', \tau + 1, a') &= (1 - \delta) \left[\mathbf{1}_{\mathcal{R}_{t+1}^{ns}}(\hat{\theta}', \tau + 1, a') + (1 - x) \mathbf{1}_{\mathcal{R}_{t+1}^{cs}}(\hat{\theta}', \tau + 1, a') \right] \\ &\quad \times \int_{\mathbb{R}} g(\hat{\theta}' | \hat{\theta}, \tau, a' - 1) \mu_{pt}(\hat{\theta}, \tau, a' - 1) d\hat{\theta}. \end{aligned} \quad (\text{OA.A.9})$$

For the initial product of a startup transitioning to age one,

$$\begin{aligned} \mu_{p,t+1}(\hat{\theta}', 1, 1) &= (1 - \delta) \left[\mathbf{1}_{\mathcal{R}_{t+1}^{ns,e}}(\hat{\theta}', 1, 1) + (1 - x) \mathbf{1}_{\mathcal{R}_{t+1}^{cs,e}}(\hat{\theta}', 1, 1) \right] \\ &\quad \times g(\hat{\theta}' | \hat{\theta}^0, 0, 0) \mu_{pt}(\hat{\theta}^0, 0, 0). \end{aligned} \quad (\text{OA.A.10})$$

Newly assigned products. For an incumbent firm transitioning to age $a' > 1$, successful search resets the product state to $(\hat{\theta}^0, 0)$. The resulting inflow into market p is

$$\begin{aligned} \mu_{p,t+1}(\hat{\theta}^0, 0, a') = (1 - \delta) \frac{x}{N - 1} \sum_{q \neq p} \left\{ \sum_{\tau \geq 0} \int_{\mathbb{R}} \int_{\mathbb{R}} \left[\mathbf{1}_{\mathcal{R}_{t+1}^{cs}}(\hat{\theta}', \tau + 1, a') \right. \right. \\ \left. \left. + (1 - p_{t+1}^e(a')) \mathbf{1}_{\mathcal{R}_{t+1}^{ds}}(\hat{\theta}', \tau + 1, a') \right] g(\hat{\theta}' | \hat{\theta}, \tau, a' - 1) \mu_{qt}(\hat{\theta}, \tau, a' - 1) d\hat{\theta} d\hat{\theta}' \right. \\ \left. + (1 - p_{t+1}^e(a')) \mu_{qt}(\emptyset, a' - 1) \right\}. \end{aligned} \quad (\text{OA.A.11})$$

An active focal line in $\mathcal{R}_{t+1}^{cs}$ reaches search because its existing product is retained. A line in $\mathcal{R}_{t+1}^{ds}$ or an already-vacant line searches only if at least one non-focal line remains active, yielding the factor $1 - p_{t+1}^e(a')$.

For startups transitioning to age one, successful search comes either from the initial active line in $\mathcal{R}_{t+1}^{cs,e}$ or from one of the $n(\varphi) - 1$ initially vacant lines, which search only when the initial product is retained:

$$\begin{aligned} \mu_{p,t+1}(\hat{\theta}^0, 0, 1) = (1 - \delta) \frac{x}{N - 1} \sum_{q \neq p} \left[\int_{\mathbb{R}} \mathbf{1}_{\mathcal{R}_{t+1}^{cs,e}}(\hat{\theta}', 1, 1) g(\hat{\theta}' | \hat{\theta}^0, 0, 0) d\hat{\theta}' \right. \\ \left. + (n(\varphi) - 1) (1 - p_{t+1}^e(1)) \right] \mu_{qt}(\hat{\theta}^0, 0, 0). \end{aligned} \quad (\text{OA.A.12})$$

If the initial product lies in $\mathcal{R}_{t+1}^{ds,e}$, the startup exits before any search takes place.

New operating startups enter at age zero with their initial products at the common prior. Restoring (φ, m) explicitly for this inflow,

$$\mu_{p,t+1}(\hat{\theta}^0, 0, 0, \varphi, m) = \frac{\mathcal{M}_{t+1}^e}{N} \frac{h_{\varphi}(\varphi) h_m(m | \varphi; m_{0,t+1}) \mathbf{1}_{\{\varphi \geq \underline{\varphi}^e\}}}{1 - H_{\varphi}(\underline{\varphi}^e)}, \quad (\text{OA.A.13})$$

where h_{φ} and $h_m(\cdot | \varphi; m_{0,t+1})$ are the densities associated with H_{φ} and H_m , respectively, and $\mathcal{M}_{t+1}^e$ denotes the mass of operating entrants. The denominator conditions the entrant type distribution on the operating decision. The corresponding gross mass of startups making the pre-entry investment is $\mathcal{M}_{t+1}^e / [1 - H_{\varphi}(\underline{\varphi}^e)]$.

Destination probabilities. For an incumbent line associated with market q , the firm's exclusion set contains $n(\varphi)$ distinct markets, so the feasible search set has size $N - n(\varphi)$. Conditional on market $p \neq q$ being feasible, a particular searching line draws p with probability $1/[N - n(\varphi)]$. This marginal probability is unchanged when several lines search and destinations are coordinated without replacement.

Conditional on the focal line being associated with q , the probability that p is not already associated with one of the firm's other $n(\varphi) - 1$ lines is

$$1 - \frac{n(\varphi) - 1}{N - 1} = \frac{N - n(\varphi)}{N - 1}.$$

Hence, the unconditional assignment probability used above is

$$\frac{1}{N - n(\varphi)} \frac{N - n(\varphi)}{N - 1} = \frac{1}{N - 1}.$$

For startups transitioning to age one, only the initial market q is excluded before their first search, so the feasible set contains $N - 1$ markets and the same marginal assignment probability applies.

Vacant product lines. For incumbent firms transitioning to age $a' > 1$, the mass of vacant lines associated with market p evolves according to

$$\begin{aligned} \mu_{p,t+1}(\emptyset, a') &= (1 - \delta) \frac{1 - x}{N - 1} \sum_{q \neq p} (1 - p_{t+1}^e(a')) \left[\sum_{\tau \geq 0} \int_{\mathbb{R}} \int_{\mathbb{R}} \mathbf{1}_{\mathcal{R}_{t+1}^{ds}}(\hat{\theta}', \tau + 1, a') \right. \\ &\quad \left. \times g(\hat{\theta}' \mid \hat{\theta}, \tau, a' - 1) \mu_{qt}(\hat{\theta}, \tau, a' - 1) d\hat{\theta} d\hat{\theta}' + \mu_{qt}(\emptyset, a' - 1) \right]. \end{aligned} \quad (\text{OA.A.14})$$

The first term inside the brackets captures active lines whose products are dropped and whose subsequent searches fail, while the second captures lines already vacant whose searches fail. Both require the firm to reach the search stage.

For startups transitioning to age one,

$$\mu_{p,t+1}(\emptyset, 1) = (1 - \delta) \frac{1 - x}{N - 1} \sum_{q \neq p} (n(\varphi) - 1) (1 - p_{t+1}^e(1)) \mu_{qt}(\hat{\theta}^0, 0, 0). \quad (\text{OA.A.15})$$

This term captures unsuccessful searches by the startup's initially vacant lines conditional on retention of its initial product. If the initial product is dropped, the startup exits before search. There are no market-specific vacant lines at age zero.

In stationary equilibrium, these laws of motion satisfy $\mu_{p,t+1} = \mu_{pt} = \mu_p$ for every product market p .

OA.A.6 Aggregate Search Costs and Price-Index Derivation

Search costs using product-line measures. We express aggregate search costs using the marginal product-line measures in Appendix OA.A.5. As in that section, we suppress (ψ, φ, m) to simplify notation; all expressions below are understood to be integrated over these states as well. For incumbent active lines, a line in $\mathcal{R}_t^{cs}$ always reaches the search stage, whereas a line in $\mathcal{R}_t^{ds}$ searches only if at least one non-focal line remains active. The same continuation requirement applies to incumbent vacant lines. For startups transitioning from age zero to age one, the initial active line searches only in $\mathcal{R}_t^{cs,e}$, while each of the $n(\varphi) - 1$ initially vacant lines searches whenever the initial product is retained. Hence,

$$\begin{aligned} \mathcal{F}_t^s = (1 - \delta) f^s \sum_{p=1}^N & \left[\sum_{a \geq 1} \sum_{\tau \geq 0} \int_{\mathbb{R}} \int_{\mathbb{R}} \left\{ \mathbf{1}_{\mathcal{R}_t^{cs}}(\hat{\theta}', \tau + 1, a + 1) + [1 - p_t^e(a + 1)] \mathbf{1}_{\mathcal{R}_t^{ds}}(\hat{\theta}', \tau + 1, a + 1) \right\} \right. \\ & \times g(\hat{\theta}' \mid \hat{\theta}, \tau, a) \mu_{p,t-1}(\hat{\theta}, \tau, a) d\hat{\theta} d\hat{\theta}' + \sum_{a \geq 1} [1 - p_t^e(a + 1)] \mu_{p,t-1}(\emptyset, a) \\ & + \int_{\mathbb{R}} \left\{ \mathbf{1}_{\mathcal{R}_t^{cs,e}}(\hat{\theta}', 1, 1) + [n(\varphi) - 1] \mathbf{1}_{\mathcal{R}_t^{ns,e} \cup \mathcal{R}_t^{cs,e}}(\hat{\theta}', 1, 1) \right\} \\ & \left. \times g(\hat{\theta}' \mid \hat{\theta}^0, 0, 0) \mu_{p,t-1}(\hat{\theta}^0, 0, 0) d\hat{\theta}' \right]. \end{aligned} \quad (\text{OA.A.16})$$

The first term counts searches by active incumbent lines, the second searches by incumbent vacant lines, and the third searches by age-one startups. In the startup term, the first component corresponds to search by the initial active line, while the second counts searches by its $n(\varphi) - 1$ initially vacant lines. If the initial product lies in $\mathcal{R}_t^{ds,e}$, the startup exits before any search. The factor $1 - \delta$ reflects that search occurs only after the exogenous death shock.

Product-level price index. For a product line with state ξ at the end of period $t - 1$ and realized demand shifter θ_{pt} , (2.9) gives

$$p_{pt}(\theta_{pt}; \xi) = \frac{\eta}{\eta - 1} \frac{w_t}{z(\varphi; z_0)} \frac{\theta_{pt}^{(\eta-1)/\eta}}{\mathcal{B}(\xi)}.$$

The product-level CES price index therefore satisfies

$$P_{pt}^{1-\eta} = \int_{\mathcal{X}} \mathbb{E} \left[\theta_{pt}^{\eta-1} p_{pt}(\theta_{pt}; \xi)^{1-\eta} \mid \xi \right] \mu_{p,t-1}(d\xi).$$

Substituting the market-clearing price gives

$$\begin{aligned} P_{pt}^{1-\eta} &= \left(\frac{\eta}{\eta - 1} w_t \right)^{1-\eta} \int_{\mathcal{X}} z(\varphi; z_0)^{\eta-1} \mathcal{B}(\xi)^{\eta-1} \mathbb{E} \left[\theta_{pt}^{\eta-1 + \frac{\eta-1}{\eta}(1-\eta)} \mid \xi \right] \mu_{p,t-1}(d\xi) \\ &= \left(\frac{\eta}{\eta - 1} w_t \right)^{1-\eta} \int_{\mathcal{X}} z(\varphi; z_0)^{\eta-1} \mathcal{B}(\xi)^{\eta} \mu_{p,t-1}(d\xi), \end{aligned}$$

where the second equality follows from $\eta - 1 + \frac{\eta-1}{\eta}(1-\eta) = \frac{\eta-1}{\eta}$ and $\mathbb{E} \left[\theta_{pt}^{(\eta-1)/\eta} \mid \xi \right] = \mathcal{B}(\xi)$.

Therefore,

$$P_{pt} = \frac{\eta}{\eta - 1} w_t \left[\int_{\mathcal{X}} \mathcal{B}(\xi)^{\eta} z(\varphi; z_0)^{\eta-1} \mu_{p,t-1}(d\xi) \right]^{-\frac{1}{\eta-1}},$$

which gives (2.36).

OA.A.7 Proof of Proposition 3

Proof. Fix the true product-specific expertise θ_{ip} , the observation window τ , and the information-processing path, and let $\hat{\sigma}^2 \equiv \hat{\sigma}_{ipt-1}^2$. All expectations below average over potential signal histories without conditioning on endogenous product retention. From (2.8),

$$\mathcal{B}_{it-1} = \exp \left\{ \frac{\eta - 1}{\eta} \hat{\theta}_{ipt-1} + \frac{1}{2} \left(\frac{\eta - 1}{\eta} \right)^2 (\hat{\sigma}^2 + \sigma_{\varepsilon}^2) \right\}.$$

If θ_{ip} were known, the corresponding conditional demand term would be

$$\mathcal{B}_{ip}^{FI} = \mathbb{E} \left[\theta_{ipt}^{(\eta-1)/\eta} \mid \theta_{ip} \right] = \theta_{ip}^{(\eta-1)/\eta} \exp \left\{ \frac{1}{2} \left(\frac{\eta - 1}{\eta} \right)^2 \sigma_{\varepsilon}^2 \right\}.$$

Hence,

$$[\mathcal{B}_{ip}^{FI}]^\eta = \theta_{ip}^{\eta-1} \exp \left\{ \frac{(\eta-1)^2}{2\eta} \sigma_\varepsilon^2 \right\},$$

which gives the denominator in (2.38). Dividing optimal output in (2.7) by its counterpart under known θ_{ip} gives the output ratio in (2.38).

It remains to characterize the sign of the expected wedge. By the Gaussian signal structure, conditional on the true $\log \theta_{ip}$ and the information-processing path,

$$\hat{\theta}_{ipt-1} \mid \theta_{ip} \sim N \left(\log \theta_{ip} + \frac{\hat{\sigma}^2}{\sigma_0^2} (\theta_0 - \log \theta_{ip}), \hat{\sigma}^2 \left(1 - \frac{\hat{\sigma}^2}{\sigma_0^2} \right) \right).$$

Using (2.38),

$$1 - \omega_{ipt} = \exp \left\{ (\eta-1) \left(\hat{\theta}_{ipt-1} - \log \theta_{ip} \right) + \frac{(\eta-1)^2}{2\eta} \hat{\sigma}^2 \right\}.$$

Applying the moment-generating function of the normal distribution therefore yields

$$\begin{aligned} \log \mathbb{E} [1 - \omega_{ipt} \mid \theta_{ip}] &= (\eta-1) \frac{\hat{\sigma}^2}{\sigma_0^2} (\theta_0 - \log \theta_{ip}) \\ &\quad + \frac{(\eta-1)^2}{2} \hat{\sigma}^2 \left(1 + \frac{1}{\eta} - \frac{\hat{\sigma}^2}{\sigma_0^2} \right). \end{aligned}$$

Since $\mathbb{E}[\omega_{ipt} \mid \theta_{ip}] = 1 - \mathbb{E}[1 - \omega_{ipt} \mid \theta_{ip}]$, the expected wedge is positive if and only if the expression above is negative. For $\hat{\sigma}^2 > 0$, rearranging gives

$$\mathbb{E} [\omega_{ipt} \mid \theta_{ip}] > 0 \iff \log \theta_{ip} > \theta_0 + \frac{\eta-1}{2} \left[\left(1 + \frac{1}{\eta} \right) \sigma_0^2 - \hat{\sigma}^2 \right],$$

with the reverse inequality implying $\mathbb{E}[\omega_{ipt} \mid \theta_{ip}] < 0$. At equality, the expected wedge is zero.

Finally, $\hat{\sigma}^2 \leq \sigma_0^2$, with strict inequality after any informative signal. Holding product tenure and other information fixed, lower information-processing ability raises $\hat{\sigma}^2$. Since the cutoff has derivative $-(\eta-1)/2 < 0$ with respect to $\hat{\sigma}^2$, lower information-processing ability lowers the cutoff and therefore expands the range of high- θ_{ip} products with positive expected output wedges. As $\hat{\sigma}^2 \rightarrow 0$, the posterior mean converges to $\log \theta_{ip}$ and $\omega_{ipt} \rightarrow 0$. $\square$

OA.A.8 Convexity and the Value of Learning

Lemma OA.2 (Convexity and the Value of Learning). *At fixed tenure and firm states, the product-line value is continuous, weakly increasing, and convex in the demand term $\mathcal{B}$. The no-search value is strictly increasing and strictly convex. Holding current demand and the future value function fixed, a larger posterior-mean innovation variance raises expected continuation value.*

Proof. The successive approximations preserve monotonicity and convexity. For the infinite-horizon limit, we use the dynamic-programming methods in [Stokey and Lucas Jr \(1989, Sections 9.1 and 9.4\)](#). The maintained parameter restrictions are $0 < \beta < 1$, $0 \leq \delta < 1$, $0 < x < 1$, $\eta > 1$, $z_i > 0$, $f^o, f^s > 0$, $\psi_{it} \geq 0$, and $\sigma_0^2, \sigma_\varepsilon^2 \in (0, \infty)$, with finite signal precisions at every finite age. Prices are symmetric and stationary, with the normalization $w = 1$. We retain the prior-reset convention and line-level payoff equivalence in [Section 2.3.7](#). The value itself is weakly increasing; strictness applies to its no-search branch.

Belief transition. At fixed tenure and firm states, $\mathcal{B}_{it}$ is strictly increasing in $\hat{\theta}$. We therefore express the existing value functions in terms of $(\mathcal{B}, \tau)$, without changing their economic definition. The time subscript retains the firm's age and its information-processing path; $t + 1$ entails age $a_{it} + 1$. For the same product carried forward,

$$v_{it}^\tau \equiv \hat{\sigma}_{ipt}^2 - \hat{\sigma}_{ipt+1}^2, \quad \mathcal{B}' = \mathcal{B} \exp \left\{ \frac{\eta - 1}{\eta} \sqrt{v_{it}^\tau} \mathbf{n} - \frac{1}{2} \left(\frac{\eta - 1}{\eta} \right)^2 v_{it}^\tau \right\}, \quad \mathbf{n} \sim N(0, 1). \quad (\text{OA.A.17})$$

Here $\mathbf{n}$ is only an integration variable. The next posterior variance corresponds to tenure $\tau + 1$. The transition follows from Gaussian updating and the definition of $\mathcal{B}$, and satisfies $\mathbb{E}_{it}[\mathcal{B}'] = \mathcal{B}$.

Unbounded payoffs and the infinite-horizon limit. Expected operating profits are

$$\bar{\pi}_{it+1}(\mathcal{B}, z_i) = \frac{(\eta - 1)^{\eta-1}}{\eta^\eta} P^\eta C z_i^{\eta-1} \mathcal{B}^\eta - f^o. \quad (\text{OA.A.18})$$

They are strictly increasing and strictly convex in $\mathcal{B} > 0$. Because they are unbounded, a contraction theorem for bounded functions cannot be applied directly. Instead, use the moment

already introduced in Appendix [OA.A.3](#),

$$H_{it}(\hat{\theta}, \tau) = \exp \left\{ (\eta - 1)\hat{\theta} + \frac{(\eta - 1)^2}{2} \hat{\sigma}_{ipt}^2 \right\}.$$

Conditional on retaining the product, $\mathbb{E}_{it} H_{i,t+1} = H_{it}$. A replacement resets the moment to the finite common-prior value H^0 . Operating and search payoffs are bounded in absolute value by a constant multiple of $1 + H_{it}$. Under any feasible policy the expected moment j periods ahead is bounded by $H_{it} + jH^0$. Hence, for finite K ,

$$\sup_{\text{feasible policies}} \mathbb{E}_{it}[\text{absolute discounted payoffs after } T] \leq K \sum_{j=T}^{\infty} \beta^j (1 + H_{it} + jH^0) \rightarrow 0. \quad (\text{OA.A.19})$$

Since posterior variance is bounded by σ_0^2 , convergence is uniform on compact sets of current $\mathcal{B}$, uniformly over firm age. Finite-horizon values with zero terminal value thus converge to the infinite-horizon value. The bound also selects the Bellman solution in the class satisfying the vanishing-tail condition; see [Stokey and Lucas Jr \(1989, Section 9.4\)](#).

Preservation of value-function properties. Start from zero terminal values for occupied and vacant lines. If the next occupied-line value is increasing and convex in $\mathcal{B}'$, its expectation under [\(OA.A.17\)](#) is increasing and convex in $\mathcal{B}$, because for every realization of $\mathbf{n}$ the next demand term is a positive multiple of the current one. Adding [\(OA.A.18\)](#) makes the no-search branch strictly increasing and strictly convex. The continue-and-search branch is a positive affine combination of the no-search values of the existing and fresh products, net of a fixed cost. The drop-and-search value is constant in the existing product's $\mathcal{B}$. Taking the maximum preserves weak monotonicity and convexity. Gaussian integration and the payoff envelope preserve continuity. Induction and [\(OA.A.19\)](#) establish these properties for the limiting value. Its no-search branch remains strictly increasing and strictly convex because of current production.

Value of information. For $v^H > v^L \geq 0$, construct the demand term associated with v^H by multiplying the one associated with v^L by an independent lognormal factor with mean one and log variance $[(\eta - 1)/\eta]^2 (v^H - v^L)$. Conditional Jensen's inequality implies a larger expectation of every integrable convex continuation function. This is the martingale-convexity argument of

Stokey (2009, Theorem 4.1, pp. 61–62). The inequality is strict if the function is nonaffine on a region reached with positive probability. Here sufficiently high demand terms lead to active production and strict curvature, and the lognormal transition has full support on $(0, \infty)$.

For a smooth convex function f , the lognormal transition also gives

$$\frac{\partial}{\partial v} \mathbb{E}[f(\mathcal{B}')] = \frac{1}{2} \left(\frac{\eta - 1}{\eta} \right)^2 \mathbb{E}[(\mathcal{B}')^2 f''(\mathcal{B}')] \geq 0.$$

Convex smoothing, or interpretation of the second derivative as a nonnegative measure, extends this result to kinked convex functions. At positive innovation variance it is strict when curvature is positive on an open set. These comparisons hold the future function and its other arguments fixed; a change in the entire ψ -path must also account for changes in future values. $\square$

OA.A.9 Proof of Proposition 4

The following expressions summarize the threshold comparisons in Proposition 4 and the additional width results. Derivatives are evaluated at regular points, as defined in the proof.

Under the information-processing conditions in part (i) and the additional sufficient condition in (OA.A.54),

$$\frac{\partial \mathcal{B}_{it}^{ds}}{\partial \psi_{it}} > 0, \quad \frac{\partial \mathcal{B}_{it}^{cs}}{\partial \psi_{it}} > 0, \quad \frac{\partial (\mathcal{B}_{it}^{cs} - \mathcal{B}_{it}^{ds})}{\partial \psi_{it}} > 0. \quad (\text{OA.A.20})$$

The two threshold signs do not require the additional width condition. The proof formalizes the sufficient path restrictions in (OA.A.41) and provides a weaker characterization through (OA.A.39).

For a given information-processing path, the fixed-cost conditions in (OA.A.50) imply

$$\frac{\partial \mathcal{B}_{it}^{ds}}{\partial z_i} < 0, \quad \frac{\partial \mathcal{B}_{it}^{cs}}{\partial z_i} > 0, \quad \frac{\partial (\mathcal{B}_{it}^{cs} - \mathcal{B}_{it}^{ds})}{\partial z_i} > 0. \quad (\text{OA.A.21})$$

At the same firm age and information-processing path, longer feasible tenure implies

$$\mathcal{B}_{it}^{ds}(\tau + 1) > \mathcal{B}_{it}^{ds}(\tau), \quad \mathcal{B}_{it}^{cs}(\tau + 1) > \mathcal{B}_{it}^{cs}(\tau). \quad (\text{OA.A.22})$$

The effect on the width of the continue-and-search region is ambiguous. Corollary OA.1 provides further sufficient conditions based on accumulated information and signal precision.

Proof. We first derive the relative-value recursion. We then compare information-processing paths, productivity, and product tenure, and finally characterize the width in demand units.

Value matching and the relative-value recursion. The original timing places next-period profits inside the discount factor. With zero payoff at exogenous exit, the branch equations are

$$V_{it}^{ns}(\mathcal{B}, \tau) = \beta \bar{\pi}_{it+1}(\mathcal{B}, z_i) + \beta(1 - \delta) \mathbb{E}_{it} V_{i,t+1}(\mathcal{B}', \tau + 1), \quad (\text{OA.A.23})$$

$$V_{it}^{cs}(\mathcal{B}, \tau) = -f^s + (1 - x)V_{it}^{ns}(\mathcal{B}, \tau) + xV_{it}^{ns}(\mathcal{B}^0, 0), \quad (\text{OA.A.24})$$

$$V_{it}^{ds}(\emptyset) = -f^s + xV_{it}^{ns}(\mathcal{B}^0, 0) + (1 - x)\beta(1 - \delta)V_{i,t+1}^{ds}(\emptyset). \quad (\text{OA.A.25})$$

The fresh product starts from the common prior, with demand term $\mathcal{B}^0$, but uses the current firm's age and future ψ -path. Product tenure resets to zero; firm age does not reset.

Define the auxiliary difference

$$U_{it}(\mathcal{B}, \tau) \equiv V_{it}^{ns}(\mathcal{B}, \tau) - \beta(1 - \delta)V_{i,t+1}^{ds}(\emptyset). \quad (\text{OA.A.26})$$

Subtracting the drop-and-search branch from the continue-and-search branch gives $(1 - x)U_{it}$. Equating the no-search and continue-and-search values gives $V_{it}^{ns}(\mathcal{B}_{it}^{cs}, \tau) = V_{it}^{ns}(\mathcal{B}^0, 0) - f^s/x$. The two threshold conditions are therefore

$$U_{it}(\mathcal{B}_{it}^{ds}(\tau), \tau) = 0, \quad (\text{OA.A.27})$$

$$U_{it}(\mathcal{B}_{it}^{cs}(\tau), \tau) = \Delta_{it}^V, \quad (\text{OA.A.28})$$

where

$$\begin{aligned} \Delta_{it}^V &\equiv V_{it}^{ns}(\mathcal{B}_{it}^{cs}(\tau), \tau) - V_{it}^{ns}(\mathcal{B}_{it}^{ds}(\tau), \tau) = V_{it}^{ns}(\mathcal{B}^0, 0) - \beta(1 - \delta)V_{i,t+1}^{ds}(\emptyset) - \frac{f^s}{x} \\ &= \frac{V_{it}^{ds}(\emptyset) - \beta(1 - \delta)V_{i,t+1}^{ds}(\emptyset)}{x}. \end{aligned} \quad (\text{OA.A.29})$$

The final equality uses (OA.A.25). In particular,

$$\mathcal{B}_{it}^{cs}(\tau) > \mathcal{B}_{it}^{ds}(\tau) \iff V_{it}^{ds}(\emptyset) > \beta(1 - \delta)V_{i,t+1}^{ds}(\emptyset), \quad (\text{OA.A.30})$$

whenever the two indifference cutoffs exist. In the stationary case, positive vacant-line value implies this inequality automatically. Thus Δ_{it}^V is the value advantage of a fresh product over vacancy, net of the search cost per successful replacement. Neither term inherits the existing product's tenure. Threshold ordering is equivalent to $\Delta_{it}^V > 0$ whenever the two indifference cutoffs exist. These are discrete-time value-matching conditions, derived directly from the choice-specific value functions. [Stokey \(2009, Chapter 7\)](#) uses the analogous adjustment-versus-inaction comparison in continuous time; we do not invoke its continuous-time verification or smooth-pasting conditions.

Subtracting vacancy from all three values gives

$$V_{it}(\mathcal{B}, \tau) - V_{it}^{ds}(\emptyset) = \max \{U_{it}(\mathcal{B}, \tau) - x\Delta_{it}^V, (1-x)U_{it}(\mathcal{B}, \tau), 0\}. \quad (\text{OA.A.31})$$

Consequently,

$$\begin{aligned} U_{it}(\mathcal{B}, \tau) &= \beta \bar{\pi}_{it+1}(\mathcal{B}, z_i) \\ &\quad + \beta(1-\delta)\mathbb{E}_{it} \max \{U_{i,t+1}(\mathcal{B}', \tau+1) - x\Delta_{i,t+1}^V, (1-x)U_{i,t+1}(\mathcal{B}', \tau+1), 0\}. \end{aligned} \quad (\text{OA.A.32})$$

The maximum is increasing and one-Lipschitz in U , weakly decreasing in Δ^V , and convex in $\mathcal{B}'$. Future replacement opportunities remain present through $\{\Delta_{i,t+j}^V\}_{j \geq 0}$.

For the order comparisons, iterate [\(OA.A.32\)](#) from zero terminal relative value, holding each economy's actual infinite-horizon sequence $\{\Delta_{i,t+j}^V\}$ fixed. The truncation error is bounded by

$$[\beta(1-\delta)]^T \mathbb{E}_{it} |U_{i,t+T}(\mathcal{B}_T, \tau+T)| \leq K'[\beta(1-\delta)]^T (1+H_{it}) \longrightarrow 0.$$

This follows from the one-Lipschitz property, the martingale bound along the original product, and the uniform bound on vacant-line values from [\(OA.A.19\)](#). These auxiliary approximations converge to the true U without requiring the same comparative-static ordering for truncated vacancy problems.

The model-implied information comparison. Fix φ_i and write the specified precision profile as $\psi_{is} = m_i^{\gamma_m} h(a_{is}; \varphi_i)$, where h is the age-and-data term in the quantitative specification. The

profile is deterministic. Locally, current ψ_{it} can index either a change in m_i at fixed age or a smooth age shift at fixed m_i , wherever its derivative in that comparison is positive. Throughout this proof, $\partial/\partial\psi_{it}$ means differentiation along that specified comparison, including its implied past and future precisions. It does not hold the other dates' precisions fixed. Integer-age comparisons and the data-profile change at age two are handled as finite comparisons. All future derivatives below use the same current ψ_{it} as index. Value-function derivatives with respect to ψ_{it} hold the displayed demand term and product tenure fixed; the response of the transition distribution is accounted for separately. Subscripts following a comma denote partial derivatives with respect to the indicated variable, as in $U_{it,\mathcal{B}}$ and V_{it,z_i}^{ns} .

For example, along the managerial comparison,

$$\frac{\partial\psi_{is}}{\partial\psi_{it}} = \frac{\psi_{is}}{\psi_{it}}, \quad \frac{\partial\mathcal{B}_{it}^j}{\partial\psi_{it}} = \frac{d\mathcal{B}_{it}^j/dm_i}{\gamma_m\psi_{it}/m_i}, \quad j \in \{ds, cs\}, \quad (\text{OA.A.33})$$

for $\gamma_m > 0$. Equal current precision need not imply equal histories across different (m_i, a_{it}) pairs. The theorem applies to both comparisons through the same path conditions; it does not identify their numerical derivatives with one another.

A precise stock-versus-flow comparison. Introduce local auxiliary notation for this calculation only:

$$S_{it}^\tau = \sum_{s=t-\tau+1}^t \psi_{is}, \quad A_{it}^\tau = \sigma_\varepsilon^2/\sigma_0^2 + \tau + S_{it}^\tau, \quad b_{it} = 1 + \psi_{i,t+1}. \quad (\text{OA.A.34})$$

Thus $\hat{\sigma}_{ipt}^2 = \sigma_\varepsilon^2/A_{it}^\tau$ and $v_{it}^\tau = \sigma_\varepsilon^2 b_{it}/[A_{it}^\tau(A_{it}^\tau + b_{it})]$. Differentiation with respect to current ψ_{it} gives the exact equivalence

$$\frac{\partial v_{it}^\tau}{\partial\psi_{it}} < 0 \quad \Longleftrightarrow \quad \frac{1}{A_{it}^\tau} \frac{\partial A_{it}^\tau}{\partial\psi_{it}} > \frac{A_{it}^\tau}{2A_{it}^\tau + b_{it}} \frac{1}{b_{it}} \frac{\partial b_{it}}{\partial\psi_{it}}. \quad (\text{OA.A.35})$$

The fraction multiplying the new-signal response lies strictly between zero and one half. Accumulated posterior precision need therefore respond by only a fraction of the proportional response of next-period signal precision. This is an exact restriction on the learning technology and information stock, with no value-function derivatives.

In the managerial comparison it is equivalently

$$\frac{\partial v_{it}^\tau}{\partial \psi_{it}} = \frac{\sigma_\varepsilon^2}{\psi_{it}} \left[\frac{S_{it}^\tau + \psi_{i,t+1}}{(A_{it}^\tau + 1 + \psi_{i,t+1})^2} - \frac{S_{it}^\tau}{(A_{it}^\tau)^2} \right]. \quad (\text{OA.A.36})$$

Corollary [OA.1](#)(i) gives a single current-state check that makes this derivative negative at every subsequent tenure. For other model-implied comparisons, use [\(OA.A.35\)](#) with their actual path responses.

Discounted accounting for the two information effects. We now weaken the earlier requirement that both effects have the desired sign at every future date. In this calculation, suppress firm indices on values and write $v_s = v_{is}^\tau$, while retaining the common differentiation variable ψ_{it} . Put $d = \beta(1 - \delta)$ and let

$$F_s(b) = \max\{U_s(b, \tau_s) - x\Delta_s^V, (1 - x)U_s(b, \tau_s), 0\}, \quad \tau_s = \tau + s - t.$$

Define the following objects under the baseline optimal policy:

$$\begin{aligned} \mathfrak{p}_s(b) &= \begin{cases} 0, & U_s(b, \tau_s) < 0, \\ 1 - x, & 0 < U_s(b, \tau_s) < \Delta_s^V, \\ 1, & U_s(b, \tau_s) > \Delta_s^V, \end{cases} \\ I_s^{ns}(b) &= \mathbf{1}\{U_s(b, \tau_s) > \Delta_s^V\}, \\ \Gamma_s(b) &= \frac{\partial}{\partial v_s} \mathbb{E}_s[F_{s+1}(\mathcal{B}') \mid \mathcal{B} = b] > 0. \end{aligned} \quad (\text{OA.A.37})$$

The derivative defining Γ_s holds F_{s+1} fixed. Its positivity follows from the lemma, including the curvature at switching boundaries. The weight $\mathfrak{p}_s$ is the probability that the same incumbent remains after that date's switching decision, conditional on exogenous firm continuation. These are local proof auxiliaries.

Let $\mathcal{B}_j$ follow the retained-product transition from $\mathcal{B}_0 = b$. Define the continuation weights by

$$Q_0 = 1, \quad Q_j = \prod_{k=1}^j \mathfrak{p}_{t+k}(\mathcal{B}_k).$$

The latter weights remove histories on which the original product is dropped or replaced. Differentiating (OA.A.32) gives

$$\begin{aligned} -\frac{\partial U_s(b, \tau_s)}{\partial \psi_{it}} = & -d \frac{\partial v_s}{\partial \psi_{it}} \Gamma_s(b) + dx \frac{\partial \Delta_{s+1}^V}{\partial \psi_{it}} \Pr\{I_{s+1}^{ns}(\mathcal{B}') = 1 \mid b\} \\ & + d\mathbb{E}_s \left[\mathbf{p}_{s+1}(\mathcal{B}') \left(-\frac{\partial U_{s+1}(\mathcal{B}', \tau_s + 1)}{\partial \psi_{it}} \right) \mid b \right]. \end{aligned} \quad (\text{OA.A.38})$$

Iterating yields the discounted loss

$$\begin{aligned} \mathcal{L}_t(b) & \equiv \sum_{j=0}^{\infty} d^{j+1} \mathbb{E}_t \left[Q_j \left\{ -\frac{\partial v_{t+j}}{\partial \psi_{it}} \Gamma_{t+j}(\mathcal{B}_j) + x \frac{\partial \Delta_{t+j+1}^V}{\partial \psi_{it}} I_{t+j+1}^{ns}(\mathcal{B}_{j+1}) \right\} \right] \\ & = -\frac{\partial U_t(b, \tau)}{\partial \psi_{it}}. \end{aligned} \quad (\text{OA.A.39})$$

Expectations include the next observation in the indicator term. The first component is the reduction in the value of further incumbent learning. The second is the additional opportunity cost of keeping an incumbent without search when replacement opportunities improve. Both are weighted by how long the original product continues.

The formula follows from the continuation envelope: away from ties,

$$\frac{\partial F_s(b)}{\partial \psi_{it}} = \mathbf{p}_s(b) \frac{\partial U_s(b, \tau_s)}{\partial \psi_{it}} - x I_s^{ns}(b) \frac{\partial \Delta_s^V}{\partial \psi_{it}}.$$

Gaussian transitions hit the two boundaries with probability zero. For this derivative representation, regularity includes absolute summability of the displayed discounted derivative terms and

$$d^J \mathbb{E}_t \left[Q_J \left| \frac{\partial U_{t+J}(\mathcal{B}_J, \tau + J)}{\partial \psi_{it}} \right| \right] \rightarrow 0.$$

It suffices that the derivatives have the same payoff-moment bound as the values, multiplied by a polynomial in J ; geometric discounting then controls the tail. The comparison between two fixed paths require no parameter differentiation or derivative-tail qualification.

The path restrictions

$$\frac{\partial v_{t+j}}{\partial \psi_{it}} \leq 0, \quad \frac{\partial \Delta_{t+j}^V}{\partial \psi_{it}} \geq 0, \quad j = 0, 1, \dots,$$

make each component nonnegative. If additionally $\partial v_t / \partial \psi_{it} < 0$, then $\mathcal{L}_t(b) > 0$ for every $b > 0$. They are sufficient, but unnecessary. More generally, only the discounted sums at the relevant margins need be favorable. A negative opportunity gain at some future ages can be offset by reduced incumbent learning, or by positive gains at other dates. This is a weaker condition than imposing separate signs on the entire two sequences.

Marginal vacancy gains. Write $g_s = \partial V_s^{ds}(\emptyset) / \partial \psi_{it}$, using the same current comparison index at every date. The identity for the value band gives

$$\frac{\partial \Delta_s^V}{\partial \psi_{it}} = (g_s - dg_{s+1})/x. \quad (\text{OA.A.40})$$

Thus nonnegative, non-increasing marginal vacancy gains suffice for nonnegative opportunity gains. The gains can also increase, as long as $g_{s+1} \leq g_s/d$. Deterministic or increasing precision does not by itself imply this restriction on the timing of value gains. When it fails, (OA.A.39) retains the actual signed opportunity gains; the comparison is not automatically lost.

Comparing information-processing paths. Use superscripts H and L to compare higher and lower information-processing paths. Align the firms at the current decision date and the same product tenure; their firm ages may differ. Compare $\psi_{is}^H \geq \psi_{is}^L$ at every corresponding date in the existing product's observation history and at every future date. All other payoff determinants and the common prior are the same. This comparison includes young and mature firms at equal product tenure when age affects payoffs only through the specified information-processing path. It is not a change in the current ψ_{it} holding an unspecified future path implicit.

The sufficient path conditions for part (i) are

$$v_{i,t+j}^{\tau+j,H} \leq v_{i,t+j}^{\tau+j,L}, \quad \Delta_{i,t+j}^{V,H} \geq \Delta_{i,t+j}^{V,L}, \quad j = 0, 1, \dots, \quad (\text{OA.A.41})$$

along continued operation of the original product. The first compares how much information remains to be resolved about that product. The second compares the net value of replacement opportunities at the ages that may be reached.

Both conditions can be checked without differentiating with respect to managerial ability.

For each path, compute

$$\hat{\sigma}_{ipt}^2 = \left[\frac{1}{\sigma_0^2} + \frac{1}{\sigma_\varepsilon^2} \sum_{s=t-\tau+1}^t (1 + \psi_{is}) \right]^{-1}, \quad v_{it}^\tau = \hat{\sigma}_{ipt}^2 - \left[\frac{1}{\hat{\sigma}_{ipt}^2} + \frac{1 + \psi_{i,t+1}}{\sigma_\varepsilon^2} \right]^{-1}. \quad (\text{OA.A.42})$$

At date $t + j$ the observation sum begins at the same adoption date. By (OA.A.29), the second condition is equivalently

$$V_{i,t+j}^{ds,H}(\emptyset) - V_{i,t+j}^{ds,L}(\emptyset) \geq \beta(1 - \delta) \left[V_{i,t+j+1}^{ds,H}(\emptyset) - V_{i,t+j+1}^{ds,L}(\emptyset) \right]. \quad (\text{OA.A.43})$$

Thus the gain in vacancy value today must be at least its discounted gain at the next age. A sufficient condition is that the gain is nonnegative and non-increasing with age. More generally, it may increase by a factor of up to $1/[\beta(1 - \delta)]$ between adjacent ages. These are restrictions on vacancy-value gains, not on the level of ψ alone. Age-increasing precision does not imply convergence of two precision paths, and convergence of the paths does not by itself imply monotone vacancy-value gains.

For completeness, a higher ψ -path weakly raises vacancy value. A vacant line carries no information about an existing product. The more informed firm can reproduce the less informed firm's signals by adding independent Gaussian noise with variance $\sigma_\varepsilon^2(1/\psi_{is}^L - 1/\psi_{is}^H)$ at each date with positive precisions; when $\psi_{is}^L = 0$, it ignores the additional signal. It can then reproduce all production and switching decisions, with the same search and exogenous exit outcomes. This feasible-policy comparison applies to the separable line-level problem, including repeated replacements. Finite-horizon comparison and (OA.A.19) imply $V_{it}^{ds,H}(\emptyset) \geq V_{it}^{ds,L}(\emptyset)$. It does not establish (OA.A.43), because the increment can differ across firm ages.

Under (OA.A.41), compare the auxiliary iterations of (OA.A.32). Their terminal relative values coincide. If future $U^H \leq U^L$, the higher- ψ continuation maximum is weakly smaller pointwise, because $\Delta^{V,H} \geq \Delta^{V,L}$. Each maximum is convex in the next demand term. Its expectation is further reduced by the weakly smaller innovation variance, by Lemma OA.2. Current profits at fixed $\mathcal{B}$ are identical. Induction and the vanishing truncation error therefore yield $U_{it}^H(\mathcal{B}, \tau) \leq U_{it}^L(\mathcal{B}, \tau)$. If the current innovation-variance comparison is strict, the inequality is strict because the next continuation maximum has a curved active region reached with positive probability.

Since each U is strictly increasing in $\mathcal{B}$, the drop-matching condition implies a weakly higher drop threshold. The search-matching condition implies a weakly higher search threshold because its left-hand side falls and its right-hand side weakly rises. Both increases are strict under a strict current variance comparison. This establishes the finite threshold comparisons in part (i). These are sufficient, not necessary, path restrictions; offsetting changes across future dates can also produce the same threshold signs.

Productivity. Holding the information-processing path fixed, the belief transition is independent of z_i . Differentiating the two value-matching conditions at regular parameter values gives

$$\frac{\partial \mathcal{B}_{it}^{ds}}{\partial z_i} = \frac{\beta(1-\delta)V_{i,t+1,z_i}^{ds}(\emptyset) - V_{it,z_i}^{ns}(\mathcal{B}_{it}^{ds}, \tau)}{V_{it,\mathcal{B}}^{ns}(\mathcal{B}_{it}^{ds}, \tau)}, \quad (\text{OA.A.44})$$

$$\frac{\partial \mathcal{B}_{it}^{cs}}{\partial z_i} = \frac{V_{it,z_i}^{ns}(\mathcal{B}^0, 0) - V_{it,z_i}^{ns}(\mathcal{B}_{it}^{cs}, \tau)}{V_{it,\mathcal{B}}^{ns}(\mathcal{B}_{it}^{cs}, \tau)}. \quad (\text{OA.A.45})$$

The denominators are strictly positive. Hence the sharp conditions for the two individual signs are respectively

$$\begin{aligned} V_{it,z_i}^{ns}(\mathcal{B}_{it}^{ds}, \tau) &> \beta(1-\delta)V_{i,t+1,z_i}^{ds}(\emptyset), \\ V_{it,z_i}^{ns}(\mathcal{B}^0, 0) &> V_{it,z_i}^{ns}(\mathcal{B}_{it}^{cs}, \tau). \end{aligned} \quad (\text{OA.A.46})$$

They need hold only at the relevant margins. They do not impose a pointwise ordering of marginal values throughout the state space.

These derivatives measure expected operating payoffs exposed to productivity. Indeed,

$$\bar{\pi}_{it+1,z_i}(\mathcal{B}, z_i) = \frac{\eta-1}{z_i} [\bar{\pi}_{it+1}(\mathcal{B}, z_i) + f^o] > 0.$$

Under a fixed switching policy, a productivity increase raises every operating payoff and leaves fixed costs and the belief transition unchanged. The envelope derivative of a branch value is therefore $(\eta-1)/z_i$ times its expected discounted gross operating payoffs, under the optimal continuation policy and the specified initial action. The first condition in (OA.A.46) says that the marginal existing product has more such exposure than postponing the vacancy problem. The second says that the fresh product has more exposure than the incumbent at the search

margin. This includes future learning and replacements, not only current production.

An equivalent formulation uses expected discounted fixed costs. Introduce only for this accounting argument $\mathcal{K}_{it}^{ns}(\mathcal{B}, \tau)$ and $\mathcal{K}_{it}^{ds}(\emptyset)$: the expected discounted operating and search fixed costs under the indicated initial branch and its optimal continuation policy. They include costs after subsequent product replacements. Operating costs carry the same one-period discount as operating profits; search costs are paid at the decision date. Thus the cost of the no-search branch begins with βf^o . Under the optimal policy, the accounting equations mirror the branch equations, with βf^o replacing $\beta \bar{\pi}$ and f^s replacing $-f^s$.

For either indicated initial branch, the envelope and accounting identity imply

$$\frac{z_i}{\eta - 1} V_{it, z_i}^{(\cdot)} = V_{it}^{(\cdot)} + \mathcal{K}_{it}^{(\cdot)}. \quad (\text{OA.A.47})$$

Here the right-hand side is expected discounted gross operating payoffs. This identity does not differentiate the optimal fixed-cost burden; it evaluates gross payoffs under the optimizing policy at the parameter point in question. It requires holding the information-processing path, search technology, and fixed costs constant when varying z_i .

Substituting value matching into (OA.A.44) and (OA.A.45) gives

$$\frac{\partial \mathcal{B}_{it}^{ds}}{\partial z_i} = \frac{\eta - 1}{z_i} \frac{\beta(1 - \delta) \mathcal{K}_{i, t+1}^{ds}(\emptyset) - \mathcal{K}_{it}^{ns}(\mathcal{B}_{it}^{ds}, \tau)}{V_{it, \mathcal{B}}^{ns}(\mathcal{B}_{it}^{ds}, \tau)}, \quad (\text{OA.A.48})$$

$$\frac{\partial \mathcal{B}_{it}^{cs}}{\partial z_i} = \frac{\eta - 1}{z_i} \frac{f^s/x + \mathcal{K}_{it}^{ns}(\mathcal{B}^0, 0) - \mathcal{K}_{it}^{ns}(\mathcal{B}_{it}^{cs}, \tau)}{V_{it, \mathcal{B}}^{ns}(\mathcal{B}_{it}^{cs}, \tau)}. \quad (\text{OA.A.49})$$

Consequently, the two conditions in (OA.A.46) are equivalent to

$$\begin{aligned} \mathcal{K}_{it}^{ns}(\mathcal{B}_{it}^{ds}, \tau) &> \beta(1 - \delta) \mathcal{K}_{i, t+1}^{ds}(\emptyset), \\ \mathcal{K}_{it}^{ns}(\mathcal{B}_{it}^{cs}, \tau) - \mathcal{K}_{it}^{ns}(\mathcal{B}^0, 0) &< \frac{f^s}{x}. \end{aligned} \quad (\text{OA.A.50})$$

At equal net values, the branch bearing a larger fixed-cost burden must also generate larger gross operating payoffs, and therefore benefits more from productivity. The second comparison allows the incumbent's fixed-cost burden to exceed that of a fresh product, but by less than the search cost per successful replacement.

The burden includes all future operating and search costs of the line, not merely the remain-

ing operating duration of its current product. Thus small f^s does not make the first comparison automatic, and longer incumbent duration alone does not determine the second. At a terminal production decision, both no-search cost burdens equal βf^o , so the second inequality holds for $f^s > 0$. With subsequent adjustment opportunities, their cost burdens can differ even at a finite horizon. As f^s approaches zero, the incumbent's excess cost burden must also vanish or become negative for the positive search-threshold sign to persist, provided the relevant burdens and cutoffs converge.

Under both inequalities in (OA.A.50), the drop threshold falls and the search threshold rises, so the region strictly widens. This proves part (ii). The width can also increase when the search threshold falls. Its exact derivative is

$$\begin{aligned} \frac{\partial(\mathcal{B}_{it}^{cs} - \mathcal{B}_{it}^{ds})}{\partial z_i} = \frac{\eta - 1}{z_i} & \left[\frac{f^s/x + \mathcal{K}_{it}^{ns}(\mathcal{B}^0, 0) - \mathcal{K}_{it}^{ns}(\mathcal{B}_{it}^{cs}, \tau)}{V_{it, \mathcal{B}}^{ns}(\mathcal{B}_{it}^{cs}, \tau)} \right. \\ & \left. + \frac{\mathcal{K}_{it}^{ns}(\mathcal{B}_{it}^{ds}, \tau) - \beta(1 - \delta)\mathcal{K}_{i, t+1}^{ds}(\emptyset)}{V_{it, \mathcal{B}}^{ns}(\mathcal{B}_{it}^{ds}, \tau)} \right]. \end{aligned} \quad (\text{OA.A.51})$$

A positive second term can dominate a negative first term; the numerical illustration below gives the corresponding finite comparison. The proposition's two individual sign conditions are therefore sufficient, but jointly unnecessary, for the width sign.

A regular parameter point is one at which the relevant value and threshold derivatives exist, the displayed expectations of derivatives are integrable, and the threshold denominators are nonzero. The last requirement follows here from strict positivity of the no-search slope. All derivative statements, including those of the corollary, are made at such points. The optional shape condition requires its additional mixed derivatives to exist on the relevant interval; the dominance condition does not. At positive innovation variance, the next demand term has a continuous density and reaches the finitely many switching boundaries at a given tenure with probability zero. The continuation envelope is therefore evaluated at its optimizing branch almost surely. No smooth-pasting assumption is used.

Product tenure. Hold current firm age and the entire information-processing path fixed. The observation history for tenure $\tau + 1$ includes one additional earlier observation relative to tenure τ . Its posterior precision is therefore strictly greater. If the original product continues, this

precision difference persists because subsequent signal precisions are the same. At fixed next signal precision,

$$\frac{\partial v_{it}^\tau}{\partial \hat{\sigma}_{ipt}^2} = 1 - \left[1 + \hat{\sigma}_{ipt}^2 \frac{1 + \psi_{i,t+1}}{\sigma_\varepsilon^2} \right]^{-2} > 0.$$

The longer-tenure product thus has strictly less remaining innovation variance at every future date before replacement.

The sequence of replacement opportunities $\{\Delta_{i,t+j}^V\}$ is unchanged. Applying the preceding convexity and iteration argument gives

$$U_{it}(\mathcal{B}, \tau + 1) < U_{it}(\mathcal{B}, \tau), \quad \mathcal{B} > 0.$$

The two right-hand sides of the threshold equations are unchanged. Strict monotonicity in $\mathcal{B}$ therefore proves both inequalities in (OA.A.22). This comparison concerns feasible adjacent tenures at the same firm age; it does not change firm age along with product tenure.

Derivatives of both thresholds and their distance. Continue to use the evolving-path comparison. At the current margins, write $\mathcal{L}_t^j = \mathcal{L}_t(\mathcal{B}_t^j)$ and $s_t^j = U_{t,\mathcal{B}}(\mathcal{B}_t^j, \tau) > 0$, for $j \in \{ds, cs\}$. Differentiating value matching gives

$$\frac{\partial \mathcal{B}_t^{ds}}{\partial \psi_{it}} = \frac{\mathcal{L}_t^{ds}}{s_t^{ds}}, \quad \frac{\partial \mathcal{B}_t^{cs}}{\partial \psi_{it}} = \frac{1}{s_t^{cs}} \left(\frac{\partial \Delta_t^V}{\partial \psi_{it}} + \mathcal{L}_t^{cs} \right). \quad (\text{OA.A.52})$$

Consequently, the necessary and sufficient conditions are $\mathcal{L}_t^{ds} > 0$ and $\frac{\partial \Delta_t^V}{\partial \psi_{it}} + \mathcal{L}_t^{cs} > 0$, respectively. Unlike an unspecified relative-value derivative, (OA.A.39) identifies their sources and weights under the model's policies.

The exact width derivative is

$$\frac{\partial(\mathcal{B}_t^{cs} - \mathcal{B}_t^{ds})}{\partial \psi_{it}} = \frac{1}{s_t^{cs}} \frac{\partial \Delta_t^V}{\partial \psi_{it}} + \frac{\mathcal{L}_t^{cs}}{s_t^{cs}} - \frac{\mathcal{L}_t^{ds}}{s_t^{ds}}. \quad (\text{OA.A.53})$$

Thus, when $\mathcal{L}_t^{ds} > 0$ and $\mathcal{L}_t^{cs} \geq 0$, the simple sufficient dominance condition is

$$\frac{1}{s_t^{cs}} \frac{\partial \Delta_t^V}{\partial \psi_{it}} > \frac{\mathcal{L}_t^{ds}}{s_t^{ds}}. \quad (\text{OA.A.54})$$

The gain from current replacement opportunities, converted into demand units at the search margin, exceeds the demand compensation required by the marginal retained product. The nonnegative incumbent adjustment at the search margin then reinforces widening. This condition does not require mixed derivatives, or a shape restriction throughout the band. It is sufficient rather than necessary because it omits that favorable search-margin adjustment. It is a condition on endogenous learning and opportunity values, not a primitive parameter inequality.

For completeness, the exact local condition for widening is

$$\frac{\partial \Delta_t^V}{\partial \psi_{it}} > \frac{s_t^{cs}}{s_t^{ds}} \mathcal{L}_t^{ds} - \mathcal{L}_t^{cs}. \quad (\text{OA.A.55})$$

This is an accounting characterization, not an independent sufficient assumption. A different sufficient restriction, useful when dominance does not hold, is

$$\frac{\partial}{\partial b} \left[\frac{\mathcal{L}_t(b)}{U_{t,\mathcal{B}}(b, \tau)} \right] \geq 0, \quad b \in [\mathcal{B}_t^{ds}, \mathcal{B}_t^{cs}], \quad (\text{OA.A.56})$$

together with $\frac{\partial \Delta_t^V}{\partial \psi_{it}} > 0$. It says that the demand compensation for lost incumbent surplus does not fall across the region. Where the mixed derivatives exist, it is equivalent to

$$\frac{\partial^2 U_t(b, \tau)}{\partial b \partial \psi_{it}} \frac{\partial U_t(b, \tau)}{\partial b} - \frac{\partial U_t(b, \tau)}{\partial \psi_{it}} \frac{\partial^2 U_t(b, \tau)}{\partial b^2} \leq 0.$$

It implies

$$\frac{\partial(\mathcal{B}_t^{cs} - \mathcal{B}_t^{ds})}{\partial \psi_{it}} \geq \frac{1}{s_t^{cs}} \frac{\partial \Delta_t^V}{\partial \psi_{it}} > 0. \quad (\text{OA.A.57})$$

The shape and dominance conditions are alternative sufficient restrictions; neither is uniformly weaker than the other.

For tenure the comparison is discrete. The value band is unchanged, but

$$\Delta_t^V = \int_{\mathcal{B}_t^{ds}}^{\mathcal{B}_t^{cs}} U_{t,\mathcal{B}}(b, \tau) db. \quad (\text{OA.A.58})$$

Convexity yields

$$\frac{\Delta_t^V}{s_t^{cs}} \leq \mathcal{B}_t^{cs} - \mathcal{B}_t^{ds} \leq \frac{\Delta_t^V}{s_t^{ds}}. \quad (\text{OA.A.59})$$

Both slopes can change with tenure. The two tenure signs therefore do not imply a sign for their distance. A finite comparison can be checked directly using the two relative-value functions, without treating integer tenure as a differentiable parameter. $\square$

OA.A.10 Sufficient Conditions from Accumulated Information

Corollary OA.1 (Sufficient Conditions from Accumulated Information). *(i) In the specified precision profile, the incumbent-learning condition holds at the current date and every subsequent tenure if*

$$\psi_{it} \geq 1, \quad \sum_{s=t-\tau+1}^t \psi_{is} > \left(\sqrt{1 + \tau + \sigma_\varepsilon^2 / \sigma_0^2} - 1 \right)^2. \quad (\text{OA.A.60})$$

This applies to managerial comparisons and to smooth age comparisons whose marginal precision gains diminish with age, as detailed in the proof. The first restriction means that the current additional signal is at least as precise as the market signal. The second requires sufficient accumulated information. It is weaker than requiring additional-signal precision to exceed prior and market precision combined. The exact condition for any model-implied comparison is (OA.A.35).

(ii) Consider any fixed pair of information-processing paths for which replacement opportunities are weakly higher at every future date and strictly higher today. The continue-and-search region is wider under the path with higher replacement opportunities when the incumbents' remaining posterior uncertainties are sufficiently small. This result allows evolving ψ and requires no curvature condition. Equation (OA.A.69) gives an explicit sufficient bound. It is a comparison between two fixed paths; it does not assert a local derivative sign at every finite tenure.

(iii) With constant ψ_i and stationary primitives,

$$\frac{\partial \Delta_i^V}{\partial \psi_i} > 0, \quad \frac{\partial \Delta_i^V}{\partial z_i} > 0, \quad \Delta_i^V(\tau + 1) = \Delta_i^V(\tau). \quad (\text{OA.A.61})$$

The two information-processing threshold signs hold whenever

$$(1 + \psi_i)\sqrt{\tau(\tau + 1)} > \frac{\sigma_\varepsilon^2}{\sigma_0^2}. \quad (\text{OA.A.62})$$

For any two fixed abilities with positive vacancy values, higher ability also gives a wider region at every sufficiently long incumbent tenure.

Proof. We first establish the stock condition, then the finite-comparison width bound, and finally the stationary specialization.

A sufficient condition on accumulated information. For the managerial comparison, abbreviate $L = \sigma_\varepsilon^2/\sigma_0^2 + \tau$, $S = S_{it}^\tau$, and $p = \psi_{i,t+1}$. Here L, S, p are auxiliary scalars for this algebraic calculation. Equation (OA.A.36) has the sign of

$$\frac{S + p}{(L + S + 1 + p)^2} - \frac{S}{(L + S)^2}.$$

At fixed L, S , the first fraction is maximized over $p \geq 0$ at $p = \max\{L + 1 - S, 0\}$. If $S < L + 1$, its maximum equals $1/[4(L + 1)]$. Hence it is smaller than the second fraction at every possible next precision exactly when

$$S > g(L) \equiv L + 2 - 2\sqrt{L + 1} = \left(\sqrt{L + 1} - 1\right)^2. \quad (\text{OA.A.63})$$

If $S \geq L + 1$, the comparison holds directly, so the same lower bound characterizes strict negativity uniformly over all $p \geq 0$. At equality the derivative can be zero for the maximizing next precision. This bound is sharp for an unrestricted next precision; the actual deterministic profile can satisfy the exact learning condition even when this sufficient bound fails.

If $\psi_{it} \geq 1$, age-monotone precision implies that each subsequent additional signal adds at least one unit to S , while L increases by one. Since $g'(L) = 1 - 1/\sqrt{L + 1} < 1$, a strict inequality in (OA.A.63) is preserved after every additional observation of the original product. This proves part (i). The earlier simple restriction $S \geq L$ implies both conditions in (OA.A.60): monotonicity gives $\psi_{it} \geq S/\tau > 1$, and $g(L) < L$. The new joint condition is therefore strictly weaker than that earlier sufficient restriction. For example, with $\tau = 1$, $\sigma_0 = 0.6$, and $\sigma_\varepsilon = 1.1$, it requires $\psi_{it} > 1.73030$, instead of $\psi_{it} \geq 4.36111$. It signs incumbent learning, not replacement

opportunities or width.

The exact test also permits simple checks for other path comparisons. For example, suppose precision is nondecreasing and its responses along the specified current- ψ_{it} comparison satisfy

$$\frac{\partial \psi_{is}}{\partial \psi_{it}} \geq \frac{\partial \psi_{i,t+1}}{\partial \psi_{it}} > 0, \quad s = t - \tau + 1, \dots, t.$$

Writing $A = A_{it}^\tau$ and $b = b_{it}$, we have

$$\frac{\partial A}{\partial \psi_{it}} \geq \tau \frac{\partial \psi_{i,t+1}}{\partial \psi_{it}}, \quad A \leq \sigma_\varepsilon^2 / \sigma_0^2 + \tau b.$$

The exact test is implied by

$$(1 + \psi_{i,t+1}) \sqrt{\tau(\tau + 1)} > \sigma_\varepsilon^2 / \sigma_0^2. \quad (\text{OA.A.64})$$

Indeed, $\tau b(2A + b)/A^2$ decreases with A , and its value at that upper bound exceeds one precisely under the displayed inequality. An increasing, concave age profile has these ordered marginal gains for a smooth age shift. The CES age-and-data profile has this concavity on its smooth age- $a \geq 2$ branch when $\kappa > 0$ and its weights are nonnegative. This application requires that the observation history and its continuation lie on that smooth branch; a comparison crossing the age-two change in the data formula must instead use the actual finite paths. This is an application of the same precision test, not a separate age theorem. The stock bound in (OA.A.60) also suffices in these smooth age comparisons. In fact,

$$\frac{\partial A / \partial \psi_{it}}{\partial b / \partial \psi_{it}} \geq \tau \geq \frac{S}{\psi_{i,t+1}},$$

so their accumulated precision response relative to the next-signal response is at least as large as in the managerial comparison at the same state. The managerial stock bound therefore implies the age comparison's exact learning condition as well, at every subsequent tenure.

A width bound for evolving information processing. We prove part (ii) without parameter derivatives. Put $d = \beta(1 - \delta)$ and write $\bar{\pi}(b, z_i) = \mathcal{A}(z_i)b^\eta - f^o$, where $\mathcal{A}(z_i) > 0$ is the coefficient in (OA.A.18). Fix an economy's actual bounded, positive sequence $\{\Delta_s^V\}_{s \geq t}$. Define $U_s^0(b)$ by

the auxiliary recursion

$$U_s^0(b) = \beta \bar{\pi}(b, z_i) + d \max\{U_{s+1}^0(b) - x \Delta_{s+1}^V, (1-x)U_{s+1}^0(b), 0\}. \quad (\text{OA.A.65})$$

Only the original incumbent has no remaining uncertainty. Fresh-product learning and all subsequent replacement opportunities retain their actual values through Δ_s^V . This is an auxiliary comparison, not a change to the prior of new products or a new equilibrium.

At the operating break-even demand term

$$b_* = (f^o / \mathcal{A}(z_i))^{1/\eta},$$

the recursion has $U_s^0(b_*) = 0$ at every date. Strict monotonicity therefore makes b_* the fully learned incumbent's drop threshold, independently of the information-processing path. Write c_t^0 for its search threshold, the unique solution of $U_t^0(c_t^0) = \Delta_t^V$. If $\Delta_s^{V,H} \geq \Delta_s^{V,L}$ at every future date, then $U_t^{0,H}(b) \leq U_t^{0,L}(b)$ by monotone iteration. If the inequality for opportunities is strict today, $c_t^{0,H} > c_t^{0,L}$. Thus the fully learned incumbents' regions are strictly ordered in width in the evolving model.

To bound the error for an incumbent with posterior variance $s = \hat{\sigma}_{ipt}^2$, let

$$\mathcal{E}(s) = \frac{\sqrt{\exp\{[(\eta-1)^2(2\eta-1)/\eta]s\} - 1}}{1-d}, \quad k(s) = [1 + \mathcal{E}(s)]^{1/\eta}. \quad (\text{OA.A.66})$$

Along continued operation, write $\mathcal{B}_j = bM_j$. The lognormal multiplier M_j has mean one and cumulative log variance at most $[(\eta-1)/\eta]^2 s$, since total future posterior-mean innovation variance cannot exceed the current posterior variance. Its moments imply

$$\begin{aligned} \mathbb{E}|M_j^\eta - 1| &\leq \sqrt{\mathbb{E}(M_j^\eta - 1)^2} \\ &\leq \sqrt{\exp\{[(\eta-1)^2(2\eta-1)/\eta]s\} - 1}. \end{aligned}$$

The second bound uses $\mathbb{E}M_j^\eta \geq 1$.

The recursion for U admits the following auxiliary policy representation. At each future date, retaining without search carries the opportunity charge $x\Delta_s^V$; continuing with search keeps the incumbent with probability $1-x$; dropping terminates its relative payoffs. Fixed operating costs

and opportunity charges are identical under the stochastic and fully learned comparisons when the same randomized action sequence is used. Only the gross operating payoffs $\beta \mathcal{A}(z_i) b^\eta M_j^\eta$ differ. Randomization cannot improve the deterministic comparison's optimum. Applying the above bound to each policy, using continuation weights at most one, and then taking suprema gives

$$0 \leq U_t(b, \tau) - U_t^0(b) \leq \beta \mathcal{A}(z_i) b^\eta \mathcal{E}(s). \quad (\text{OA.A.67})$$

The lower bound follows independently from convexity and monotone iteration. The payoff tail bound justifies the infinite sum, uniformly over policies and future precision paths with this current variance.

Both functions are strictly increasing. Moreover, $U_t^0(b_2) - U_t^0(b_1) \geq \beta \mathcal{A}(z_i)(b_2^\eta - b_1^\eta)$ for $b_2 \geq b_1$. Applying (OA.A.67) at either matching root therefore gives

$$\frac{b_*}{k(s)} \leq \mathcal{B}_t^{ds} \leq b_*; \quad \frac{c_t^0}{k(s)} \leq \mathcal{B}_t^{cs} \leq c_t^0. \quad (\text{OA.A.68})$$

For a pair of paths, set $k_H = k(\hat{\sigma}_{ipt,H}^2)$ and $k_L = k(\hat{\sigma}_{ipt,L}^2)$. A sufficient condition for strictly greater width under the higher-opportunity path is

$$\frac{c_t^{0,H}}{k_H} - c_t^{0,L} > b_* \left(1 - \frac{1}{k_L} \right). \quad (\text{OA.A.69})$$

Indeed, the higher-path width is at least $c_t^{0,H}/k_H - b_*$, whereas the lower-path width is at most $c_t^{0,L} - b_*/k_L$. As both posterior variances tend to zero, the difference between these bounds tends to $c_t^{0,H} - c_t^{0,L} > 0$. This proves part (ii). The bound is conservative and still uses replacement opportunities, but requires no threshold derivatives, mixed derivatives, or restriction on the sign of every incumbent innovation-variance change. It can be evaluated for the actual pair of paths. Sufficiently small uncertainty need not be attainable at a given young firm age, and no uniform bound is asserted as the pair of paths approaches an infinitesimal comparison.

The stationary specialization. With constant ψ_i and stationary line-level primitives, the vacant-line problem is independent of firm age. Product tenure and beliefs still evolve for occupied

lines. Equation (OA.A.29) therefore simplifies to

$$\Delta_i^V = \frac{1 - \beta(1 - \delta)}{x} V_i^{ds}(\emptyset). \quad (\text{OA.A.70})$$

Since the stationary vacant-line value is positive, (OA.A.70) implies $\Delta_i^V > 0$. The relative-value recursion extends continuously to $U_i(0, \tau) = -\beta f^o$, while $U_i(\mathcal{B}, \tau) \rightarrow \infty$ as $\mathcal{B} \rightarrow \infty$. Strict monotonicity therefore gives two unique, finite, positive, ordered matching cutoffs. No separate ordering assumption is needed here.

The policy comparison in Appendix OA.A.9 shows that more precise signals raise vacancy value. The increase is strict in this model. Following successful search and continued operation, there is positive probability of retaining a product with a nondegenerate posterior. The more informed firm can preserve the less informed switching policy but choose output using the more precise belief. Since output does not affect signal precision, this yields a strict gain from the strictly convex optimized production payoff. For a small precision increase, the additional posterior-mean innovation variance is proportional to that increase, so the same argument on a compact set of active histories gives a strictly positive marginal gain at regular parameters. Hence

$$\frac{\partial \Delta_i^V}{\partial \psi_i} > 0. \quad (\text{OA.A.71})$$

This value-band result requires no stock condition.

For the existing product,

$$\hat{\sigma}_{ip, \tau}^2 = [\sigma_0^{-2} + \tau(1 + \psi_i)/\sigma_\varepsilon^2]^{-1}, \quad v_i^\tau = \frac{(1 + \psi_i)/\sigma_\varepsilon^2}{[\sigma_0^{-2} + \tau(1 + \psi_i)/\sigma_\varepsilon^2][\sigma_0^{-2} + (\tau + 1)(1 + \psi_i)/\sigma_\varepsilon^2]}.$$

Differentiating gives

$$\frac{\partial v_i^\tau}{\partial \psi_i} = \frac{1}{\sigma_\varepsilon^2} \frac{\sigma_0^{-4} - \tau(\tau + 1)[(1 + \psi_i)/\sigma_\varepsilon^2]^2}{[\sigma_0^{-2} + \tau(1 + \psi_i)/\sigma_\varepsilon^2]^2 [\sigma_0^{-2} + (\tau + 1)(1 + \psi_i)/\sigma_\varepsilon^2]^2}. \quad (\text{OA.A.72})$$

This derivative is negative exactly under (OA.A.62). The required precision in that condition falls with tenure. If it holds at the current tenure, it holds at every subsequent tenure of the same product. Meanwhile, (OA.A.71) implies that replacement opportunities improve at every date. The two path conditions in (OA.A.41) therefore hold for a local increase in ψ_i , with a

strict current variance effect. The proof of the proposition gives

$$U_{i,\psi_i}(\mathcal{B}, \tau) < 0, \quad \frac{\partial \mathcal{B}_i^{ds}(\tau)}{\partial \psi_i} > 0, \quad \frac{\partial \mathcal{B}_i^{cs}(\tau)}{\partial \psi_i} > 0.$$

Future product replacements are accounted for by the increasing value band. The comparison concerns retention relative to vacancy; the no-search value itself can rise with information-processing ability.

Under the additional curvature condition (OA.A.56), (OA.A.57) proves the third inequality in (OA.A.20). No curvature restriction of this kind was needed to prove the two individual threshold signs.

Sufficiently long incumbent tenure. There is also a sufficient regime for a finite ability comparison that does not impose (OA.A.56). Fix any two finite abilities $\psi_i^H > \psi_i^L$ with positive vacant-line values. For each ability, $v_i^{\tau+j} \rightarrow 0$ as $\tau \rightarrow \infty$, for every fixed j . The actual stationary value band remains fixed as incumbent tenure increases; fresh products continue to start from the original prior.

Consider the auxiliary iterations of (OA.A.32) with this fixed value band. For each finite number of iterations, the retained product's transition converges to $\mathcal{B}' = \mathcal{B}$. The payoff moment bound permits passage through the finite expectations, uniformly on compact demand intervals. The truncation error already proved for U is uniform in tenure on such intervals. Taking the tenure limit first for each finite truncation, and then letting the truncation grow, therefore shows that $U_i(\mathcal{B}, \tau)$ converges locally uniformly to the unique scalar solution of

$$U_i(\mathcal{B}, \infty) = \beta \bar{\pi}(\mathcal{B}, z_i) + \beta(1 - \delta) \max\{U_i(\mathcal{B}, \infty) - x \Delta_i^V, (1 - x)U_i(\mathcal{B}, \infty), 0\}. \quad (\text{OA.A.73})$$

Uniqueness follows because the right-hand side is a contraction in its scalar continuation argument, with modulus $\beta(1 - \delta) < 1$.

The limiting drop and search roots are unique and strictly crossed. Local uniform convergence therefore gives convergence of both cutoffs. Evaluating (OA.A.73) at its matching levels yields

$$\bar{\pi}(\mathcal{B}_i^{ds}(\infty), z_i) = 0,$$

$$\bar{\pi}(\mathcal{B}_i^{cs}(\infty), z_i) = \frac{1 - \beta(1 - \delta)(1 - x)}{\beta} \Delta_i^V. \quad (\text{OA.A.74})$$

The limiting drop cutoff is independent of ψ_i . The limiting search cutoff is strictly increasing in Δ_i^V , hence strictly increasing between the two abilities. Consequently,

$$\lim_{\tau \rightarrow \infty} [\mathcal{B}_i^{cs,H}(\tau) - \mathcal{B}_i^{ds,H}(\tau) - \mathcal{B}_i^{cs,L}(\tau) + \mathcal{B}_i^{ds,L}(\tau)] > 0. \quad (\text{OA.A.75})$$

The width difference is therefore positive at every sufficiently large finite tenure. Economically, once little information remains to be learned about the incumbent, higher ability mainly improves fresh-product opportunities; the drop margin approaches the same operating break-even cutoff. This result compares fixed ability levels. It does not interchange an ability derivative with the tenure limit, impose monotonicity of width in tenure, or provide a common tenure bound across all ability pairs. It applies to the stationary benchmark and does not imply that the required tenure is feasible at any given young firm age.

Finally, the stationary benchmark also implies $\partial_{z_i} \Delta_i^V > 0$ without an additional low-search-cost condition. A vacant line searches and produces after success with positive probability; holding its switching policy fixed, higher z_i strictly raises operating payoffs. Therefore $V_{i,z_i}^{ds} > 0$, and (OA.A.70) proves the claim. This value-band sign still does not replace the two marginal-payoff conditions needed for the individual productivity-threshold signs in the proposition. $\square$

Thus greater information-processing ability can make firms more selective about both retention and search. The threshold effects remain distinct from its effect on beliefs: observed switching depends on policy thresholds together with the distribution of demand terms.

OA.A.11 Returns to Pre-Entry Investment

Proposition OA.1 (Returns to Pre-Entry Investment). *Hold prices and the joint distribution of normalized types $(\varphi_i, \tilde{m}_i)$ fixed, with $z_i = z_0 \varphi_i$ and $m_i = m_0 \tilde{m}_i$. Let $\mathcal{E}(m_0, z_0) \equiv \mathbb{E}_\varphi[V^e(\varphi)]$ denote expected entry value before investment costs. Under optimal policies, write $\mathcal{E} = \mathcal{G} - \mathcal{K}$, where $\mathcal{G}$ denotes expected discounted gross operating payoffs and $\mathcal{K}$ denotes expected discounted operating and search fixed costs.¹*

¹Gross operating payoffs are revenues less variable production costs. Both components include subsequent product replacements and exclude pre-entry investment costs. The endogenous-exit payoff $V^{ds}(\emptyset)$ is decomposed into

At regular points with finite, integrable derivatives, $\partial\mathcal{E}/\partial m_0 \geq 0$ and

$$\frac{\partial\mathcal{E}}{\partial z_0} = \frac{\eta - 1}{z_0} \mathcal{G} > 0, \quad (\text{OA.A.76})$$

$$\frac{\partial^2\mathcal{E}}{\partial z_0 \partial m_0} = \frac{\eta - 1}{z_0} \frac{\partial\mathcal{G}}{\partial m_0} = \frac{\eta - 1}{z_0} \left(\frac{\partial\mathcal{E}}{\partial m_0} + \frac{\partial\mathcal{K}}{\partial m_0} \right), \quad (\text{OA.A.77})$$

where the strict productivity sign requires production with positive probability. Thus, the capabilities are locally complementary if and only if higher managerial capability weakly raises expected discounted gross operating payoffs. A sufficient condition for strict complementarity is

$$\frac{\partial\mathcal{E}}{\partial m_0} > 0, \quad \frac{\partial\mathcal{K}}{\partial m_0} \geq 0. \quad (\text{OA.A.78})$$

Proof. Higher m_0 increases signal precision along the model-implied information-processing path. The firm can reproduce the lower-precision signals by adding independent noise and follow the associated decisions. Optimizing with the additional information therefore weakly raises expected entry value.

Fix a feasible operating and switching policy, with quantities chosen optimally conditional on information. Its expected payoff has the form $z_0^{\eta-1}H - \mathcal{K}$, where H and $\mathcal{K}$ are independent of z_0 when the policy and information-processing path are held fixed. For endogenous-exit payoffs, use their equivalent vacant-line continuation representation for the initial line at age one and all lines from age two onward. The envelope theorem then gives (OA.A.76). Changes in decision regions contribute no additional boundary terms because the competing firm-level payoffs coincide at indifference, including the age-one retention–exit margin and the preproduction operating–abandonment margin. The maintained integrability conditions justify passage to the infinite horizon. Differentiating (OA.A.76) with respect to m_0 and using $\mathcal{G} = \mathcal{E} + \mathcal{K}$ establishes (OA.A.77) and the sufficient condition. $\square$

Condition (OA.A.78) requires managerial improvements to raise value without reducing the discounted fixed-cost burden. It is sufficient, not necessary: complementarity can also hold when fixed costs fall, provided gross operating payoffs increase. The condition concerns en-

its corresponding gross operating payoffs and fixed costs under vacant-line continuation. This applies to the initial line at age one and all lines from age two onward. Abandonment before production contributes zero continuation payoff.

ogenous operating and search decisions rather than primitives alone.

OA.B Model Extension

OA.B.1 Persistent Product-Market Shocks

This appendix collects the persistent-shock equations under the baseline timing, entry, and exit conventions. Calendar-time subscripts are suppressed only for stationary aggregates.

Demand and information. With the prior $\log \theta_{ip} \sim N(\theta_0, \sigma_0^2)$, realized demand satisfies

$$\log \theta_{ipt} = \log \theta_{ip} + \varepsilon_{ipt}, \quad (\text{OA.B.1})$$

$$\varepsilon_{ip,t+1} \mid \varepsilon_{ipt} \sim N(\rho \varepsilon_{ipt}, (1 - \rho^2) \sigma_\varepsilon^2), \quad 0 \leq \rho < 1. \quad (\text{OA.B.2})$$

Market innovations are independent across periods and lines and independent of expertise and private-signal errors. New products start with stationary shocks $\varepsilon \sim N(0, \sigma_\varepsilon^2)$. The additional signal remains

$$\log s_{ipt} = \log \theta_{ip} + v_{ipt}, \quad v_{ipt} \sim N\left(0, \frac{\sigma_\varepsilon^2}{\psi_{it}}\right). \quad (\text{OA.B.3})$$

Signal errors retain their baseline independence assumptions. Beliefs condition on $\mathcal{I}_{ipt} = \{\theta_{ips}, s_{ips}\}_{s=t-\tau+1}^t$, where $\tau \equiv \tau_{ipt}$.

After the first observation, market outcomes enter through

$$\log \theta_{ip,t+1} - \rho \log \theta_{ipt} = (1 - \rho) \log \theta_{ip} + \varepsilon_{ip,t+1} - \rho \varepsilon_{ipt}. \quad (\text{OA.B.4})$$

The innovation has variance $(1 - \rho^2) \sigma_\varepsilon^2$. Market precision is $1/\sigma_\varepsilon^2$ for the first observation and $(1 - \rho)/[(1 + \rho) \sigma_\varepsilon^2]$ thereafter.

Posterior beliefs. For $\tau \geq 1$, posterior moments are

$$\begin{aligned} \hat{\theta}_{ipt} = \hat{\sigma}_{ipt}^2 & \left[\frac{\theta_0}{\sigma_0^2} + \frac{\log \theta_{ip,t-\tau+1}}{\sigma_\varepsilon^2} \right. \\ & \left. + \sum_{s=t-\tau+2}^t \frac{\log \theta_{ips} - \rho \log \theta_{ip,s-1}}{(1 + \rho) \sigma_\varepsilon^2} + \sum_{s=t-\tau+1}^t \frac{\psi_{is} \log s_{ips}}{\sigma_\varepsilon^2} \right], \end{aligned} \quad (\text{OA.B.5})$$

$$\hat{\sigma}_{ipt}^2 = \left[\frac{1}{\sigma_0^2} + \frac{1}{\sigma_\varepsilon^2} + (\tau - 1) \frac{1 - \rho}{(1 + \rho)\sigma_\varepsilon^2} + \sum_{s=t-\tau+1}^t \frac{\psi_{is}}{\sigma_\varepsilon^2} \right]^{-1}. \quad (\text{OA.B.6})$$

The first sum is empty at $\tau = 1$. For continuing products with positive tenure,

$$\hat{\sigma}_{ip,t+1}^2 = \left[\frac{1}{\hat{\sigma}_{ipt}^2} + \frac{1 - \rho}{(1 + \rho)\sigma_\varepsilon^2} + \frac{\psi_{i,t+1}}{\sigma_\varepsilon^2} \right]^{-1}, \quad (\text{OA.B.7})$$

$$\hat{\theta}_{ip,t+1} = \hat{\sigma}_{ip,t+1}^2 \left[\frac{\hat{\theta}_{ipt}}{\hat{\sigma}_{ipt}^2} + \frac{\log \theta_{ip,t+1} - \rho \log \theta_{ipt}}{(1 + \rho)\sigma_\varepsilon^2} + \frac{\psi_{i,t+1} \log s_{ip,t+1}}{\sigma_\varepsilon^2} \right]. \quad (\text{OA.B.8})$$

From the common prior (θ_0, σ_0^2) , the first update is

$$\hat{\sigma}_{ip,t+1}^2 = \left[\frac{1}{\sigma_0^2} + \frac{1 + \psi_{i,t+1}}{\sigma_\varepsilon^2} \right]^{-1}, \quad (\text{OA.B.9})$$

$$\hat{\theta}_{ip,t+1} = \hat{\sigma}_{ip,t+1}^2 \left[\frac{\theta_0}{\sigma_0^2} + \frac{\log \theta_{ip,t+1}}{\sigma_\varepsilon^2} + \frac{\psi_{i,t+1} \log s_{ip,t+1}}{\sigma_\varepsilon^2} \right]. \quad (\text{OA.B.10})$$

These initial updates apply to entrants' and replacement products, using signal precision at the observation date.

State and joint transition. A positive-tenure line has state

$$\xi_{ipt} = (\hat{\theta}_{ipt}, \log \theta_{ipt}, \tau_{ipt}; a_{it}, \psi_{it}, \varphi_i, m_i). \quad (\text{OA.B.11})$$

Posterior variance is determined by tenure and the information history. Current realized demand forecasts the persistent shock; the equivalent numerical coordinates are $(\hat{\theta}_{ipt}, \log \theta_{ipt} - \hat{\theta}_{ipt})$.

Conditional on retention,

$$\log \theta_{ip,t+1} \mid \mathcal{I}_{ipt} \sim N\left(\rho \log \theta_{ipt} + (1 - \rho) \hat{\theta}_{ipt}, (1 - \rho)^2 \hat{\sigma}_{ipt}^2 + (1 - \rho^2) \sigma_\varepsilon^2\right). \quad (\text{OA.B.12})$$

The pair $(\hat{\theta}_{ip,t+1}, \log \theta_{ip,t+1})$ is jointly Gaussian, conditional on current information and firm

states, with moments

$$\mathbb{E}_{it} \begin{pmatrix} \hat{\theta}_{ip,t+1} \\ \log \theta_{ip,t+1} \end{pmatrix} = \begin{pmatrix} \hat{\theta}_{ipt} \\ \rho \log \theta_{ipt} + (1 - \rho) \hat{\theta}_{ipt} \end{pmatrix}, \quad (\text{OA.B.13})$$

$$\text{Var}_{it} \begin{pmatrix} \hat{\theta}_{ip,t+1} \\ \log \theta_{ip,t+1} \end{pmatrix} = \begin{pmatrix} \hat{\sigma}_{ipt}^2 - \hat{\sigma}_{ip,t+1}^2 & (1 - \rho) \hat{\sigma}_{ipt}^2 \\ (1 - \rho) \hat{\sigma}_{ipt}^2 & (1 - \rho)^2 \hat{\sigma}_{ipt}^2 + (1 - \rho^2) \sigma_\varepsilon^2 \end{pmatrix}. \quad (\text{OA.B.14})$$

The cross-covariance uses

$$\text{Cov}_{it}(\hat{\theta}_{ip,t+1}, \log \theta_{ip,t+1}) = \text{Cov}_{it}(\log \theta_{ip}, \log \theta_{ip,t+1}).$$

For a tenure-zero product, the initial pair is jointly Gaussian with moments

$$\mathbb{E}_{it} \begin{pmatrix} \hat{\theta}_{ip,t+1} \\ \log \theta_{ip,t+1} \end{pmatrix} = \begin{pmatrix} \theta_0 \\ \theta_0 \end{pmatrix}, \quad (\text{OA.B.15})$$

$$\text{Var}_{it} \begin{pmatrix} \hat{\theta}_{ip,t+1} \\ \log \theta_{ip,t+1} \end{pmatrix} = \begin{pmatrix} \sigma_0^2 - \hat{\sigma}_{ip,t+1}^2 & \sigma_0^2 \\ \sigma_0^2 & \sigma_0^2 + \sigma_\varepsilon^2 \end{pmatrix}, \quad (\text{OA.B.16})$$

where (OA.B.9) gives the next posterior variance. Let $g_{it}(d\xi' \mid \xi)$ include these Gaussian laws and $a_{i,t+1} = a_{it} + 1$, $\tau_{ip,t+1} = \tau_{ipt} + 1$. Write $\xi^0(a; \varphi, m)$ for the common-prior state, without realized demand, and $\mathbb{E}_{it}^0$ for its initial transition.

Production. The demand term for positive tenure is

$$\begin{aligned} \mathcal{B}_{it}(\xi) &\equiv \mathbb{E}_{it} \left[\theta_{ip,t+1}^{(\eta-1)/\eta} \right], \\ \log \mathcal{B}_{it}(\xi) &= \frac{\eta-1}{\eta} \left[\rho \log \theta_{ipt} + (1 - \rho) \hat{\theta}_{ipt} \right] \\ &\quad + \frac{1}{2} \left(\frac{\eta-1}{\eta} \right)^2 \left[(1 - \rho)^2 \hat{\sigma}_{ipt}^2 + (1 - \rho^2) \sigma_\varepsilon^2 \right]. \end{aligned} \quad (\text{OA.B.17})$$

For a tenure-zero product,

$$\mathcal{B}^0 = \exp \left\{ \frac{\eta-1}{\eta} \theta_0 + \frac{1}{2} \left(\frac{\eta-1}{\eta} \right)^2 (\sigma_0^2 + \sigma_\varepsilon^2) \right\}. \quad (\text{OA.B.18})$$

Quantity, production labor, realized price, and expected profit satisfy

$$y_{ip,t+1}(\xi) = \left(\frac{\eta - 1}{\eta} \right)^\eta P_{p,t+1}^\eta C_{p,t+1} \mathcal{B}_{it}(\xi)^\eta \left(\frac{z_i}{w_{t+1}} \right)^\eta, \quad (\text{OA.B.19})$$

$$\ell_{ip,t+1}(\xi) = \left(\frac{\eta - 1}{\eta} \right)^\eta P_{p,t+1}^\eta C_{p,t+1} \mathcal{B}_{it}(\xi)^\eta z_i^{\eta-1} w_{t+1}^{-\eta}, \quad (\text{OA.B.20})$$

$$p_{ip,t+1} = \frac{\eta}{\eta - 1} \frac{w_{t+1}}{z_i} \frac{\theta_{ip,t+1}^{(\eta-1)/\eta}}{\mathcal{B}_{it}(\xi)}, \quad (\text{OA.B.21})$$

$$\bar{\pi}_{it+1}(\xi) = \frac{(\eta - 1)^{\eta-1}}{\eta^\eta} \left(\frac{z_i}{w_{t+1}} \right)^{\eta-1} P_{p,t+1}^\eta C_{p,t+1} \mathcal{B}_{it}(\xi)^\eta - w_{t+1} f^o. \quad (\text{OA.B.22})$$

At tenure zero, use $\mathcal{B}^0$. Expected price remains $\eta w_{t+1}/[(\eta - 1)z_i]$.

Product-line values. Under the baseline timing and exit payoffs, the branch values satisfy

$$V_{it}^{ns}(\xi) = \beta \bar{\pi}_{it+1}(\xi) + \beta(1 - \delta) \mathbb{E}_{it}[V_{i,t+1}(\xi') \mid \xi], \quad (\text{OA.B.23})$$

$$V_{it}^{cs}(\xi) = -w_t f^s + (1 - x) V_{it}^{ns}(\xi) + x V_{it}^{ns}(\xi^0), \quad (\text{OA.B.24})$$

$$V_{it}^{ds}(\emptyset) = -w_t f^s + x V_{it}^{ns}(\xi^0) + (1 - x) \beta(1 - \delta) V_{i,t+1}^{ds}(\emptyset). \quad (\text{OA.B.25})$$

Common firm states are suppressed. Here ξ^0 is the fresh-product state at the current firm age, evaluated using $\mathcal{B}^0$ and the initial transition.

For firms aged two and older,

$$V_{it}(\xi) = \max\{V_{it}^{ns}(\xi), V_{it}^{cs}(\xi), V_{it}^{ds}(\emptyset)\}. \quad (\text{OA.B.26})$$

Let $\mathcal{R}_{it}^{ns}$, $\mathcal{R}_{it}^{cs}$, and $\mathcal{R}_{it}^{ds}$ denote the maximizing regions. Where ordered cutoffs exist, they depend on $(\log \theta_{ipt}, \tau_{ipt}; a_{it}, \psi_{it}, \varphi_i, m_i)$ and satisfy

$$V_{it}^{cs}(\hat{\theta}^{ds}, \log \theta_{ipt}, \tau_{ipt}) = V_{it}^{ds}(\emptyset), \quad (\text{OA.B.27})$$

$$V_{it}^{ns}(\hat{\theta}^{cs}, \log \theta_{ipt}, \tau_{ipt}) = V_{it}^{cs}(\hat{\theta}^{cs}, \log \theta_{ipt}, \tau_{ipt}). \quad (\text{OA.B.28})$$

Equal demand terms need not imply equal continuation values, so both continuous state coordinates are retained.

Incumbent firm value is

$$V(\Theta_{it}; a_{it}, \psi_{it}, \varphi_i, m_i) = \sum_{k=1}^{n_i} V_{it}(\xi_{ik,t}), \quad a_{it} \geq 2, \quad (\text{OA.B.29})$$

with $V_{it}(\xi_{ik,t}) = V_{it}^{ds}(\emptyset)$ for vacant lines.

Startups and pre-entry decisions. Upon endogenous exit at age one, the initial line receives $V_{t+1}^{ds}(\emptyset)$, while the initially vacant lines receive zero. Its first retention decision has value

$$V_{t+1}^{e,1}(\hat{\theta}, \log \theta; \psi, \varphi, m) = \max \left\{ V_{t+1}^{ns}(\hat{\theta}, \log \theta, 1) + (n(\varphi) - 1)V_{t+1}^{ds}(\emptyset), \right. \\ \left. V_{t+1}^{cs}(\hat{\theta}, \log \theta, 1) + (n(\varphi) - 1)V_{t+1}^{ds}(\emptyset), V_{t+1}^{ds}(\emptyset) \right\}. \quad (\text{OA.B.30})$$

Firm states are $(1, \psi, \varphi, m)$, with maximizing regions $\mathcal{R}_{t+1}^{ns,e}$, $\mathcal{R}_{t+1}^{cs,e}$, and $\mathcal{R}_{t+1}^{ds,e}$. Retention requires $\max\{V^{ns}, V^{cs}\} + (n(\varphi) - 1)V^{ds} \geq V^{ds}$; conditional on retention, the initial line searches if $V^{cs} > V^{ns}$. Exit precedes all search.

Before m is revealed, operating value conditional on φ is

$$\tilde{V}_t^e(\varphi) = \beta \bar{\pi}_{t+1}(\xi^0(0; \varphi)) \\ + \beta(1 - \delta) \mathbb{E}_{m|\varphi} \left[\mathbb{E}_t^0 V_{t+1}^{e,1}(\hat{\theta}', \log \theta'; \psi_{i,t+1}, \varphi, m) \right]. \quad (\text{OA.B.31})$$

The operating option and interior cutoff satisfy

$$V_t^e(\varphi) = \max\{\tilde{V}_t^e(\varphi), 0\}, \quad \tilde{V}_t^e(\underline{\varphi}_t^e) = 0. \quad (\text{OA.B.32})$$

Pre-entry investments solve

$$(a_t^m, a_t^z) \in \arg \max_{a_t^m, a_t^z} \{ \mathbb{E}_\varphi[V_t^e(\varphi)] - w_t f_t^e(a_t^m, a_t^z) \}, \quad (\text{OA.B.33})$$

using the baseline capability technologies and costs. Free entry requires

$$\mathbb{E}_\varphi[V_t^e(\varphi)] = w_t f_t^e(a_t^m, a_t^z). \quad (\text{OA.B.34})$$

In stationarity, $M = m_0$, $Z = z_0$, $a^m = m_0$, and $a^z = z_0$.

Product-line measures. Let μ_{pt} record lines after search, with vacant lines indexed by their latest search destination. Initially vacant startup lines enter this measure after their first search. Let $\mathcal{X}_a$ include active states at age a , including the prior atom; $\mu_{pt}(\emptyset, a)$ is vacant-line mass. Firm states (ψ, φ, m) are suppressed and permanent types are integrated in aggregates.

The post-drop vacancy probability in symmetric equilibrium is

$$q_{t+1}^\emptyset(a') = \frac{\sum_q [\mu_{qt}(\emptyset, a' - 1) + \int_{\mathcal{X}_{a'-1}} \int \mathbf{1}_{\mathcal{R}_{t+1}^{ds}}(\xi') g_t(d\xi' | \xi) \mu_{qt}(d\xi)]}{\sum_q [\mu_{qt}(\emptyset, a' - 1) + \mu_{qt}(\mathcal{X}_{a'-1})]}, \quad a' \geq 2. \quad (\text{OA.B.35})$$

Conditional independence of the $n(\varphi) - 1$ non-focal vacancy indicators gives the approximation

$$p_{t+1}^e(a') = \left[q_{t+1}^\emptyset(a') \right]^{n(\varphi)-1}, \quad a' \geq 2. \quad (\text{OA.B.36})$$

Here p^e is the probability that all other lines are vacant; set $p^e = 1$ when $n(\varphi) = 1$. Vacant and dropped lines therefore reach search with probability $1 - p^e$, while retained lines reach search directly.

For startups, endogenous-exit probability conditional on no exogenous death is exactly

$$p_{t+1}^e(1) = \int \mathbf{1}_{\mathcal{R}_{t+1}^{ds,e}}(\xi') g_t(d\xi' | \xi^0(0)). \quad (\text{OA.B.37})$$

Each initially vacant line reaches search with probability $1 - p_{t+1}^e(1)$.

Continuing products and new assignments. For measurable $A \subseteq \mathcal{X}_{a+1}$ containing only positive-tenure states,

$$\mu_{p,t+1}(A) = (1 - \delta) \int_{\mathcal{X}_a} \int_A \left[\mathbf{1}_{\mathcal{R}_{t+1}^{ns}}(\xi') + (1 - x) \mathbf{1}_{\mathcal{R}_{t+1}^{cs}}(\xi') \right] g_t(d\xi' | \xi) \mu_{pt}(d\xi). \quad (\text{OA.B.38})$$

At $a = 0$, use the startup regions $\mathcal{R}_{t+1}^{ns,e}$, $\mathcal{R}_{t+1}^{cs,e}$ and the initial transition.

Successful searches from incumbent source ages $a \geq 1$ generate

$$\mu_{p,t+1}(\xi^0(a+1)) = \frac{(1 - \delta)x}{N - 1} \sum_{q \neq p} \left\{ \int_{\mathcal{X}_a} \int \left[\mathbf{1}_{\mathcal{R}_{t+1}^{cs}}(\xi') \right]$$

$$\begin{aligned}
& + (1 - p_{t+1}^e(a+1)) \mathbf{1}_{\mathcal{R}_{t+1}^{ds}}(\xi') \Big] g_t(d\xi' \mid \xi) \mu_{qt}(d\xi) \\
& + (1 - p_{t+1}^e(a+1)) \mu_{qt}(\emptyset, a) \Big\}. \tag{OA.B.39}
\end{aligned}$$

For startups' first adjustment,

$$\begin{aligned}
\mu_{p,t+1}(\xi^0(1)) = & \frac{(1-\delta)x}{N-1} \sum_{q \neq p} \left[\int \mathbf{1}_{\mathcal{R}_{t+1}^{cs,e}}(\xi') g_t(d\xi' \mid \xi^0(0)) \right. \\
& \left. + (n(\varphi) - 1)(1 - p_{t+1}^e(1)) \right] \mu_{qt}(\xi^0(0)). \tag{OA.B.40}
\end{aligned}$$

Under the baseline assignment mechanism and symmetry, destination probability $1/[N - n(\varphi)]$ over feasible markets averages to $1/(N - 1)$ for each $p \neq q$.

The density of operating entrants is

$$\mu_{p,t+1}(\xi^0(0; \varphi, m)) = \frac{\mathcal{M}_{t+1}^e}{N} \frac{h_\varphi(\varphi) h_m(m \mid \varphi; m_{0,t+1}) \mathbf{1}_{\{\varphi \geq \underline{\varphi}_{t+1}^e\}}}{1 - H_\varphi(\underline{\varphi}_{t+1}^e)}. \tag{OA.B.41}$$

Here $\mathcal{M}_{t+1}^e$ is operating entrant mass. Without a single operating cutoff, replace the indicator by $\mathbf{1}_{\{\tilde{V}_{t+1}^e(\varphi) \geq 0\}}$ and the denominator by its probability.

Vacant-line measures. For incumbent source ages $a \geq 1$,

$$\begin{aligned}
\mu_{p,t+1}(\emptyset, a+1) = & \frac{(1-\delta)(1-x)}{N-1} \sum_{q \neq p} (1 - p_{t+1}^e(a+1)) \\
& \times \left[\mu_{qt}(\emptyset, a) + \int_{\mathcal{X}_a} \int \mathbf{1}_{\mathcal{R}_{t+1}^{ds}}(\xi') g_t(d\xi' \mid \xi) \mu_{qt}(d\xi) \right]. \tag{OA.B.42}
\end{aligned}$$

Vacant lines are indexed by their failed-search destinations; failed replacement searches leave retained lines occupied.

For startups transitioning to age one,

$$\mu_{p,t+1}(\emptyset, 1) = \frac{(1-\delta)(1-x)}{N-1} \sum_{q \neq p} (n(\varphi) - 1)(1 - p_{t+1}^e(1)) \mu_{qt}(\xi^0(0)). \tag{OA.B.43}$$

Search resources and aggregation. Aggregate search labor is

$$\begin{aligned} \mathcal{F}_{t+1}^s = (1 - \delta) f^s \sum_p \left\{ \sum_{a \geq 1} \left[\int_{\mathcal{X}_a} \int \left[\mathbf{1}_{\mathcal{R}_{t+1}^{cs}}(\xi') \right. \right. \right. \\ \left. \left. \left. + (1 - p_{t+1}^e(a+1)) \mathbf{1}_{\mathcal{R}_{t+1}^{ds}}(\xi') \right] g_t(d\xi' | \xi) \mu_{pt}(d\xi) \right. \right. \\ \left. \left. + (1 - p_{t+1}^e(a+1)) \mu_{pt}(\emptyset, a) \right] \right. \\ \left. + \mu_{pt}(\xi^0(0)) \left[\int \mathbf{1}_{\mathcal{R}_{t+1}^{cs,e}}(\xi') g_t(d\xi' | \xi^0(0)) \right. \right. \\ \left. \left. + (n(\varphi) - 1)(1 - p_{t+1}^e(1)) \right] \right\}. \end{aligned} \quad (\text{OA.B.44})$$

The expression integrates over suppressed types and excludes firms exiting before search.

Let $\mathcal{X}$ include all active states and prior atoms. Operating and entry resource costs are

$$\mathcal{F}_{t+1}^o = f^o \sum_p \mu_{pt}(\mathcal{X}), \quad \mathcal{F}_{t+1}^e = f_{t+1}^e \frac{\mathcal{M}_{t+1}^e}{1 - H_{\varphi}(\underline{\varphi}_{t+1}^e)}. \quad (\text{OA.B.45})$$

Entry costs apply to gross potential entrant mass, including types that abandon before production.

In stationary symmetric equilibrium, let μ denote the active measure in one product market, including permanent types. Then

$$L_p = \left(\frac{\eta - 1}{\eta} \right)^\eta P^\eta C w^{-\eta} \int_{\mathcal{X}} \mathcal{B}(\xi)^\eta z(\varphi; z_0)^{\eta-1} \mu(d\xi), \quad (\text{OA.B.46})$$

$$P = \frac{\eta}{\eta - 1} w \left[\int_{\mathcal{X}} \mathcal{B}(\xi)^\eta z(\varphi; z_0)^{\eta-1} \mu(d\xi) \right]^{-1/(\eta-1)}, \quad (\text{OA.B.47})$$

$$C = \frac{\bar{L} - \mathcal{F}^o - \mathcal{F}^s - \mathcal{F}^e}{N} \left[\int_{\mathcal{X}} \mathcal{B}(\xi)^\eta z(\varphi; z_0)^{\eta-1} \mu(d\xi) \right]^{1/(\eta-1)}. \quad (\text{OA.B.48})$$

Labor-market clearing requires $\bar{L} = N L_p + \mathcal{F}^o + \mathcal{F}^s + \mathcal{F}^e$, and $\bar{C} = N^{\nu/(\nu-1)} C$. Normalize $w = 1$.

The exact firm distribution follows the baseline portfolio transition on the expanded state space, with Γ^e normalized conditional on operating. The marginal laws above instead use (OA.B.36). Stationarity requires an invariant distribution, optimal decisions, free entry, and

market clearing.

At $\rho = 0$, the model reduces to the baseline, and realized demand is no longer a necessary decision-state coordinate.

OA.B.2 Mapping Shipment Fixed Effects to Expertise

Under full information about permanent expertise, its contribution to $\log \mathcal{B}_{ip,t-1}$ has coefficient $(\eta - 1)/\eta$. Equation (2.11) therefore implies a total loading of $\eta - 1$ on $\log \theta_{ip}$ in log real shipments. After accounting for firm productivity and aggregate components, this gives the benchmark mapping $\alpha_{ip}^{PV} \simeq (\eta - 1) \log \theta_{ip}$. Demand-shock persistence affects shipment dynamics but does not change this full-information loading. We use this mapping as a benchmark approximation, with the prior mean anchored by the empirical fixed-effect normalization.

OA.C Additional Quantitative Results

OA.C.1 External Validation

This subsection compares untargeted model moments with published evidence on firm-size variation, demand and productivity over the firm life cycle, and within-firm sales concentration.

Sources of Firm-Size Variation. Table OA.1 compares the model’s sources of firm-size variation with Hottman et al. (2016) (HRW). Demand-related components account for 64.44% of log deflated-sales variance, compared with a 75.80% contribution from firm appeal in HRW. Productivity and product scope jointly account for the remaining 35.56% in the model.

Demand and Productivity over the Firm Life Cycle. Table OA.2 compares demand and productivity gaps relative to old nonexitors with Foster et al. (2016) (FHS). The model reproduces entrants’ larger disadvantage in demand than in productivity and the narrowing demand gap with age. It overstates the demand gaps, particularly among exiting firms, and generates a small entrant productivity disadvantage rather than the positive empirical coefficient.

Ex Ante Heterogeneity and Firm Growth. Using the employment-based statistical decomposition of Sterk et al. (2021) (SSP), Table OA.3 reports the share of log-employment variance

Table OA.1: Sources of firm-size variation

Component	Contribution (%)
<i>A. Model</i>	
Productivity	26.67
Product scope	8.89
Product-specific expertise	6.58
Persistent demand	56.96
Imperfect learning	-0.37
Demand realization	1.27
<i>Grouped contributions</i>	
Productivity and product scope	35.56
Demand-related components	64.44
<i>B. HRW</i>	
Firm appeal	75.80
Product scope	21.03
Average marginal cost	-3.70
Cost dispersion	7.06
Markup	-0.20

Notes: Model contributions are each component's covariance with log deflated sales divided by the variance of log deflated sales. Demand-related components comprise product-specific expertise, persistent demand, imperfect learning, and demand realization. Productivity and potential scope both depend on φ_i ; the grouped contributions summarize the components above. HRW values are from Table X, the all-firm cross-sectional decomposition. Their firm-appeal measure encompasses several model components, while the model has no markup heterogeneity. Values are rounded.

Table OA.2: Demand and productivity gaps relative to old nonexitters

Group	Demand		Productivity	
	Model	FHS	Model	FHS
Entrants (ages 0–4)	-1.39	-0.89	-0.05	0.02
Young (ages 5–9)	-0.27	-0.48	-0.03	0.01
Medium (ages 10–14)	-0.12	-0.38	-0.02	0.00
Exiters before next census	-2.19	-0.59	-0.08	-0.02

Notes: Entries are log-point coefficients relative to old nonexitters. Empirical estimates are from FHS Table 1. The model uses firm observations and date effects; FHS use plant observations and product-year effects. Model productivity is $\log z_i$; empirical productivity is estimated TFPQ. The demand measures likewise follow their respective model and empirical constructions. Age groups follow FHS.

attributable to ex ante heterogeneity. The model is close to the manufacturing estimate at age five, but its contribution remains high as firms age, whereas the empirical contribution declines.

Table OA.3: Ex ante shares of log-employment variance

Firm age	Model (%)	SSP manufacturing (%)
5	67.83	62.06
10	68.67	49.47
20	70.60	39.31

Notes: Both comparisons use balanced panels and SSP's statistical decomposition of employment. Manufacturing estimates are from SSP Appendix Table C.1, column (2). The model's age-20 value extrapolates beyond the age-0–19 estimation panel.

Table OA.4: Largest-product sales share by product scope

Product scope	Model (%)	Bernard et al. (%)
2	87.62	80.00
3	82.50	70.00
4	80.16	63.00
5	78.95	58.00
6	75.03	54.00
7	77.70	52.00
8	76.64	50.00
9	75.71	48.00
10	75.64	46.00

Notes: Entries are the mean sales share of the largest product across firms with the stated product scope. Products are ranked by sales within each firm. Empirical values are from Bernard et al. (2010), Table 10.

Within-Firm Sales Concentration. Table OA.4 compares the largest product's share of firm sales with Bernard et al. (2010). In both the model and data, the largest product generally accounts for a smaller share as product scope increases. The model nevertheless generates greater sales concentration, with larger differences among firms producing more products.

OA.C.2 Selection and Underlying Productive Potential

Firm Continuation and Size. Table OA.5 reports continuation and large-firm status by productivity and managerial ability. Among high-productivity firms, the joint probability of continuing and being large rises from 4.39% to 5.59% between the low- and high-managerial-ability groups. Thus, the higher large-firm share among continuing firms also translates into a higher probability relative to original entrants.

Current Output and Productive Potential. Table OA.6 compares high- and low- φ_i firms with comparable output at initial ages five and ten. We form 40 weighted output bins at each initial

Table OA.5: Continuation and large-firm status at age 20 (%)

Group	Continuation	Large continuation	Continuing and large
Low z , low m	24.21 (1.56)	0.82 (0.58)	0.20 (0.14)
Low z , middle m	21.97 (1.49)	0.66 (0.36)	0.15 (0.08)
Low z , high m	15.77 (1.22)	1.46 (1.00)	0.23 (0.16)
Middle z , low m	6.02 (1.11)	1.22 (0.69)	0.07 (0.04)
Middle z , middle m	6.56 (1.04)	1.06 (0.77)	0.07 (0.05)
Middle z , high m	5.04 (0.67)	2.06 (1.10)	0.10 (0.06)
High z , low m	34.68 (2.07)	12.67 (0.41)	4.39 (0.32)
High z , middle m	35.40 (2.14)	14.09 (0.52)	4.99 (0.33)
High z , high m	30.07 (1.87)	18.61 (0.80)	5.59 (0.41)

Notes: Groups are defined by productivity z and managerial ability m and fixed at entry. Age-20 statistics pool firm-age masses over ages 18–22. Continuation and the joint probability are measured relative to original entrants; the conditional probability is measured among continuing firms. Large firms have employment at or above the age-specific 90th percentile among active firms, including ties. Parentheses report block Monte Carlo standard errors in percentage points.

Table OA.6: Current output and future outcomes by productive potential

Outcome	Initial age 5	Initial age 10
Current output gap (%)	4.50	1.72
Cumulative future-output gap (%)	2597.42 (3771.16)	439.30 (1098.84)
Continuation gap (pp)	4.43 (4.23)	5.68 (4.70)
Active and in top output decile (pp)	10.14 (1.35)	12.37 (3.49)
Common-support coverage (%)	100.00	100.00

Notes: Comparisons are high minus low φ among firms active at each initial age, using common weights across 40 current-output bins and preserving ties. The current-output gap is based on standardized mean logs. Follow-up spans ten years; cumulative output is undiscounted and includes zeros after exit. Continuation and top-decile status are measured at the end of the horizon. Top-decile status is based on output, whereas Table OA.5 uses employment. Managerial ability is not held fixed. Parentheses report block Monte Carlo standard errors.

age and apply common bin weights to the two groups. Since φ_i determines both productivity and potential scope, the comparison captures their combined association with subsequent outcomes.²

High- φ_i firms are more likely to be active and in the top output decile ten years later, with gaps of 10.14 and 12.37 percentage points at initial ages five and ten, respectively. Cumulative future-output gaps are also positive but have substantial Monte Carlo uncertainty.

²The age-zero comparison is omitted because common-support coverage is only 1.81% and the current-output gap is 101.46%.

Table OA.7: Output per original entrant at baseline prices

Age	Annual output		Cumulative output	
	Baseline	Higher m_0	Baseline	Higher m_0
5	7.77 (1.79)	7.77 (1.88)	29.78 (7.18)	30.11 (7.31)
10	45.42 (27.21)	49.52 (31.46)	164.45 (41.70)	168.37 (46.72)
20	13.50 (2.97)	14.04 (3.21)	822.91 (332.09)	834.52 (319.87)
30	28.01 (11.77)	34.67 (14.32)	1008.57 (345.89)	1022.74 (334.27)
50	5.67 (1.70)	8.03 (3.50)	1435.06 (389.03)	1520.14 (392.61)

Notes: Both paths follow the same baseline entrants at baseline prices, with zero output after exit. Cumulative output sums ages zero through the displayed age without discounting. Parentheses report block Monte Carlo standard errors of the levels. Standard errors of paired differences also require the covariance between the two estimates and cannot be inferred from the reported standard errors alone.

OA.C.3 Paired Policy Outcomes and Simulation Uncertainty

Table OA.7 supplements Figure 6 with annual and cumulative output per original entrant at baseline prices. The paired paths hold initial product-specific expertise and underlying random draws fixed while allowing decisions to adjust. Through age 20, cumulative output rises from 822.91 to 834.52, an increase of 1.41%.

OA.C.4 Aggregate Accounting and Resource Allocation

Accounting Decomposition. We decompose the equilibrium change in aggregate deflated sales into contributions from the firm distribution and within-group performance. Groups are defined by firm age, φ , and baseline managerial ability. Let μ_g^k and $\bar{r}_g^k$ denote firm mass and mean deflated sales per firm in group g , where $k = 0$ denotes the baseline and $k = 1$ the policy economy. For groups present in both economies,

$$\sum_g (\mu_g^1 \bar{r}_g^1 - \mu_g^0 \bar{r}_g^0) = \underbrace{\sum_g \frac{\bar{r}_g^0 + \bar{r}_g^1}{2} (\mu_g^1 - \mu_g^0)}_{\text{Firm distribution}} + \underbrace{\sum_g \frac{\mu_g^0 + \mu_g^1}{2} (\bar{r}_g^1 - \bar{r}_g^0)}_{\text{Within-group performance}}. \quad (\text{OA.C.1})$$

Using averages of baseline and policy values allocates half of the interaction term to each component. We apply the same identity to separate firm mass from group shares within the distribution contribution, and employment per firm from deflated sales per worker within the performance contribution. These refinements preserve the two broad contributions. Within-group perfor-

Table OA.8: Decomposition of the equilibrium consumption change

Component	Contribution (pp)
Firm mass	-8.01
Firm composition	6.26
Employment per firm	1.66
Deflated sales per worker	0.63
Newly populated groups	0.00
Groups no longer populated	0.00
Numerical residual	0.00
Total	0.54

Notes: Contributions are percentage points of baseline consumption. Firm mass and composition sum to the distribution contribution; employment per firm and deflated sales per worker sum to within-group performance. Groups present in only one economy are reported separately. The residual reconciles group-level contributions with the aggregate change. Values are rounded.

mance includes changes in the composition of firms within groups.

Table OA.8 reports the resulting contributions. Employment per firm and deflated sales per worker contribute 1.66 and 0.63 percentage points, respectively, accounting for the 2.29 percentage-point contribution from within-group performance in Table 7. With N fixed, consumption and aggregate deflated sales change proportionally, yielding identical contributions when normalized by their baseline values.

For the comparison holding firm mass fixed, we replace group masses with $\mathcal{M}^0 s_g^k$, where $\mathcal{M}^0$ is baseline firm mass and s_g^k is group g 's firm share in economy k , and recompute the decomposition. This normalization measures changes in mean deflated sales per firm. Aggregate deflated sales rise by 8.90%, with composition and within-group performance contributing 6.51 and 2.38 percentage points, respectively. These contributions differ from those above because the normalization changes the weights and interaction terms.

Labor Allocation. Table OA.9 reports how the fixed labor endowment is allocated across production, operating costs, search, entry, and the program. The program uses 0.11% of the labor endowment, accompanied by small reductions in production, search, and private entry labor.

Table OA.9: Labor allocation in the baseline and policy economies

Labor use	Baseline (%)	Policy (%)	Change (pp)
Production	89.54	89.48	-0.06
Operating costs	0.07	0.07	0.00
Product search	0.17	0.16	-0.01
Private entry	10.21	10.19	-0.02
Public program	0.00	0.11	0.11
Total	100.00	100.00	0.00

Notes: Shares are percentages of the fixed labor endowment. Firm employment includes production, operating, and search labor. Private entry resources include costs incurred by unsuccessful potential entrants. Program resources equal 0.1% of baseline consumption expenditure, corresponding to 0.11% of the labor endowment. Values are rounded.

OA.C.5 Full Information at Fixed Parameters

The full-information benchmark reveals product-specific expertise upon acquisition, before the product's first production. Future demand shocks and the expertise of undiscovered products remain unknown, and operating and search costs are unchanged. We hold model parameters, (m_0, z_0) , and the entry cost fixed, while policies, the entry cutoff, entrant mass, and prices adjust. Stationary consumption is 1.67% lower than in the baseline.³

To distinguish production adjustments from changes in product switching and firm continuation, Table OA.10 compares cumulative deflated sales at baseline prices along three paths: baseline learning (L), informed production along the baseline product-portfolio and exit history (S), and reoptimized full-information decisions (D). The production component, $S - L$, holds portfolios and continuation fixed. The dynamic component, $D - S$, captures subsequent product-switching and firm-exit responses, including their interactions with production. Both components are normalized by cumulative baseline deflated sales and sum to the total change. These comparisons hold prices fixed, whereas the stationary consumption comparison also incorporates equilibrium adjustments.

OA.C.6 Restricted Planner: Formulation and Proofs

This appendix formalizes the restricted planner introduced in Section 4.8. The planner maximizes household welfare subject to physical feasibility and the CE's valuation restrictions on

³This benchmark retains the model's market structure and exit-payoff assumptions and does not provide a welfare upper bound.

Table OA.10: Full information and cumulative deflated sales at baseline prices

Years	Contribution (% of baseline)			MCSE (pp)
	Production	Dynamic	Total	
5	5.84	-1.23	4.61	1.86
10	5.44	-2.25	3.19	5.74
20	-2.35	-3.30	-5.65	9.33
50	1.60	4.89	6.50	10.44

Notes: Contributions are percentages of cumulative baseline deflated sales per original entrant at baseline prices. A horizon of H years includes ages zero through $H - 1$, whereas Figure 6 includes the displayed age. The final column reports the Monte Carlo standard error of the total change in percentage points. Longer-horizon estimates have substantial simulation uncertainty. Values are rounded.

product switching, startup operation, and entry. These restrictions preserve the assigned endogenous-exit values that make mature firms' product-line values additive. Assigned exit values create no resources; welfare depends on household consumption.⁴

OA.C.6.1 Objective and maintained restrictions

Definition OA.1 (Restricted planner). *Given the initial firm distribution, inherited capabilities, and predetermined quantities, the restricted planner chooses pre-entry investments, production, product switching, startup operation, and entry to maximize*

$$W^{\text{SP}} \equiv \max \sum_{t=0}^{\infty} \beta^t \log \bar{C}_t^{\text{SP}}, \quad (\text{OA.C.2})$$

over physically feasible allocations satisfying conditional production efficiency and Assumption OA.2 in the environment below. The fixed-capability problem holds capabilities and associated acquisition costs fixed. The stationary problem requires constant aggregate quantities and distributions.

Planner objects carry the superscript SP. Let ξ collect product-line information, tenure, and common firm characteristics. Operating returns and discounting are defined in (OA.C.14) and (OA.C.17). The product-line and startup value functions below support decisions under the assigned exit-value convention; they are distinct from the planner's welfare value W^{SP} .

Assumption OA.2 (Valuation restrictions). *Conditional on the aggregate path associated with the*

⁴For a related benchmark, Acemoglu et al. (2018) restrict the planner to decentralized production and pricing while allowing control over entry, exit, and R&D.

planner's allocation, product switching decisions, startup operation, and attempted entry maximize net present values in labor units, subject to:

- (i) At ages two and above, each line is evaluated independently of its effects on firm exit and other lines. Upon endogenous exit, each potential line is assigned its vacant-line value $V_t^{ds,SP}(\emptyset)$. At age one, only the initial line receives this value; initially vacant lines receive zero.
- (ii) Startup operation and free entry use these value functions, including assigned exit values. Each entry attempt incurs the resource cost $f_t^{e,SP}$.

Assigned exit values enter these value functions only. Welfare and feasibility depend on actual consumption and resource use.

The aggregate path is endogenous and must be consistent with the resulting decisions and resource constraints.

Maintained environment. The CE primitives, information structure, aggregation, and timing apply, including $0 < \beta < 1$, $1 < \nu < \eta$, and symmetric product markets. Policies use firms' information and a common measurable rule for indifference. For each admissible allocation, idiosyncratic uncertainty satisfies an exact law of large numbers conditional on production-planning information, with finite aggregation integrals. Quantities remain measurable with respect to firms' information. The precision path ψ follows the CE law, independently of other lines' retention decisions and aggregate entry. Exogenous death occurs after production with zero scrap value. After updating beliefs, firms dropping all active products exit before search and incur no further search costs; eligible vacant lines of continuing firms search, with positive vacant-line value as in the CE. At age one, retention preserves the other $n(\varphi) - 1$ lines' search opportunities. The initial firm distribution and inherited capabilities are given; incumbents retain their birth-cohort characteristics. Initial quantities y_{ip0} are predetermined by firms' decisions at $t = -1$.

OA.C.6.2 Resources and production

Pre-entry investment. Pre-entry investments determine managerial and productivity capabilities through

$$\log m_{0t}^{\text{SP}} = (1 - \zeta_m) \log a_t^{m,\text{SP}} + \zeta_m \log M_{t-1}^{\text{SP}}, \quad (\text{OA.C.3})$$

$$\log z_{0t}^{\text{SP}} = (1 - \zeta_z) \log a_t^{z,\text{SP}} + \zeta_z \log Z_{t-1}^{\text{SP}}, \quad (\text{OA.C.4})$$

$$M_{t-1}^{\text{SP}} = m_{0,t-1}^{\text{SP}}, \quad Z_{t-1}^{\text{SP}} = z_{0,t-1}^{\text{SP}}, \quad (\text{OA.C.5})$$

$$f_t^{e,\text{SP}} = \bar{f}^e + \alpha_m^e (a_t^{m,\text{SP}})^\chi + \alpha_z^e (a_t^{z,\text{SP}})^\chi. \quad (\text{OA.C.6})$$

At stationarity, $a^{m,\text{SP}} = m_0^{\text{SP}}$ and $a^{z,\text{SP}} = z_0^{\text{SP}}$. Fixed-capability comparisons hold acquisition costs fixed and use consistent inherited capabilities. The planner chooses pre-entry investments without imposing their private optimality conditions.

Entry and firm distributions. Let $\mathcal{E}_t^{\text{SP}}$ denote attempted entry and $q_t^{e,\text{SP}}$ the probability of operating under (OA.C.24). Entry attempts at t pay $f_t^{e,\text{SP}}$ and, conditional on operation, first produce at $t + 1$. Operating entry and entry resources satisfy

$$\mathcal{M}_t^{e,\text{SP}} = q_t^{e,\text{SP}} \mathcal{E}_t^{\text{SP}}, \quad \mathcal{F}_t^{e,\text{SP}} = f_t^{e,\text{SP}} \mathcal{E}_t^{\text{SP}}. \quad (\text{OA.C.7})$$

With a single operating cutoff and continuous H_φ , $q_t^{e,\text{SP}} = 1 - H_\varphi(\underline{\varphi}_t^{e,\text{SP}})$; otherwise it is the probability of the operating set under the common indifference rule. Let $\Gamma_t^{e,\text{SP}}$ be the entrant distribution conditional on operation. Physical firm measures satisfy

$$\Gamma_t^{\text{SP}}(A) = \int_{\mathcal{X}^f} \mathcal{Q}_t^{f,\text{SP}}(A \mid \xi^f) \Gamma_{t-1}^{\text{SP}}(d\xi^f) + \mathcal{M}_t^{e,\text{SP}} \Gamma_t^{e,\text{SP}}(A), \quad (\text{OA.C.8})$$

where Γ_t^{SP} records portfolios after period- t updating, exit, search, and entry planning; its active-line projection $\mu_{p,t}^{\text{SP}}$ determines which lines produce at $t + 1$. The kernel $\mathcal{Q}_t^{f,\text{SP}}$ uses the CE's physical law of motion, including both exit mechanisms. The entrant term is zero when $q_t^{e,\text{SP}} = 0$. Accounting retains portfolios and predetermined cohort characteristics without enlarging admissible policy arguments.

Labor feasibility. Operating and search resources equal their unit costs times the mass of producing lines and expected actual searches:

$$\mathcal{F}_t^{o,SP} = f^o \sum_{p=1}^N \mu_{p,t-1}^{SP}(\mathcal{X}), \quad \mathcal{F}_t^{s,SP} = f^s \int \mathcal{N}_t^{s,SP}(\xi^f) \Gamma_{t-1}^{SP}(d\xi^f), \quad (\text{OA.C.9})$$

where $\mathcal{N}_t^{s,SP}(\xi^f)$ counts expected period- t searches per inherited firm after exogenous death and endogenous exit. All three resource costs use period- t labor; the inherited measures index the firms facing current production and subsequent search decisions. Assigned exit values do not enter these resource equations.

Define variable production labor and the productive index of a representative product market by

$$R_t^{SP} \equiv \bar{L} - \mathcal{F}_t^{o,SP} - \mathcal{F}_t^{s,SP} - \mathcal{F}_t^{e,SP}, \quad (\text{OA.C.10})$$

$$S_t^{SP} \equiv \int_{\mathcal{X}} \mathcal{B}(\xi)^\eta z_i^{\eta-1} \mu_{p,t-1}^{SP}(d\xi). \quad (\text{OA.C.11})$$

Here $\mathcal{B}(\xi) \equiv \mathbb{E}[\theta_{ipt}^{(\eta-1)/\eta} \mid \xi]$ is the predictive demand term for production at t , including demand shocks; ξ contains the information available when quantities are chosen. Productivity z_i uses the firm's birth-cohort capability, with $z_i = z_0 \varphi_i$ at fixed capabilities. Require $R_t^{SP} > 0$ and $0 < S_t^{SP} < \infty$.

Proposition OA.2 (Conditional production and operating returns). *At dates when quantities can be chosen, given active-line measures and variable production labor, the consumption-maximizing allocation satisfies*

$$\ell_{pt}^{SP}(\xi) = \frac{R_t^{SP}}{N} \frac{\mathcal{B}(\xi)^\eta z_i^{\eta-1}}{S_t^{SP}}, \quad (\text{OA.C.12})$$

$$C_{pt}^{SP} = \frac{R_t^{SP}}{N} (S_t^{SP})^{1/(\eta-1)}, \quad \bar{C}_t^{SP} = N^{\nu/(\nu-1)} C_{pt}^{SP}. \quad (\text{OA.C.13})$$

The direct marginal contribution of an active line, net of its operating fixed cost and expressed in labor units, is

$$\bar{\pi}_t^{SP}(\xi) \equiv \frac{R_t^{SP}}{N(\eta-1)S_t^{SP}} \mathcal{B}(\xi)^\eta z_i^{\eta-1} - f^o = \frac{\ell_{pt}^{SP}(\xi)}{\eta-1} - f^o. \quad (\text{OA.C.14})$$

At a CE allocation, $\bar{\pi}_t^{\text{SP}}(\xi) = \bar{\pi}_t(\xi)/w_t$.

Proof. Suppress SP. Since quantities use inherited information, the maintained exact law of large numbers gives the realized aggregator almost surely:

$$C_{pt}^{(\eta-1)/\eta} = \int \mathcal{B}(\xi) [z_i \ell_{pt}(\xi)]^{(\eta-1)/\eta} \mu_{p,t-1}(d\xi). \quad (\text{OA.C.15})$$

Given variable labor L_{pt} in market p , concavity and the first-order condition imply

$$\ell_{pt}(\xi) = \frac{L_{pt} \mathcal{B}(\xi)^\eta z_i^{\eta-1}}{S_{pt}}, \quad C_{pt} = L_{pt} S_{pt}^{1/(\eta-1)}.$$

Symmetry and concavity of the upper CES aggregator give $L_{pt} = R_t/N$, proving (OA.C.12)–(OA.C.13).

At symmetry, variable labor has marginal log-consumption value $1/R_t$. Holding search and entry resources fixed, an additional unit mass of type- ξ lines raises the market's productive index by $\mathcal{B}(\xi)^\eta z_i^{\eta-1}$ and uses f^o units of labor. Its marginal log-consumption contribution is therefore

$$\frac{\mathcal{B}(\xi)^\eta z_i^{\eta-1}}{N(\eta-1)S_t} - \frac{f^o}{R_t}. \quad (\text{OA.C.16})$$

Dividing by $1/R_t$ proves (OA.C.14). The CE quantity first-order condition gives expected revenue $\eta w_t \ell_{pt}(\xi)/(\eta-1)$. Subtracting labor costs yields $\bar{\pi}_t(\xi)/w_t = \ell_{pt}(\xi)/(\eta-1) - f^o$. This equality is related to the alignment of private and social incentives under CES demand in [Dhinda and Morrow \(2019\)](#). $\square$

The direct marginal return excludes effects on firm exit and other valuation restrictions. Its use in the dynamic decision criteria follows from Assumption OA.2. Initial quantities remain fixed at their predetermined levels.

OA.C.6.3 Decision rules under the valuation restrictions

Valuation in labor units. Along a deterministic aggregate path, the value functions use direct marginal-consumption weights $1/R_t^{\text{SP}}$ to discount labor returns:

$$\beta_t^{\text{SP}} \equiv \beta \frac{R_t^{\text{SP}}}{R_{t+1}^{\text{SP}}}. \quad (\text{OA.C.17})$$

This normalization equals β at stationarity. It represents direct marginal-consumption valuation, rather than the costate of the full constrained problem. Appendix OA.C.6.5 relates it to household intertemporal valuation.

Product switching decisions. Under Assumption OA.2, the choice-specific value functions and, at ages two and above, the Bellman equation are:

$$V_t^{ns,SP}(\xi) = \beta_t^{SP} [\bar{\pi}_{t+1}^{SP}(\xi) + (1 - \delta)\mathbb{E}[V_{t+1}^{SP}(\xi') \mid \xi]] , \quad (\text{OA.C.18})$$

$$V_t^{cs,SP}(\xi) = -f^s + (1 - x)V_t^{ns,SP}(\xi) + xV_t^{ns,SP}(\xi^0), \quad (\text{OA.C.19})$$

$$V_t^{ds,SP}(\emptyset) = -f^s + xV_t^{ns,SP}(\xi^0) + (1 - x)\beta_t^{SP}(1 - \delta)V_{t+1}^{ds,SP}(\emptyset), \quad (\text{OA.C.20})$$

$$V_t^{SP}(\xi) = \max \left\{ V_t^{ns,SP}(\xi), V_t^{cs,SP}(\xi), V_t^{ds,SP}(\emptyset) \right\} . \quad (\text{OA.C.21})$$

The fresh-product state ξ^0 has the common prior and the same firm characteristics and age. Common states advance in every continuation term, including vacancy, using the CE information transitions. Search costs are incurred after exogenous death and carry no additional $1 - \delta$ factor.

Dropping is valued as vacancy continuation whether the firm continues or exits; actual search requires continuation. At ages two and above, the total assigned exit value is $n(\varphi)V_t^{ds,SP}(\emptyset)$.

Startup operation. At age one, retention preserves the other $n(\varphi) - 1$ vacancy opportunities, while exit assigns only the initial line its vacant-line value:

$$V_t^{e,1,SP}(\xi) = \max \left\{ V_t^{ns,SP}(\xi) + [n(\varphi) - 1]V_t^{ds,SP}(\emptyset), \right. \\ \left. V_t^{cs,SP}(\xi) + [n(\varphi) - 1]V_t^{ds,SP}(\emptyset), V_t^{ds,SP}(\emptyset) \right\}. \quad (\text{OA.C.22})$$

Common states are suppressed; the choice-specific values use age one with continuation at age two.

Before initial production, φ is observed and m is not. Startup operating value and the option to abandon before production satisfy

$$\tilde{V}_t^{e,SP}(\varphi) = \beta_t^{SP} \left[\bar{\pi}_{t+1}^{SP}(\xi^0(0; \varphi)) + (1 - \delta)\mathbb{E}_{m|\varphi}\mathbb{E}^0 V_{t+1}^{e,1,SP}(\xi') \right], \quad (\text{OA.C.23})$$

$$V_t^{e,SP}(\varphi) = \max\{\tilde{V}_t^{e,SP}(\varphi), 0\}. \quad (\text{OA.C.24})$$

Managerial ability is revealed after initial production; $\mathbb{E}^0$ integrates the subsequent first update. Zero denotes abandonment before production. A single operating cutoff, when it exists, satisfies $\tilde{V}_t^{e,SP}(\varphi_t^{e,SP}) = 0$.

Entry. Conditional on the aggregate path and capability levels, Assumption OA.2(ii) requires

$$\mathcal{E}_t^{SP} \in \arg \max_{\mathcal{E} \geq 0} \mathcal{E} \left[\mathbb{E}_\varphi V_t^{e,SP}(\varphi) - f_t^{e,SP} \right]. \quad (\text{OA.C.25})$$

Linearity in attempted entry implies the complementary slackness condition

$$0 \leq \mathcal{E}_t^{SP} \perp f_t^{e,SP} - \mathbb{E}_\varphi V_t^{e,SP}(\varphi) \geq 0. \quad (\text{OA.C.26})$$

A positive net value makes attempted entry unbounded; a negative value implies zero entry. General equilibrium consistency and feasibility determine entry mass. This condition follows from imposing the valuation criterion on entry, independently of line separability; exit values create no consumption.

OA.C.6.4 Regularity and verification

Assume Borel state spaces, measurable returns and transition kernels, and finite-horizon expected absolute returns uniformly bounded over admissible policies from each initial state and date. Let r_s^{SP} denote the expected next-period operating return discounted by β_s^{SP} , net of current search costs, as in (OA.C.18)–(OA.C.20). The process used to evaluate product-line values includes vacancy continuation after endogenous exit and an absorbing exogenous-death state with zero subsequent returns. For every initial state ξ and date t , assume

$$\lim_{T \rightarrow \infty} \sup_g \mathbb{E}_{t,\xi}^g \left[\sum_{s=T}^{\infty} \beta^{s-t} \frac{R_t^{SP}}{R_s^{SP}} |r_s^{SP}| \right] = 0. \quad (\text{OA.C.27})$$

Here g ranges over admissible policies for this process. Ex ante startup values, including vacancy opportunities, are integrable, and admissible aggregate paths satisfy

$$\sum_{t \geq 0} \beta^t |\log \bar{C}_t^{SP}| < \infty.$$

We assume that the maximum is attained for each planner problem considered.

Lemma OA.3 (Verification of value functions). *For a fixed aggregate path satisfying these conditions, the Bellman equations have a finite measurable value function attained by a measurable policy. It is unique among solutions whose discounted expected terminal values vanish under every admissible policy.*

Proof. Following the finite-horizon approach in [Stokey and Lucas Jr \(1989, Sections 9.1 and 9.4\)](#), backward induction with zero terminal values gives measurable finite-horizon value functions. Since

$$\prod_{s=t}^{T-1} \beta_s^{\text{SP}} = \beta^{T-t} \frac{R_t^{\text{SP}}}{R_T^{\text{SP}}}, \quad (\text{OA.C.28})$$

condition [\(OA.C.27\)](#) makes the truncation error vanish uniformly over policies and permits passage to the limit in the Bellman equations. The finite action set admits a measurable maximizing policy.

For any solution satisfying the terminal condition, iterating the Bellman inequality and taking limits shows that it weakly exceeds every policy's expected return. A maximizing policy gives equality. For the constructed value function, discounted expected terminal magnitude is bounded by the maximal expected absolute remaining return, which vanishes by [\(OA.C.27\)](#). This establishes attainment and uniqueness. Integrability then determines the startup values. $\square$

The verification argument concerns value functions for a given aggregate path; it does not establish equilibrium existence or uniqueness.

OA.C.6.5 Replication and capability choice

The following proposition compares the restricted planner with the CE when capabilities and their acquisition costs are fixed. The CE in this comparison retains the startup-operation and free-entry conditions but does not impose private optimality of capability investment.

Proposition OA.3 (Replication and restricted efficiency). *Under the maintained restrictions and regularity conditions:*

- (i) For fixed (m_0, z_0) and associated costs, the planner's admissible stationary allocations coincide with stationary CE allocations of the economy with capabilities fixed.
- (ii) For a fixed capability path and associated costs, both problems admit identical real allocation paths, provided they start from the same inherited state and predetermined quantities and decentralized firm values use household intertemporal valuation.

The restricted planner therefore selects a welfare-maximizing CE allocation. If the equilibrium is unique, or the same equilibrium-selection rule is imposed on both problems, the corresponding CE solves the restricted planner problem. In case (ii), along the common allocation path,

$$\bar{C}_t^{\text{SP}} = \bar{C}_t^{\text{CE}} \quad \text{for all } t \geq 0, \quad W^{\text{SP}} = W^{\text{CE}}. \quad (\text{OA.C.29})$$

Proof. Stationary allocations. At a stationary CE with fixed capabilities, Proposition OA.2 gives the same production allocation and $\bar{\pi}^{\text{SP}} = \bar{\pi}/w$. Dividing CE value equations by the constant wage gives (OA.C.18)–(OA.C.24), with identical discounting, transitions, costs, and rules for assigning exit values. Lemma OA.3 implies

$$V^{j,\text{SP}} = V^j/w, \quad j \in \{ns, cs, ds, e\}, \quad V^{\text{SP}} = V/w, \quad (\text{OA.C.30})$$

with the same normalization for $V^{e,1}$ and $\tilde{V}^e$. The common indifference rule gives identical line policies and startup operation, including $\mathcal{B}^{cs}$ and $\mathcal{B}^{ds}$ whenever these thresholds exist. Dividing CE free entry by w gives (OA.C.26). Common entry masses, physical transitions, and resource costs support the same invariant distribution, establishing admissibility for the planner.

Conversely, take an admissible stationary planner allocation and any constant wage $w > 0$. Quantities $z_i \ell_p^{\text{SP}}(\xi)$ are supported by demand-clearing prices

$$p_{ipt} = \frac{\eta}{\eta - 1} \frac{w}{z_i} \frac{\theta_{ipt}^{(\eta-1)/\eta}}{\mathcal{B}(\xi)}, \quad P_p = \frac{\eta}{\eta - 1} w (S^{\text{SP}})^{-1/(\eta-1)}. \quad (\text{OA.C.31})$$

These prices satisfy the CE quantity first-order condition and give expected profit $w\bar{\pi}^{\text{SP}}$. Multiplying these value functions by w recovers the CE line, startup, and entry conditions. The physical distribution and labor feasibility imply market clearing; consolidating household-owned firms gives the household budget identity. Thus the stationary real allocation sets coincide.

Capabilities are fixed, so their private investment condition is excluded.

Sequential allocations. CE firm and entry values must use the household stochastic discount factor, which is deterministic along the aggregate path. With log utility, the factor for nominal payoffs is $\beta \bar{P}_t \bar{C}_t / (\bar{P}_{t+1} \bar{C}_{t+1})$, where $\bar{P}_t$ is the CE consumption price index. Symmetry, the constant markup, and conditional production imply

$$\bar{P}_t \bar{C}_t = N P_{pt} C_{pt} = \frac{\eta}{\eta - 1} w_t R_t. \quad (\text{OA.C.32})$$

Converting next-period labor returns into current labor units therefore gives

$$\beta \frac{\bar{P}_t \bar{C}_t}{\bar{P}_{t+1} \bar{C}_{t+1}} \frac{w_{t+1}}{w_t} = \beta \frac{R_t}{R_{t+1}}. \quad (\text{OA.C.33})$$

Thus the labor-unit CE discount factor is $\beta R_t / R_{t+1}$; applying a constant β to unadjusted labor-unit returns generally defines a different transition problem.⁵ Equivalently, multiplying labor-unit payoffs and values at each date t by $1/R_t$ gives a recursion with constant β .

Under household intertemporal valuation, the stationary normalization and price construction apply date by date. Common terminal conditions ensure value uniqueness. With the same inherited state, predetermined quantities, cohort characteristics, and physical law of motion, both problems admit the same real allocation paths.

Restricted efficiency. If the equilibrium is unique, or a common selection rule restricts both problems to the same allocation or path, the CE solves the corresponding restricted problem. Along a common path, consumption agrees at every date and discounted welfare converges, giving (OA.C.29). Stationary welfare is $\log \bar{C} / (1 - \beta)$. With multiple admissible equilibria, the planner can select a welfare-maximizing equilibrium if one exists; efficiency of every CE need not follow. $\square$

Corollary OA.2 (The value of capability choice). *Starting from a stationary CE's firm distribution, inherited capabilities $(M_{-1}, Z_{-1}) = (m_0^{\text{CE}}, z_0^{\text{CE}})$, and predetermined quantities, the restricted planner with capability choice attains $W^{\text{SP}} \geq W^{\text{CE}}$.*

Proof. Set $(a_t^{m, \text{SP}}, a_t^{z, \text{SP}}) = (m_0^{\text{CE}}, z_0^{\text{CE}})$ for every $t \geq 0$. Equations (OA.C.3)–(OA.C.5) reproduce

⁵For comparison, Arkolakis et al. (2018, Appendix F) specify a constant β in the transition Bellman equation with the wage normalized to one; their replication code uses the same convention.

the benchmark capabilities at every date, and (OA.C.6) reproduces their acquisition cost. Retaining the CE entry, production, and product switching allocation preserves physical feasibility and, by Proposition OA.3(i), the valuation conditions. Starting from the benchmark distribution and predetermined quantities, this stationary path is therefore admissible. Maximizing welfare over a set containing that path gives the result without requiring CE uniqueness. $\square$

Scope of the comparison. The fixed-capability replication result concerns efficiency under the maintained restrictions; it does not establish unrestricted efficiency or eliminate the production and selection distortions caused by imperfect learning. The planner respects firms' information and retains the assigned exit-value convention. Its assigned exit values can differ from their CE counterparts across allocations or capabilities; equality holds at a replicated allocation. Using these values for entry preserves the decision structure without establishing equivalence to joint portfolio optimization with zero physical scrap or welfare invariance to the exit convention. Replication uses the same physical law of motion in both problems; under a common numerical approximation, it applies to the approximated system.

Allowing capability choice measures the welfare gain from changing pre-entry investment within this restricted decision structure. Corollary OA.2 establishes a weak welfare gain because the baseline CE remains feasible; it does not determine the gain's magnitude or the direction of changes in managerial and productivity investment.

OA.D Empirical Appendix

OA.D.1 Data Construction

Data Sources Our main data source is the U.S. Census Bureau's quinquennial Census of Manufactures (CMF), linked to the Longitudinal Business Database (LBD). Our sample covers eight census years from 1977 to 2012. The CMF covers U.S. manufacturing establishments with at least one paid employee and reports establishment-level employment, payroll, worker hours, input costs, value added, capital expenditures, and inventories. Crucially for our analysis, it reports product codes—five-digit SIC or seven-digit NAICS codes—and the value of shipments

by product.⁶ See [Bernard et al. \(2010\)](#), [Kehrig et al. \(2011\)](#), and [Kehrig and Vincent \(2021\)](#) for further details.

The LBD provides annual information on U.S. private nonfarm establishments with paid employees from 1976 onward, including employment, payroll, industry codes, and location. We link CMF establishments to the LBD and use firm identifiers to aggregate establishment-level observations to the firm level. The linked data allow us to measure firm size, age, productivity, employment growth, and entry and exit. See [Jarmin and Miranda \(2002\)](#) and [Chow et al. \(2021\)](#) for details.

We use industry-level variables from the NBER-CES Manufacturing Industry Database described in [Becker et al. \(2013\)](#). The version we use covers 1958–2018 and provides annual measures of shipments, value added, employment, payroll, input costs, investment, total factor productivity, and price deflators. Industries are defined by four-digit SIC or six-digit NAICS codes.

In addition, we use the Management and Organizational Practices Survey (MOPS), an establishment-level survey of management practices and organizational structure in U.S. manufacturing, with waves for 2010, 2015, and 2021.⁷ We construct three measures of management and information processing. The first is the average structured management score developed by [Bloom et al. \(2019\)](#). The other two capture data availability (Q27) and the intensity of data use in decision making (Q28). We aggregate all three establishment-level measures to the firm level using value-of-shipments weights.

Product Code Concordances We define products at the five-digit SIC level, using 1987 SIC codes throughout the sample.⁸ For 1977–1997, we use product codes that the Census Bureau concords to 1987 SIC codes. To extend this consistent product definition to subsequent census years, we use the 1997 CMF, which reports 1987 SIC and 1997 NAICS product codes.

Starting with the concordance between 1987 SIC and 1997 NAICS product codes, we use the 2002 Census bridge file to link 1997 and 2002 NAICS codes. We repeat this procedure for 2007

⁶For example, SIC industry 3651 includes radios and radio combinations (36511), television receivers (36512), and consumer high-fidelity components (36514).

⁷See [Bloom et al. \(2019\)](#) for details on the survey and the structured management score.

⁸We obtain product-level information from the CMF product trailers, using five-digit SIC codes through 1997 and seven-digit NAICS codes thereafter. For 1977–1997, the files contain both “as-collected” product codes and codes concorded to the 1987 SIC revision.

and 2012, mapping product codes in each year to 1987 SIC codes. To maintain consistency within firms over 2002–2012, we match firm-product pairs to the preceding census and retain their SIC codes. For pairs without a match, we assign SIC codes using the concordance. When a NAICS product code maps to multiple SIC codes, we assign the SIC code most frequently matched to that NAICS code.

Firm Age and Longitudinal Identifiers We use firm age constructed by the Census Bureau following [Haltiwanger et al. \(2013\)](#). When a firm first appears, its age equals that of its oldest establishment; firm age increases annually thereafter. We define young firms as those aged five or less. Because the LBD begins in 1976, firm ages are left-censored for firms operating that year. We use the 1972 CMF to adjust these ages and reduce misclassification of young firms.⁹ We measure firm size by total employment or total value of shipments. All nominal values are deflated to 1997 dollars using industry-level deflators for the relevant variables.

Our baseline analysis uses the Census Bureau’s firm identifier, `lbd fid`, which we refer to as the “firm identifier” throughout. As described in [Chow et al. \(2021\)](#), this identifier addresses the recycling of firm identifiers in the legacy LBD but does not fully resolve breaks associated with firm reorganizations. Such breaks can affect our ability to track firms’ product portfolios over time. To assess the robustness of our results, we construct alternative longitudinal firm identifiers following [Dent et al. \(2018\)](#).

OA.D.2 Summary Statistics

Table [OA.11](#) summarizes the firm characteristics, product-specific expertise, and management measures used in the empirical analysis. Panel A distinguishes young firms, aged five or less, from mature firms. Panels B and C report the distributions of expertise and management measures in their respective samples.

OA.D.3 Additional Descriptive Evidence

Table [OA.12](#) reports the shares of firms engaging in product switching and their output shares by age group, both including and excluding firm entry and exit. To distinguish product switch-

⁹For LBD firms owning establishments observed in the 1972 CMF, we add four years to firm age if beginning-of-year finished goods inventories (FIB) equal zero and five years if they are positive.

Table OA.11: Summary Statistics

	All firms	Young firms	Mature firms
<i>Panel A: Firm characteristics</i>			
Employment	118.8 (1,469.0)	32.5 (77.8)	131.3 (1,571.0)
Firm age	14.74 (8.42)	2.91 (1.62)	16.44 (7.59)
Number of products	2.189 (2.896)	1.877 (1.509)	2.233 (3.041)
Observations	741,000	93,000	648,000
	Mean	Std. dev.	Observations
<i>Panel B: Product-specific expertise</i>			
Normalized expertise, q_{ip}	0.366	0.244	1,622,000
<i>Panel C: Management measures</i>			
Structured management score	0.589	0.159	46,000
Data availability	0.452	0.285	46,000
Intensity of data use	0.432	0.250	46,000

Notes: Panel A reports means and standard deviations (in parentheses) across firm-year observations, including entrants and exiters. The analysis sample consists of firms with estimable product-specific expertise and observations in at least two census years. Young firms are aged five or less; mature firms are aged six or more. Products are defined by five-digit 1987 SIC codes. Panel B reports statistics for normalized expertise across firm-product-year observations in the product-dropping regression sample. Panel C reports firm-level management measures constructed from MOPS responses using value-of-shipments weights. All three management measures range from zero to one.

Table OA.12: Share of Product Switching Firms

variables	Including entry and exit			Continuers only		
	All	Age 0-5	Age 6 +	All	Age 0-5	Age 6 +
Product adding only						
Share of firms	0.094	0.100	0.093	0.128	0.130	0.128
Output share	0.058	0.098	0.057	0.067	0.139	0.066
Product dropping only						
Share of firms	0.398	0.329	0.407	0.182	0.143	0.187
Output share	0.282	0.404	0.280	0.169	0.146	0.169
Product churning						
Share of firms	0.248	0.310	0.239	0.336	0.394	0.327
Output share	0.582	0.284	0.588	0.674	0.409	0.678

Notes: Firms in the manufacturing sector from 1982 to 2007.

Table OA.13: Share of Multi-Product Firms

Classification	Share of firms			Output share		
	All	Age 0-5	Age 6 +	All	Age 0-5	Age 6 +
sic5	0.450	0.432	0.452	0.891	0.615	0.896
sic4	0.332	0.301	0.336	0.838	0.451	0.845
sic2	0.183	0.153	0.187	0.702	0.267	0.711

Notes: Firms in the manufacturing sector from 1982 to 2007.

ing within continuing firms from changes associated with entry and exit, we focus on the last three columns. In the pooled sample, 65% of continuing firms switch products over a five-year interval, accounting for 91% of their total output. These patterns underscore the prevalence and economic importance of product switching, consistent with [Bernard et al. \(2010\)](#).

Product switching often involves both adding and dropping products, which we refer to as product churning. Approximately 52% of product-switching firms (34% out of 65%) both add and drop products; the remainder either add or drop products. Product churning is more prevalent among young firms, whereas dropping products without adding others is less common than among mature firms.

Table [OA.13](#) reports the prevalence of multi-product firms at three aggregation levels: five-digit SIC products, four-digit SIC industries, and two-digit SIC sectors. Under our baseline five-digit definition, multi-product firms constitute 45% of firms but account for 89% of output. Multi-product firms also span multiple industries and sectors. They are prevalent even among

Table OA.14: Number of Products Produced/Added/Dropped

	Including entry and exit			Continuers only		
	All	Age 0-5	Age 6 +	All	Age 0-5	Age 6 +
Number of products firms produce						
Mean	2.189	1.877	2.233	2.192	1.645	2.269
Stdev.	(2.896)	(1.509)	(3.041)	(3.153)	(1.377)	(3.321)
Number of products firms add						
Mean	1.27	1.831	1.189	1.244	1.605	1.194
Stdev.	(1.831)	(1.518)	(1.858)	(1.880)	(1.374)	(1.935)
Number of products firms drop						
Mean	1.273	1.218	1.281	0.941	0.790	0.963
Stdev.	(1.981)	(1.482)	(2.043)	(1.841)	(1.101)	(1.922)

Notes: Firms in the manufacturing sector from 1982 to 2007.

young firms, accounting for 43% of firms and 62% of output in this age group. Single-product firms, by contrast, constitute 55% of firms but account for 11% of output. Firms that produce a single product and never switch products during the sample represent 14.8% of firms and 2.8% of output.

Table OA.14 reports the means and standard deviations of the number of products firms produce, add, and drop, for continuing firms, including single-product firms. On average, firms produce 2.2 products and add 1.2 and drop 0.9 products over five years.¹⁰ Young firms produce fewer products than mature firms (1.6 versus 2.3), but add more (1.6 versus 1.2) and drop fewer (0.8 versus 1.0).

OA.D.4 Product Additions over the Firm Life cycle

We examine how the probability of adding products and the expertise of these additions vary with firm age. Let $\mathcal{I}_{ijt}^{add}$ equal one if firm i adds at least one product between census years t and $t + 5$, following the definition in Section 3.1. To assess the expertise of these additions, we define $\mathcal{I}_{ijt}^{add,high}$ to equal one if the average expertise of newly added products observed at $t + 5$ exceeds the average expertise of the firm's products at t . For each outcome, we estimate:

$$Y_{ijt} = \alpha + \beta \mathcal{I}_{ijt}^{young} + X_{ijt} \gamma + \delta_i + \delta_{jt} + \varepsilon_{ijt}, \quad (\text{OA.D.1})$$

¹⁰Bernard et al. (2010) report an average of 3.5 products among multi-product firms.

Table OA.15: Product Add and Retention

	$Add_{t:t+5}$	$Add_{t:t+5}^{high}$
Young Firm $_{ijt}$	0.148*** (0.005)	-0.112*** (0.017)
Observations	682,000	66,000
Fixed effects	i, jt	i, jt
Controls	Full	Full

Notes: Estimates for industry-year, firm fixed effects, and the constant are suppressed. Robust standard errors clustered at the firm's modal five-digit SIC product level are reported in parentheses. Observations are unweighted, and observation counts are rounded due to the Census Bureau disclosure avoidance procedures. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

where $Y_{ijt} \in \{\mathcal{I}_{ijt}^{add}, \mathcal{I}_{ijt}^{add,high}\}$, and δ_{jt} denotes an industry-year fixed effect.

Column 1 of Table OA.15 shows that young firms are 14.8 percentage points more likely to add a product over the next five years. Column 2 shows that, in the sample used to assess addition quality, young firms are 11.2 percentage points less likely to have additions whose average expertise exceeds that of their existing portfolio.¹¹

Together, these results suggest that young firms experiment with products but exhibit weaker selection on expertise. These patterns are consistent with the model's learning mechanism: lower information-processing ability limits firms' ability to distinguish high-expertise from low-expertise products early in the firm life cycle.

OA.D.5 Product Additions, Drops, and Firm Performance

To examine how product portfolio adjustments relate to firm performance, we estimate:

$$\Delta y_{ijt} = \alpha + \beta_1 \mathcal{I}_{ijt}^{drop} + \beta_2 \mathcal{I}_{ijt}^{add} + X_{ijt} \gamma + \delta_i + \delta_{jt} + \varepsilon_{ijt}, \quad (\text{OA.D.2})$$

where Δy_{ijt} denotes the change in log real value added, employment, total factor productivity, or labor productivity between t and $t + 5$.¹² The indicators $\mathcal{I}_{ijt}^{drop}$ and $\mathcal{I}_{ijt}^{add}$ equal one if firm i drops or adds at least one product over the same interval, respectively.

¹¹Full regression results appear in Appendix OA.D.8.

¹²Value added equals Total Value of Shipments + Final Inventory Investment + Work-in-Progress Goods - Resales - Material Inputs - Energy Expenditures. Labor productivity is real value of shipments divided by total employment. We use the Census Bureau's TFP beta 8 measure available in the CMF.

Table OA.16: Product Add/Drop and Firm Performance

	ΔV_{add}	ΔEmp	ΔTFP	ΔLP
$Add_{t:t+5}$	0.830*** (0.019)	0.802*** (0.019)	0.735*** (0.019)	0.762*** (0.019)
$Drop_{t:t+5}$	-0.754*** (0.019)	-0.731*** (0.018)	-0.686*** (0.017)	-0.713*** (0.018)
Observations	682,000	682,000	648,000	648,000
Fixed effects	i, jt	i, jt	i, jt	i, jt
Controls	Full	Full	Full	Full

Notes: Estimates for industry-year, firm fixed effects, and the constant are suppressed. Robust standard errors clustered at the firm's modal five-digit SIC product level are reported in parentheses. Observations are unweighted, and observation counts are rounded due to the Census Bureau disclosure avoidance procedures. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table OA.17: Firm-Level Output and Long-run Performance Growth Decomposition

	Output			Long-run Performance (θ_{ip})		
	all	age 0-5	age 6 +	all	age 0-5	age 6 +
Total changes	0.112	0.662	0.104	0.017	0.130	0.016
Within share	0.543	0.444	0.545	0	0	0
Between share	0.008	0.002	0.009	0.318	0.126	0.320
Cross share	0.009	-0.005	0.010	0	0	0
Addition share	0.424	0.554	0.422	0.574	0.853	0.572
Drop share	0.015	0.005	0.015	0.107	0.022	0.108

Notes: Firms in the manufacturing sector from 1982 to 2007.

Table OA.16 shows that product additions are positively associated with growth across all four performance measures, while product drops are negatively associated with growth.¹³ Under this additive specification, firms that both add and drop products have higher predicted growth than firms that do neither. The implied differences, $\hat{\beta}_1 + \hat{\beta}_2$, are 0.076 log points for real value added, 0.071 for employment, and 0.049 for both total factor productivity and labor productivity.¹⁴

OA.D.6 Decomposition of Firm-Level Growth in Output and Product Expertise

We conduct an accounting decomposition of changes in the shipment-weighted average of log real product shipments and product-specific expertise. This decomposition distinguishes changes within continuing firm-product pairs, reallocation across continuing products, and product additions and drops.

Define within-firm output share of product p at t for firm i as $\omega_{ipt} = \frac{PV_{ipt}}{\sum_{p \in \mathcal{P}_{it}} PV_{ipt}}$, where PV_{ipt} denotes the real product value of shipments of product p by firm i at t , and $\mathcal{P}_{it}$ is the set of products in firm i 's product portfolio at t . Let y_{ipt} denote either the log of real product value of shipments, $\log(PV_{ipt})$, or the firm-product long-run performance, θ_{ip} . We compute the shipment-weighted average log product shipments or expertise as $y_{it} = \sum_{p \in \mathcal{P}_{it}} \omega_{ipt} y_{ipt}$.

We then decompose the change in each shipment-weighted average as:

$$\begin{aligned} \Delta y_{it} = & \underbrace{\sum_{p \in \mathcal{C}_{it}} \omega_{ip,t-5} \Delta y_{ipt}}_{\text{within term}} + \underbrace{\sum_{p \in \mathcal{C}_{it}} (y_{ip,t-5} - y_{it-5}) \Delta \omega_{ipt}}_{\text{between term}} + \underbrace{\sum_{p \in \mathcal{C}_{it}} \Delta \omega_{ipt} \Delta y_{ipt}}_{\text{cross term}} \\ & + \underbrace{\sum_{p \in \mathcal{N}_{it}} \omega_{ipt} (y_{ipt} - y_{it-5})}_{\text{addition term}} - \underbrace{\sum_{p \in \mathcal{X}_{it}} \omega_{ip,t-5} (y_{ip,t-5} - y_{it-5})}_{\text{drop term}}. \end{aligned} \quad (\text{OA.D.3})$$

$\Delta X_{.t} = X_{.t} - X_{.t-5}$ is the change in any variable $X_{.t}$ between $t - 5$ and t , $\mathcal{C}_{it}$ is the set of continuing products that firm i produces in both $t - 5$ and t , $\mathcal{N}_{it}$ is the set of products newly added by firm i in t , and $\mathcal{X}_{it}$ is the set of products dropped by firm i in t .

Table OA.17 presents decomposition results for shipment-weighted average log product shipments and expertise (θ_{ip}). We normalize each term by the total change to quantify its contribution to growth. We refer to these values as shares. The first three columns show that growth in the shipment-based measure is primarily driven by within-product growth (within share) and new product addition (addition term). For young firms, however, new product addition plays a more prominent role than within-product growth. The combined contribution of product-switching components, addition and drop shares, is 28% higher for young firms than for mature firms (0.554 + 0.005 vs. 0.422 + 0.015).

¹³Appendix Table OA.26 reports full regression results.

¹⁴For related evidence linking product switching to firm growth and productivity, see Bernard et al. (2010) and Argente et al. (2018).

OA.D.7 Full Tables: Product Dropping

The main text presents the product-dropping regressions through the interacted specification in equation (3.41). The specifications for overall selection on performance and average age differences are, respectively,

$$\mathcal{I}_{ipjt}^{drop} = \alpha + \beta Q_{ipt} + X_{ipjt}\gamma_1 + X_{ijt}\gamma_2 + \delta_i + \delta_{pt} + \varepsilon_{ipjt}. \quad (\text{OA.D.4})$$

$$\mathcal{I}_{ipjt}^{drop} = \alpha + \beta \mathcal{I}_{ijt}^{young} + X_{ipjt}\gamma_1 + X_{ijt}\gamma_2 + \delta_i + \delta_{pt} + \varepsilon_{ipjt}, \quad (\text{OA.D.5})$$

where the performance measures, controls, and fixed effects are defined in the main text. For management and information processing, we estimate

$$\mathcal{I}_{ipjt}^{drop} = \alpha + \beta_1 q_{ip} + \beta_2 m_{ijt} + \beta_3 q_{ip} \times m_{ijt} + X_{ipjt}\gamma_1 + X_{ijt}\gamma_2 + \delta_i + \delta_{pt} + \varepsilon_{ipjt}, \quad (\text{OA.D.6})$$

where m_{ijt} denotes the structured management score, data availability (Q27), or data use intensity (Q28). The tables below report full results.

Table OA.18: Probability of Product Drop (PV)

VARIABLES	<i>Drop_{t:t+5}</i>	<i>Drop_{t:t+5}</i>	<i>Drop_{t:t+5}</i>	<i>Drop_{t:t+5}</i>	<i>Drop_{t:t+5}</i>
$\log(PV_{ipt})$	-0.055*** (0.002)	-0.054*** (0.002)	-0.053*** (0.002)	-0.053*** (0.002)	-0.052*** (0.002)
$\log(\text{Emp})$			-0.020*** (0.002)	-0.009*** (0.002)	-0.012*** (0.002)
$\log(\text{Total Assets})$			0.003*** (0.001)	0.019*** (0.004)	0.019*** (0.004)
# establishments			-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)
# products			0.007*** (0.000)	0.007*** (0.000)	0.006*** (0.000)
$\log(\text{CA})$			-0.002*** (0.001)	-0.002*** (0.001)	-0.002*** (0.001)
$\log(\text{CS})$			-0.004*** (0.001)	-0.004*** (0.001)	-0.003*** (0.001)
TFP			-0.004* (0.002)	-0.006** (0.002)	-0.007*** (0.003)
$\log(\text{Prod. Emp})$				-0.009*** (0.001)	-0.010*** (0.001)
$\log(\text{Equipment})$				-0.028*** (0.004)	-0.029*** (0.004)
$\log(\text{Structure})$				0.011*** (0.003)	0.011*** (0.003)
$\log(\text{Material})$				-0.004*** (0.001)	-0.004*** (0.001)
$\log(\text{Energy})$				0.004*** (0.001)	0.003*** (0.001)
Tenure		0.004*** (0.000)			0.004*** (0.001)
Tenure squared		-0.000*** (0.000)			-0.000*** (0.000)
Constant	0.943*** (0.010)	0.942*** (0.010)	0.965*** (0.015)	0.978*** (0.016)	1.004*** (0.017)

Notes: Estimates for product-year, firm fixed effects, and the constant are suppressed. Observations are unweighted.
 * p<0.1, ** p<0.05, *** p<0.01.

Table OA.19: Probability of Product Drop (θ_{ip})

VARIABLES	$Drop_{t:t+5}$	$Drop_{t:t+5}$	$Drop_{t:t+5}$	$Drop_{t:t+5}$	$Drop_{t:t+5}$
θ_{ip}	-1.212*** (0.029)	-1.028*** (0.022)	-1.189*** (0.029)	-1.187*** (0.029)	-1.074*** (0.021)
log(Emp)			-0.035*** (0.002)	-0.015*** (0.002)	-0.016*** (0.002)
log(Total Assets)			0.002* (0.001)	0.023*** (0.004)	0.021*** (0.004)
# establishments			-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)
# products			0.005*** (0.000)	0.005*** (0.000)	0.005*** (0.000)
log(CA)			0 (0.001)	0 (0.001)	0.002*** (0.001)
log(CS)			-0.005*** (0.001)	-0.005*** (0.001)	-0.002*** (0.001)
TFP			-0.011*** (0.002)	-0.020*** (0.002)	-0.013*** (0.002)
log(Prod. Emp)				-0.009*** (0.001)	-0.010*** (0.001)
log(Equipment)				-0.025*** (0.004)	-0.025*** (0.004)
log(Structure)				0.005** (0.003)	0.006** (0.003)
log(Material)				-0.013*** (0.001)	-0.008*** (0.001)
log(Energy)				0.001 (0.001)	0.002* (0.001)
log(PV_{ipt})		-0.021*** (0.001)			-0.017*** (0.001)
Tenure		0.041*** (0.001)			0.042*** (0.001)
Tenure squared		-0.001*** (0.000)			-0.001*** (0.000)
Constant	1.053*** (0.015)	0.963*** (0.014)	1.164*** (0.021)	1.221*** (0.022)	1.111*** (0.022)

Notes: Estimates for product-year, firm fixed effects, and the constant are suppressed. Observations are unweighted.

* p<0.1, ** p<0.05, *** p<0.01.

Table OA.20: Probability of Product Drop (Young Firm)

VARIABLES	<i>Drop_{t:t+5}</i>	<i>Drop_{t:t+5}</i>	<i>Drop_{t:t+5}</i>	<i>Drop_{t:t+5}</i>	<i>Drop_{t:t+5}</i>
Young Firm _{it}	-0.084*** (0.004)	-0.073*** (0.003)	-0.092*** (0.004)	-0.093*** (0.004)	-0.075*** (0.003)
log(Emp)			-0.050*** (0.002)	-0.025*** (0.002)	-0.015*** (0.002)
log(Total Assets)			-0.008*** (0.001)	0.015*** (0.005)	0.019*** (0.004)
# establishments			-0.004*** (0.000)	-0.004*** (0.000)	-0.002*** (0.000)
# products			0.011*** (0.001)	0.011*** (0.001)	0.006*** (0.000)
log(CA)			-0.003*** (0.001)	-0.002*** (0.001)	-0.001** (0.001)
log(CS)			-0.002** (0.001)	-0.002** (0.001)	-0.003*** (0.001)
TFP			-0.014*** (0.002)	-0.027*** (0.003)	-0.008*** (0.003)
log(Prod. Emp)				-0.008*** (0.001)	-0.010*** (0.001)
log(Equipment)				-0.032*** (0.004)	-0.029*** (0.004)
log(Structure)				0.011*** (0.003)	0.011*** (0.003)
log(Material)				-0.019*** (0.001)	-0.005*** (0.001)
log(Energy)				-0.002* (0.001)	0.003*** (0.001)
log(<i>PV_{ipt}</i>)		-0.054*** (0.002)			-0.052*** (0.002)
Tenure		0.002*** (0.000)			0.003*** (0.000)
Tenure squared		-0.000*** (0.000)			-0.000*** (0.000)
Constant	0.591*** (0.000)	0.951*** (0.011)	0.828*** (0.015)	0.918*** (0.016)	1.038*** (0.018)

Notes: Estimates for product-year, firm fixed effects, and the constant are suppressed. Observations are unweighted.

* p<0.1, ** p<0.05, *** p<0.01.

Table OA.21: Probability of Product Drop (PV $\times$ Young)

VARIABLES	<i>Drop_{t:t+5}</i>	<i>Drop_{t:t+5}</i>	<i>Drop_{t:t+5}</i>	<i>Drop_{t:t+5}</i>	<i>Drop_{t:t+5}</i>
$\log(PV_{ipt})$	-0.056*** (0.002)	-0.055*** (0.002)	-0.054*** (0.002)	-0.054*** (0.002)	-0.052*** (0.002)
Young Firm _{it}	-0.163*** (0.008)	-0.142*** (0.007)	-0.159*** (0.008)	-0.160*** (0.008)	-0.141*** (0.007)
$\log(PV_{ipt}) \times \text{Young}_{it}$	0.012*** (0.001)	0.011*** (0.001)	0.011*** (0.001)	0.011*** (0.001)	0.010*** (0.001)
$\log(\text{Emp})$			-0.025*** (0.002)	-0.013*** (0.002)	-0.015*** (0.002)
$\log(\text{Total Assets})$			0.001 (0.001)	0.019*** (0.004)	0.019*** (0.004)
# establishments			-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)
# products			0.007*** (0.000)	0.007*** (0.000)	0.006*** (0.000)
$\log(\text{CA})$			-0.002*** (0.001)	-0.001* (0.001)	-0.001* (0.001)
$\log(\text{CS})$			-0.003*** (0.001)	-0.003*** (0.001)	-0.003*** (0.001)
TFP			-0.005** (0.002)	-0.008*** (0.003)	-0.009*** (0.003)
$\log(\text{Prod. Emp})$				-0.009*** (0.001)	-0.010*** (0.001)
$\log(\text{Equipment})$				-0.029*** (0.004)	-0.029*** (0.004)
$\log(\text{Structure})$				0.011*** (0.003)	0.011*** (0.003)
$\log(\text{Material})$				-0.004*** (0.001)	-0.005*** (0.001)
$\log(\text{Energy})$				0.004*** (0.001)	0.003*** (0.001)
Tenure		0.003*** (0.000)			0.003*** (0.000)
Tenure squared		-0.000*** (0.000)			-0.000*** (0.000)
Constant	0.960*** (0.011)	0.957*** (0.011)	1.012*** (0.016)	1.027*** (0.017)	1.044*** (0.018)

Notes: Estimates for product-year, firm fixed effects, and the constant are suppressed. Observations are unweighted.
 * p<0.1, ** p<0.05, *** p<0.01.

Table OA.22: Probability of Product Drop ($\theta_{ip} \times \text{Young}$)

VARIABLES	$Drop_{t:t+5}$	$Drop_{t:t+5}$	$Drop_{t:t+5}$	$Drop_{t:t+5}$	$Drop_{t:t+5}$
θ_{ip}	-1.221*** (0.030)	-1.032*** (0.022)	-1.196*** (0.030)	-1.194*** (0.030)	-1.081*** (0.022)
Young Firm _{it}	-0.411*** (0.022)	-0.371*** (0.023)	-0.402*** (0.022)	-0.403*** (0.022)	-0.368*** (0.022)
$\theta_{ip} \times \text{Young}_{it}$	0.395*** (0.044)	0.514*** (0.043)	0.367*** (0.043)	0.368*** (0.043)	0.497*** (0.043)
log(Emp)			-0.040*** (0.002)	-0.020*** (0.002)	-0.018*** (0.002)
log(Total Assets)			0 (0.001)	0.022*** (0.004)	0.020*** (0.004)
# establishments			-0.002*** (0.000)	-0.002*** (0.000)	-0.001*** (0.000)
# products			0.005*** (0.000)	0.005*** (0.000)	0.005*** (0.000)
log(CA)			0 (0.001)	0.001* (0.001)	0.002*** (0.001)
log(CS)			-0.004*** (0.001)	-0.004*** (0.001)	-0.002*** (0.001)
TFP			-0.012*** (0.002)	-0.021*** (0.002)	-0.014*** (0.002)
log(Prod. Emp)				-0.009*** (0.001)	-0.010*** (0.001)
log(Equipment)				-0.026*** (0.004)	-0.026*** (0.004)
log(Structure)				0.005* (0.003)	0.006** (0.003)
log(Material)				-0.014*** (0.001)	-0.008*** (0.001)
log(Energy)				0 (0.001)	0.002* (0.001)
log(PV_{ipt})		-0.021*** (0.001)			-0.017*** (0.001)
Tenure		0.039*** (0.001)			0.039*** (0.001)
Tenure squared		-0.001*** (0.000)			-0.001*** (0.000)
Constant	1.071*** (0.015)	0.980*** (0.014)	1.222*** (0.021)	1.280*** (0.023)	1.146*** (0.022)

Notes: Estimates for product-year, firm fixed effects, and the constant are suppressed. Observations are unweighted.
 * p<0.1, ** p<0.05, *** p<0.01.

Table OA.23: Probability of Product Drop ($\theta_{ip} \times$ Management)

VARIABLES	$Drop_{t:t+5}$	$Drop_{t:t+5}$	$Drop_{t:t+5}$	$Drop_{t:t+5}$	$Drop_{t:t+5}$	$Drop_{t:t+5}$
Management	0.356*** (0.040)	0.465*** (0.041)	0.341*** (0.033)	0.406*** (0.033)	0.348*** (0.035)	0.427*** (0.036)
$\theta_{ip} \times$ MOPS	-0.983*** (0.080)	-1.036*** (0.082)	-0.768*** (0.061)	-0.832*** (0.062)	-0.837*** (0.066)	-0.912*** (0.068)
θ_{ip}	-1.211*** (0.045)	-1.055*** (0.041)	-1.292*** (0.037)	-1.127*** (0.033)	-1.289*** (0.038)	-1.119*** (0.035)
log(Emp)		-0.015*** (0.002)		-0.015*** (0.002)		-0.015*** (0.002)
log(Total Assets)		0.021*** (0.004)		0.020*** (0.004)		0.020*** (0.004)
# establishments		-0.001*** (0.000)		-0.002*** (0.000)		-0.002*** (0.000)
# products		0.005*** (0.000)		0.005*** (0.000)		0.005*** (0.000)
log(CA)		0.002*** (0.001)		0.002*** (0.001)		0.002*** (0.001)
log(CS)		-0.001 (0.001)		-0.001 (0.001)		-0.001 (0.001)
TFP		-0.012*** (0.002)		-0.012*** (0.002)		-0.012*** (0.002)
log(Prod. Emp)		-0.011*** (0.001)		-0.011*** (0.001)		-0.011*** (0.001)
log(Equipment)		-0.025*** (0.004)		-0.025*** (0.004)		-0.025*** (0.004)
log(Structure)		0.005** (0.003)		0.005** (0.003)		0.005** (0.003)
log(Material)		-0.007*** (0.001)		-0.007*** (0.001)		-0.007*** (0.001)
log(Energy)		0.002*** (0.001)		0.002*** (0.001)		0.002*** (0.001)
log(PV_{ipt})		-0.017*** (0.001)		-0.017*** (0.001)		-0.017*** (0.001)
Tenure		0.041*** (0.001)		0.041*** (0.001)		0.041*** (0.001)
Tenure squared		-0.001*** (0.000)		-0.001*** (0.000)		-0.001*** (0.000)
Constant	1.014*** (0.022)	1.048*** (0.028)	0.998*** (0.019)	1.056*** (0.025)	1.012*** (0.019)	1.064*** (0.026)

Notes: Managerial ability is measured using the Management and Organizational Practices Survey (MOPS). Columns (1)–(2) use the overall management index as in Bloom et al. (2019), columns (3)–(4) use the data-availability measure (Q27), and columns (5)–(6) use the data use intensity measure (Q28). Estimates for firm and product-year fixed effects are suppressed. Standard errors clustered at the SIC5 level are reported in parentheses. Observations are unweighted. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

OA.D.8 Full Tables: Firm-Level Evidence

Table OA.24: Product Add and Retention (Young Firm)

	$Add_{t:t+5}$	$Add_{t:t+5}$	$Add_{t:t+5}$
Young Firm _{it}	0.140*** (0.005)	0.148*** (0.005)	0.148*** (0.005)
log(Emp)		0.032*** (0.003)	0.029*** (0.002)
log(Total Assets)		0.010*** (0.001)	-0.005 (0.004)
# establishments		0.012*** (0.004)	0.012*** (0.003)
# products		-0.027*** (0.002)	-0.027*** (0.002)
log(CA)		-0.003*** (0.001)	-0.003** (0.001)
log(CS)		-0.006*** (0.001)	-0.007*** (0.001)
TFP		-0.001 (0.003)	0.008*** (0.003)
log(Prod. Emp)			-0.008*** (0.001)
log(Equipment)			0.021*** (0.003)
log(Structure)			-0.008*** (0.003)
log(Material)			0.010*** (0.001)
log(Energy)			0.006*** (0.001)
Constant	0.350*** (0.002)	0.220*** (0.015)	0.174*** (0.018)

Notes: Estimates for industry-year, firm fixed effects, and the constant are suppressed. Observations are unweighted.
 * p<0.1, ** p<0.05, *** p<0.01.

Table OA.25: High-Performing Product Add and Retention (Young Firm)

	$Add_{t:t+5}^{high}$	$Add_{t:t+5}^{high}$	$Add_{t:t+5}^{high}$
Young Firm _{it}	-0.115*** (0.016)	-0.114*** (0.017)	-0.112*** (0.017)
log(Emp)		-0.018** (0.008)	-0.018** (0.009)
log(Total Assets)		-0.007 (0.005)	0.037*** (0.012)
# establishments		-0.007*** (0.003)	-0.007*** (0.003)
# products		0.020*** (0.003)	0.020*** (0.003)
log(CA)		0.003 (0.003)	0.002 (0.003)
log(CS)		-0.004 (0.003)	-0.003 (0.003)
TFP		-0.020** (0.008)	-0.034*** (0.009)
log(Prod. Emp)			0.021*** (0.003)
log(Equipment)			-0.025*** (0.009)
log(Structure)			-0.015* (0.009)
log(Material)			-0.016*** (0.004)
log(Energy)			-0.013*** (0.005)
Constant	0.347*** (0.003)	0.463*** (0.041)	0.516*** (0.049)

Notes: Estimates for industry-year, firm fixed effects, and the constant are suppressed. Robust standard errors clustered at the firm's modal five-digit SIC product level are reported in parentheses. Observations are unweighted, and observation counts are rounded due to the Census Bureau disclosure avoidance procedures. * p<0.1, ** p<0.05, *** p<0.01.

Table OA.26: Product Add/Drop and Firm Performance

	ΔV_{add}	ΔEmp	ΔTFP	ΔLP
$Add_{t:t+5}$	0.830*** (0.019)	0.802*** (0.019)	0.735*** (0.019)	0.762*** (0.019)
$Drop_{t:t+5}$	-0.754*** (0.019)	-0.731*** (0.018)	-0.686*** (0.017)	-0.713*** (0.018)
log(Emp)	-0.187*** (0.009)	-0.604*** (0.008)	0.099*** (0.007)	0.200*** (0.007)
log(Total Assets)	-0.071*** (0.006)	-0.021*** (0.006)	-0.026** (0.012)	-0.018 (0.011)
# establishments	-0.027*** (0.008)	-0.024*** (0.007)	-0.026*** (0.008)	-0.028*** (0.008)
# products	0.077*** (0.006)	0.079*** (0.005)	0.070*** (0.005)	0.072*** (0.006)
log(CA)	-0.003* (0.002)	0.006*** (0.002)	0.002 (0.002)	0.003 (0.002)
log(CS)	0.014*** (0.002)	0.014*** (0.002)	0.011*** (0.002)	0.012*** (0.002)
TFP	-0.561*** (0.012)	0.130*** (0.006)	-0.369*** (0.022)	-0.088*** (0.026)
log(Prod. Emp)	0.021*** (0.003)	0.028*** (0.003)	0.019*** (0.003)	0.001 (0.003)
log(Equipment)	0.030*** (0.005)	0.075*** (0.005)	0.059*** (0.007)	0.029*** (0.006)
log(Structure)	-0.046*** (0.005)	-0.028*** (0.005)	-0.028*** (0.008)	-0.035*** (0.007)
log(Material)	-0.049*** (0.006)	0.068*** (0.003)	0.023*** (0.006)	-0.042*** (0.008)
log(Energy)	-0.030*** (0.004)	0.042*** (0.002)	0.010*** (0.003)	0.035*** (0.004)
Constant	2.252*** (0.052)	0.424*** (0.047)	-0.415*** (0.081)	-0.633*** (0.100)
Observations	682,000	682,000	648,000	648,000
Fixed effects	i, jt	i, jt	i, jt	i, jt
Controls	Full	Full	Full	Full

Notes: Estimates for industry-year, firm fixed effects, and the constant are suppressed. Robust standard errors clustered at the firm's modal five-digit SIC product level are reported in parentheses. Observations are unweighted, and observation counts are rounded due to the Census Bureau disclosure avoidance procedures. * p<0.1, ** p<0.05, *** p<0.01.

OA.D.9 Robustness

First, differences in observation histories may affect the precision of estimated expertise, particularly for young firms. We therefore re-estimate expertise using only the first two observations per firm-product pair, holding the number of observations constant across pairs. We also use 5-, 10-, and 15-year rolling estimation windows to assess sensitivity to the horizon over which expertise is assumed constant. The interaction between expertise and the young-firm indicator remains positive and statistically significant across these specifications, although its magnitude varies.

Second, expertise estimates may capture firm-level determinants of shipments not fully accounted for by baseline controls. We vary firm-level controls, including a specification without them. We also consider replacing these controls with firm-year fixed effects, which absorb time-varying conditions common to a firm's products. Alternatively, we consider an AKM-style specification (Abowd et al., 1999) and construct product-specific expertise as the time average of residuals within firm-product pairs.¹⁵ The positive age interaction persists when expertise is estimated without firm-level controls.

Third, we assess sensitivity to expertise normalization. We replace the baseline measure with standardized estimates and percentile rankings within industry-year or product-year cells. These alternatives measure expertise relative to firms and products within these cells. The age difference in selection persists across these measures.

Fourth, we assess statistical inference using bootstrapped standard errors. The expertise coefficient remains negative and the interaction with the young-firm indicator positive, with both statistically significant.

Finally, product disappearance may reflect firm exit rather than portfolio adjustment within continuing firms. We re-estimate the product-dropping regressions after excluding firms that subsequently exit. The main age-related selection pattern persists.

¹⁵Appendix OA.D.11 discusses these specifications.

Table OA.27: Probability of Product Drop (Alternative θ_{ip} Estimates)

VARIABLES	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
θ_{ip}	0.037 (0.037)	-0.013 (0.037)	-0.608*** (0.016)	-0.617*** (0.016)	-0.491*** (0.015)	-0.495*** (0.015)	-0.313*** (0.016)	-0.318*** (0.016)
Young Firm		-0.453*** (0.025)		-0.344*** (0.016)		-0.303*** (0.017)		-0.130*** (0.020)
$\theta_{ip} \times \text{Young}$		0.439*** (0.042)		0.376*** (0.025)		0.287*** (0.026)		0.083*** (0.030)
Constant	0.781*** (0.021)	0.849*** (0.020)	0.946*** (0.021)	0.983*** (0.021)	0.897*** (0.019)	0.935*** (0.019)	0.939*** (0.016)	0.974*** (0.017)
Observations	1,622,000	1,622,000	1,622,000	1,622,000	1,622,000	1,622,000	1,622,000	1,622,000
Controls	Full	Full	Full	Full	Full	Full	Full	Full
θ_{ip} estimation	2obs	2obs	5yr rolling	5yr rolling	10yr rolling	10yr rolling	15yr rolling	15yr rolling
VARIABLES	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)
θ_{ip}	-0.050*** (0.001)	-0.050*** (0.001)	-0.158*** (0.005)	-0.160*** (0.005)	-0.049*** (0.001)	-0.050*** (0.001)	-0.160*** (0.005)	-0.163*** (0.005)
Young Firm		-0.108*** (0.004)		-0.148*** (0.005)		-0.108*** (0.004)		-0.148*** (0.005)
$\theta_{ip} \times \text{Young}$		0.025*** (0.002)		0.076*** (0.007)		0.027*** (0.002)		0.076*** (0.007)
Constant	0.553*** (0.020)	0.584*** (0.020)	0.646*** (0.020)	0.678*** (0.020)	0.555*** (0.021)	0.585*** (0.020)	0.646*** (0.020)	0.677*** (0.020)
Observations	1,622,000	1,622,000	1,622,000	1,622,000	1,622,000	1,622,000	1,622,000	1,622,000
Adjusted R^2	0.495	0.497	0.494	0.496	0.495	0.497	0.494	0.496
Controls	Full	Full	Full	Full	Full	Full	Full	Full
θ_{ip} estimation	Norm (jt)	Norm (jt)	Percentile (jt)	Percentile (jt)	Norm (pt)	Norm (pt)	Percentile (pt)	Percentile (pt)

Notes: Columns are based on alternative specification of estimating θ_{ip} . Columns (1)-(2) are based on the first 2 observations for each product-firm pair, (3)-(4) are estimates with 5 year rolling-year basis, (5)-(6) are with 10 year rolling-year basis, (7)-(8) are with 15 year rolling-year basis, (9)-(10) are based on standardization with the industry-year distribution, (11)-(12) are based on the percentile rankings within industry-year, (13)-(14) are based on standardization with the product-year distribution, and (15)-(16) are based on the percentile rankings within product-year. Full-control specifications include the same set of controls as in the baseline regressions. Estimates for product-year and firm fixed effects are suppressed. Observations are unweighted. Standard errors clustered at the SIC5 level are reported in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table OA.28: Probability of Product Drop (Alternative θ_{ip} Estimates)

VARIABLES	$Drop_{t:t+5}$	$Drop_{t:t+5}$
θ_{ip}	-0.563*** (0.016)	-0.580*** (0.017)
Young Firm		-0.317*** (0.014)
$\theta_{ip} \times \text{Young}$		0.394*** (0.027)
Constant	0.855*** (0.024)	0.892*** (0.024)
Observations	1,622,000	1,622,000
Adjusted R^2	0.493	0.495
Controls	Full	Full
θ_{ip} estimation	No controls	No controls

Notes: Results are based on alternative specification of estimating θ_{ip} without firm controls. Estimates for product-year and firm fixed effects are suppressed. Observations are unweighted. Standard errors clustered at the SIC5 level are reported in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table OA.29: Probability of Product Drop (θ_{ip} , bootstrapped standard errors)

VARIABLES	$Drop_{t:t+5}$	$Drop_{t:t+5}$	$Drop_{t:t+5}$	$Drop_{t:t+5}$	$Drop_{t:t+5}$
θ_{ip}	-1.212*** (0.094)	-1.028*** (0.119)	-1.189*** (0.095)	-1.187*** (0.095)	-1.074*** (0.120)
log(Emp)			-0.035*** (0.001)	-0.015*** (0.001)	-0.016*** (0.001)
log(Total Assets)			0.002*** (0.000)	0.023*** (0.001)	0.021*** (0.001)
# establishments			-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)
# products			0.005*** (0.000)	0.005*** (0.000)	0.005*** (0.000)
log(CA)			0 (0.000)	0 (0.000)	0.002*** (0.000)
log(CS)			-0.005*** (0.000)	-0.005*** (0.000)	-0.002*** (0.000)
TFP			-0.011*** (0.001)	-0.020*** (0.001)	-0.013*** (0.001)
log(Prod. Emp)				-0.009*** (0.001)	-0.010*** (0.000)
log(Equipment)				-0.025*** (0.001)	-0.025*** (0.001)
log(Structure)				0.005*** (0.001)	0.006*** (0.001)
log(Material)				-0.013*** (0.001)	-0.008*** (0.001)
log(Energy)				0.001 (0.001)	0.002*** (0.001)
log(PV_{ipt})		-0.021*** (0.002)			-0.017*** (0.002)
Tenure		0.041*** (0.000)			0.042*** (0.000)
Tenure squared		-0.001*** (0.000)			-0.001*** (0.000)

Notes: Estimates for product-year and firm fixed effects are suppressed. Observations are unweighted. Bootstrapped standard errors are reported in parentheses. * p<0.1, ** p<0.05, *** p<0.01.

Table OA.30: Probability of Product Drop ($\theta_{ip} \times \text{Young}$, bootstrapped standard errors)

VARIABLES	$Drop_{t:t+5}$	$Drop_{t:t+5}$	$Drop_{t:t+5}$	$Drop_{t:t+5}$	$Drop_{t:t+5}$
θ_{ip}	-1.221*** (0.091)	-1.032*** (0.118)	-1.196*** (0.092)	-1.194*** (0.092)	-1.081*** (0.120)
Young Firm _{it}	-0.411*** (0.055)	-0.371*** (0.042)	-0.402*** (0.056)	-0.403*** (0.056)	-0.368*** (0.043)
$\theta_{ip} \times \text{Young}_{it}$	0.395*** (0.096)	0.514*** (0.072)	0.367*** (0.098)	0.368*** (0.099)	0.497*** (0.073)
log(Emp)			-0.040*** (0.001)	-0.020*** (0.001)	-0.018*** (0.001)
log(Total Assets)			0 (0.000)	0.022*** (0.001)	0.020*** (0.001)
# establishments			-0.002*** (0.000)	-0.002*** (0.000)	-0.001*** (0.000)
# products			0.005*** (0.000)	0.005*** (0.000)	0.005*** (0.000)
log(CA)			0 (0.000)	0.001** (0.000)	0.002*** (0.000)
log(CS)			-0.004*** (0.000)	-0.004*** (0.000)	-0.002*** (0.000)
TFP			-0.012*** (0.001)	-0.021*** (0.001)	-0.014*** (0.001)
log(Prod. Emp)				-0.009*** (0.001)	-0.010*** (0.000)
log(Equipment)				-0.026*** (0.001)	-0.026*** (0.001)
log(Structure)				0.005*** (0.001)	0.006*** (0.001)
log(Material)				-0.014*** (0.001)	-0.008*** (0.001)
log(Energy)				0 (0.000)	0.002*** (0.001)
log(PV_{ipt})		-0.021*** (0.002)			-0.017*** (0.002)
Tenure		0.039*** (0.000)			0.039*** (0.000)
Tenure squared		-0.001*** (0.000)			-0.001*** (0.000)

Notes: Estimates for product-year and firm fixed effects are suppressed. Observations are unweighted. Bootstrapped standard errors are reported in parentheses. * p<0.1, ** p<0.05, *** p<0.01.

Table OA.31: Probability of Product Drop ($\theta_{ip} \times \text{Firm Age}$)

VARIABLES	$\text{Drop}_{t:t+5}$	$\text{Drop}_{t:t+5}$
θ_{ip}	-0.816*** (0.038)	-0.628*** (0.033)
$\theta_{ip} \times \text{Firm Age}_{it}$	-0.022*** (0.002)	-0.024*** (0.002)
log(Emp)		-0.015*** (0.002)
log(Total Assets)		0.021*** (0.004)
# establishments		-0.002*** (0.000)
# products		0.005*** (0.000)
log(CA)		0.001** (0.001)
log(CS)		-0.004*** (0.001)
TFP		-0.013*** (0.002)
log(Prod. Emp)		-0.010*** (0.001)
log(Equipment)		-0.025*** (0.004)
log(Structure)		0.006** (0.003)
log(Material)		-0.008*** (0.001)
log(Energy)		0.002* (0.001)
log(PV_{ipt})		-0.018*** (0.001)
Tenure		0.041*** (0.001)
Tenure squared		-0.001*** (0.000)
Constant	1.018*** (0.014)	1.070*** (0.022)

Notes: Estimates for product-year, firm fixed effects, and the constant are suppressed. Firm Age is measured as the age of the firm in years. Observations are unweighted. * p<0.1, ** p<0.05, *** p<0.01.

Table OA.32: Number of Input Materials Used per Product by Firms

	Including entry and exit			Continuers only		
	all	age 0-5	age 6 +	all	age 0-5	age 6 +
Mean	20.30	19.45	20.43	21.33	21.50	21.31
Stdev.	(20.29)	(19.86)	(20.35)	(21.00)	(21.34)	(20.95)

Notes: Firms in the manufacturing sector from 1982 to 2007.

OA.D.10 Product-Firm Proximity

Up to this point, we have focused on the relationship between product dropping and adding, as well as the relative expertise of added and dropped products. We now turn to examining the proximity between newly added products and a firm's existing product portfolio. The literature finds that firms tend to produce products close to each other in terms of various measures, such as industry classification or the composition of material inputs used (e.g., [Bernard et al., 2010](#); [Boehm et al., 2022](#)). Nonetheless, we also observe many instances in which firms expand into entirely new industries.

To examine whether the proximity of added products to a firm's existing product portfolio varies over the firm life-cycle, we use the share of input materials used to produce a newly added product that are also used for the firm's existing products as a proxy for product proximity.^{16,17} Let $MaterialCloseness_{ijt}^{add}$ denote the average of the proximity measure of products added by firm i between t and $t + 5$. The mean (standard deviation) of the material-based closeness of added products in the churning case is 0.708 (0.271). Table OA.32 shows the number of input materials used per product by firms across age groups. Focusing on the continuing firms, we find that young firms use slightly more variety of input materials to produce each product, and the variance is also larger compared to mature firms.

We then estimate the following regression model:

$$MaterialCloseness_{ijt}^{add} = \alpha + \beta T_{ijt}^{young} + X_{ijt}\gamma + \delta_i + \delta_{jt} + \varepsilon_{ijt}. \quad (OA.D.7)$$

¹⁶The CMF provides establishment-level data on the list of material inputs used and associated expenditures. By focusing on single-product establishments, we identify the set of material inputs used in the production of each product.

¹⁷We are in the process of exploring an expenditure-weighted version of the material input-based proximity measure, following [Boehm et al. \(2022\)](#).

Table OA.33: Material-Based Closeness of Added Products (Young Firm)

	$Mat_{t:t+5}^{add}$	$Mat_{t:t+5}^{add}$	$Mat_{t:t+5}^{add}$
Young Firm _{it}	-0.005 (0.005)	-0.003 (0.004)	-0.003 (0.004)
log(Emp)		-0.012*** (0.002)	-0.003 (0.003)
log(Total Assets)		-0.004*** (0.001)	-0.001 (0.004)
# establishments		0.005*** (0.002)	0.005*** (0.002)
# products		-0.024*** (0.002)	-0.024*** (0.002)
log(CA)		-0.003*** (0.001)	-0.003*** (0.001)
log(CS)		-0.007*** (0.001)	-0.007*** (0.001)
TFP		-0.003 (0.002)	-0.007*** (0.002)
# materials		0.005*** (0.000)	0.005*** (0.000)
log(Prod. Emp)			-0.001 (0.001)
log(Equipment)			-0.005* (0.003)
log(Structure)			0.004* (0.002)
log(Material)			-0.005*** (0.001)
log(Energy)			-0.007*** (0.001)
Constant	0.520*** (0.003)	0.489*** (0.011)	0.521*** (0.013)
Observations	198,000	198,000	198,000
Fixed effects	<i>i, jt</i>	<i>i, jt</i>	<i>i, jt</i>

Notes: Estimates for industry-year, firm fixed effects, and the constant are suppressed. Robust standard errors clustered at the firm's modal five-digit SIC product level are reported in parentheses. Observations are unweighted, and observation counts are rounded due to the Census Bureau disclosure avoidance procedures. * p<0.1, ** p<0.05, *** p<0.01.

Table OA.33 shows that relative to mature firms, young firms tend to add and retain products that are more distant from their existing product portfolio. Results are also consistent with industry-based proximity.

OA.D.11 Discussion on Product-Firm Expertise Estimation

One alternation of the expertise estimation is to use product-year and firm-year fixed effects as follows, instead of controlling for the time-varying firm and industry characteristics:

$$y_{ipjt} = \theta_{ip} + X_{ipjt}\gamma + \delta_{pt} + \delta_{it} + \varepsilon_{ipjt}, \quad (\text{OA.D.8})$$

where δ_{pt} is a product-year fixed effect, and δ_{it} is a firm-year fixed effect.

δ_{pt} controls for product specific characteristics as well as product-year specific shocks. Again, this is because each product is subject to different production technology or demand structure. Therefore, it is possible that there are several other effects affecting the value of shipment, attributed to product-year specific technology or demand shocks.

In a similar fashion, firms or entrepreneurs have different base sets of available resources or ability to manufacture products (i.e. a financing constraint, customer capital, brand values, and marketing resources, etc.). The value of shipment for each product can also be influenced by these firm-level properties. Furthermore, any types of industry-specific characteristics that can give a substantial impact on the total value of shipment are absorbed by the firm-year fixed effect. Therefore, equation (OA.D.8) could be one alternative of the baseline quality estimation in (3.40).

However, estimation based on (OA.D.8) faces several limitations due to the inclusion of multiple fixed effects. In particular, estimating firm-product and firm-year fixed effects simultaneously leads to the exclusion of a non-negligible portion of the sample, as some products appear in only one firm or some firms produce only a single product in a given year. Therefore, an alternative approach is to replace the firm-year fixed effect δ_{it} with an extensive set of firm-level controls X_{ijt} , which is our baseline specification (3.40).

Alternatively, we could implement an AKM-style specification (Abowd et al., 1999) and estimate the firm-product match quality by taking the time average of the residuals. These specifications include

$$y_{ipjt} = X_{ipjt}\gamma + \delta_{pt} + \delta_{it} + \varepsilon_{ipjt} \quad (\text{OA.D.9})$$

$$y_{ipjt} = X_{ipjt}\gamma_1 + X_{ijt}\gamma_2 + X_{pt}\gamma_3 + \delta_p + \delta_i + \delta_t + \varepsilon_{ipjt} \quad (\text{OA.D.10})$$

$$y_{ipjt} = X_{ipjt}\gamma_1 + X_{ijt}\gamma_2 + \delta_{pt} + \delta_i + \delta_t + \varepsilon_{ipjt}. \quad (\text{OA.D.11})$$

A critical limitation of this approach is the lack of clarity regarding what the time-varying residuals capture, and whether their average offers a more accurate measure of firm-product match quality than the baseline model.

References

- Abowd, John M, Francis Kramarz, and David N Margolis**, “High wage workers and high wage firms,” *Econometrica*, 1999, 67 (2), 251–333.
- Acemoglu, Daron, Ufuk Akcigit, Harun Alp, Nicholas Bloom, and William Kerr**, “Innovation, reallocation, and growth,” *American Economic Review*, 2018, 108 (11), 3450–91.
- Argente, David, Munseob Lee, and Sara Moreira**, “Innovation and product reallocation in the great recession,” *Journal of Monetary Economics*, 2018, 93, 1–20.
- Arkolakis, Costas, Theodore Papageorgiou, and Olga A Timoshenko**, “Firm learning and growth,” *Review of Economic Dynamics*, 2018, 27, 146–168.
- Becker, Randy, Wayne Gray, and Jordan Marvakov**, “NBER-CES manufacturing industry database: Technical notes,” *NBER Working Paper*, 2013, 5809.
- Bernard, Andrew B, Stephen J Redding, and Peter K Schott**, “Multiple-product firms and product switching,” *American Economic Review*, 2010, 100 (1), 70–97.
- Bloom, Nicholas, Erik Brynjolfsson, Lucia Foster, Ron Jarmin, Megha Patnaik, Itay Saporta-Eksten, and John Van Reenen**, “What drives differences in management practices?,” *American Economic Review*, 2019, 109 (5), 1648–1683.
- Boehm, Johannes, Swati Dhingra, and John Morrow**, “The comparative advantage of firms,” *Journal of Political Economy*, 2022, 130 (12), 3025–3100.
- Chow, Melissa C, Teresa C Fort, Christopher Goetz, Nathan Goldschlag, James Lawrence, Elisabeth Ruth Perlman, Martha Stinson, and T Kirk White**, “Redesigning the Longitudinal Business Database,” Technical Report, National Bureau of Economic Research 2021.
- Dent, Robert C, Benjamin W Pugsley, Harrison Wheeler et al.**, “Longitudinal Linking of Enterprises in the LBD and SSL,” Technical Report, Center for Economic Studies, US Census Bureau 2018.
- Dhingra, Swati and John Morrow**, “Monopolistic Competition and Optimum Product Diversity under Firm Heterogeneity,” *Journal of Political Economy*, 2019, 127 (1), 196–232.
- Foster, Lucia, John Haltiwanger, and Chad Syverson**, “The slow growth of new plants: learning about demand?,” *Economica*, 2016, 83 (329), 91–129.
- Haltiwanger, John, Ron S Jarmin, and Javier Miranda**, “Who creates jobs? Small versus large versus young,” *Review of Economics and Statistics*, 2013, 95 (2), 347–361.
- Hottman, Colin J., Stephen J. Redding, and David E. Weinstein**, “Quantifying the Sources of Firm Heterogeneity,” *Quarterly Journal of Economics*, 2016, 131 (3), 1291–1364.
- Jarmin, Ron S and Javier Miranda**, “The longitudinal business database,” *Available at SSRN* 2128793, 2002.
- Kehrig, Matthias and Nicolas Vincent**, “The micro-level anatomy of the labor share decline,” *Quarterly Journal of Economics*, 2021, 136 (2), 1031–1087.
- **et al.**, “The cyclical productivity dispersion,” *US Census Bureau Center for Economic Studies Paper No. CES-WP-11-15*, 2011.
- Sterk, Vincent, Petr Sedláček, and Benjamin Pugsley**, “The nature of firm growth,” *American Economic Review*, 2021, 111 (2), 547–579.
- Stokey, Nancy L.**, *The Economics of Inaction: Stochastic Control Models with Fixed Costs*, Princeton, NJ: Princeton University Press, 2009.
- Stokey, Nancy L and Robert E Lucas Jr**, *Recursive methods in economic dynamics*, Harvard University Press, 1989.